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PLANNING DEVELOPMENT PROJECT BUR 3402

MAGADI ROAD CORRIDOR LINER DEVELOPMENT AND MANAGEMENT PLAN 2024-2034

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DECLARATION

STUDENT DECLARATION

I **CLEVARENCE OTIENO OSANO** hereby declare that this Development Project is my original work and has not been submitted to any other university or institution of higher learning for academic award.

SIGNED..... DATE.....

SUPERVISOR DECLARATION

This Development Project has been submitted for examination with the approval of the University supervisor.

PROF. JEREMIAH AYONGA

SIGNED..... DATE.....

Dedication

I dedicate this research project to my late grandfather, Boaz Ogwen Odiyo, and my father, Erustus Osano Ogwen. Their unwavering encouragement to persevere through all circumstances, reminding me that there is always light at the end of the tunnel, has been a source of immense inspiration. Their words of wisdom and motivation have carried me through many struggles and moments of doubt, enabling me to reach the level I am at today.

Acknowledgment

I wish to express my profound gratitude to the Almighty God for blessing me with good health and the necessary resources to undertake this academic work. I am deeply thankful to my supervisor, Prof. Jeremiah Ayonga, for his continuous constructive contributions and unwavering support throughout the research process. His guidance and encouragement have been instrumental in bringing this work to fruition.

Additionally, I extend my appreciation to my classmates like Job Wafula, Dismike Mike, Viveck Ochieng, Jame Oboiko, and many others for their camaraderie and encouragement, which have been invaluable throughout this journey. Lastly, I am deeply grateful to my entire family for their unwavering support and encouragement, without which this achievement would not have been possible.

Abstract

The Magadi Road Corridor Linear Development and Management Plan outlines a comprehensive strategy for the sustainable development and efficient management of the Magadi Road corridor. This corridor, a vital transportation artery connecting urban and peri-urban areas in [specific region or city], plays a crucial role in regional connectivity and economic growth. The plan integrates urban planning principles with environmental sustainability goals to enhance transportation efficiency, promote economic activities, and improve the quality of life for residents along the corridor.

Key components of the plan include infrastructure improvements such as road widening, junction upgrades, and the implementation of integrated transportation systems to alleviate traffic congestion and improve mobility. Environmental considerations focus on minimizing the corridor's ecological footprint through green infrastructure, stormwater management, and landscaping initiatives. Social aspects emphasize enhancing public spaces, promoting mixed-use development, and fostering community engagement through participatory planning processes.

The management framework includes strategies for governance, maintenance, and monitoring to ensure the long-term sustainability of infrastructure investments. Economic development strategies aim to leverage the corridor's strategic location to attract investments, stimulate business activities, and create employment opportunities. Overall, the Magadi Road Corridor Linear Development and Management Plan aims to transform the corridor into a model of sustainable urban development, integrating infrastructure improvements with environmental stewardship and community well-being.

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CHAPTER ONE: INTRODUCTION

1.1 Background to the Planning Research Project

Magadi Road Corridor is a vital transportation route connecting Nairobi, the capital city of Kenya, with several satellite towns and regions to the southwest, including Ongata Rongai. Ongata Rongai is a rapidly growing peri-urban town located approximately 17 kilometres from Nairobi's central business district. The corridor significantly facilitates commuter traffic, freight movement, and access to various amenities for residents along its path.

AN ANALYSIS OF THE IMPACT OF LINEAR URBAN PATTERN ON THE PROVISION OF SERVICES AND INFRASTRUCTURE

The study probed into the pivotal role of transportation infrastructure, specifically Magadi Road, in shaping the linear urban pattern observed along its route. This pattern is not merely coincidental but rather a result of deliberate interactions between transport infrastructure planning and urban growth dynamics. Magadi Road serves as a primary axis of urban development, attracting significant commercial and residential developments due to its layout and accessibility. While this concentration of activities enhances accessibility and economic opportunities, it also exacerbates challenges in service provision, such as strains on utilities and congestion, necessitating careful planning and infrastructure development to sustain the demands of the linear urban pattern.

The socioeconomic outcomes of the linear pattern along Magadi Road are profound and multidimensional, fostering economic development but also exacerbating social disparities. While the concentration of businesses and services brings prosperity, it also leads to variations in property values, income levels, and community cohesion within the linear corridor. Rapid urbanization and population growth have strained the Magadi road corridor, leading to significant challenges like encroachment of road reserves, traffic congestion and even encroachment of green space areas. To address these challenges, a comprehensive approach is essential, integrating transportation planning measures, spatial planning interventions, and infrastructure upgrades.

Recommendations

The main recommendation for addressing the challenges along the Magadi Road Corridor involves implementing comprehensive infrastructure upgrades and transportation management strategies. This includes expanding road capacity and bypass, improving maintenance, and enhancing public transportation options to alleviate traffic congestion and improve accessibility for commuters and residents. Additionally, integrated land use planning and environmental conservation efforts are necessary to mitigate land use conflicts, preserve natural habitats, and reduce pollution along the corridor. Moreover, fostering social equity and community engagement through inclusive development policies is crucial to ensuring that all residents benefit from the corridor's economic opportunities while addressing disparities in housing, services, and employment.

1.2 Background of the Planning Development Project

1.2.1 MAGADI ROAD CORRIDOR LINEAR DEVELOPMENT AND MANAGEMENT PLAN

The MAGADI ROAD CORRIDOR DEVELOPMENT AND MANAGEMENT PLAN for 2024-2034 aim to capitalize on the findings regarding the linear urban pattern along Magadi Road. Building upon

the analysis highlighting the significance of Magadi Road as a primary axis of urban development, the project seeks to optimize land use patterns and enhance transportation infrastructure to foster sustainable and inclusive growth. By integrating transportation planning measures, such as improving public transit and pedestrian infrastructure, with spatial planning interventions promoting mixed-use development and optimized land use patterns, the plan aims to mitigate congestion, improve accessibility, and address social disparities within the linear corridor.

Through targeted mitigation strategies and investments in infrastructure upgrades and service delivery improvements, the Magadi Road Corridor Integrated Physical and Land Use Plan endeavours to harness the benefits of the linear pattern while minimizing its negative repercussions. By adopting a holistic approach that integrates transportation planning and infrastructure development, the project aims to pave the way for sustainable and inclusive urban development along Magadi Road and similar transportation corridors, ultimately enhancing the quality of life for residents and fostering economic prosperity for businesses along the corridor.

1.2.2 Problem statement

The rapid urbanization and population growth experienced along the Magadi Road corridor have led to significant challenges that impede development projects, as evidenced by research findings. Key among these challenges is traffic congestion, exacerbated by narrow lanes and inadequate bypasses, causing delays and inefficiencies during peak hours. Moreover, infrastructure deficiencies, such as poor road maintenance and limited public transportation options, compromise accessibility and safety for both commuters and residents. These pressing issues underscore the urgent need for comprehensive infrastructure upgrades and transportation management strategies to effectively address the growing demands placed on the corridor.

The corridor faces considerable obstacles stemming from land use conflicts. Competing interests between residential, commercial, and industrial stakeholders often result in zoning disputes, disparities, and regulatory challenges, hindering the timely implementation of much-needed infrastructure improvements. Furthermore, environmental concerns loom large as urbanization encroaches upon natural habitats, green space areas, and agricultural lands. The escalation in vehicular traffic further exacerbates air and noise pollution, posing threats to both environmental sustainability and public health. In light of these multifaceted challenges, a concerted effort is imperative to devise and implement sustainable solutions that mitigate the adverse impacts of rapid development along the Magadi Road corridor.

1.2.3 Justification

The MAGADI ROAD CORRIDOR LINEAR DEVELOPMENT AND MANAGEMENT PLAN addresses the urgent challenges as per the research project findings which included of rapid urbanization and population growth along the corridor, such as traffic congestion, narrow lanes, and inadequate infrastructure. By implementing comprehensive upgrades and transportation strategies, the project aims to improve accessibility, safety, and quality of life for residents and businesses, thereby sustaining economic growth. The plan also tackles land use conflicts by fostering stakeholder collaboration to streamline development and expedite infrastructure improvements. Proactively addressing environmental concerns, the project promotes sustainable development to mitigate urbanization impacts on natural habitats and green spaces. Ultimately, the plan integrates

transportation and spatial planning to create a resilient, efficient urban environment adaptable to future growth, ensuring long-term prosperity for the Magadi Road corridor.

1.2.4 Location and area coverage of the planning area

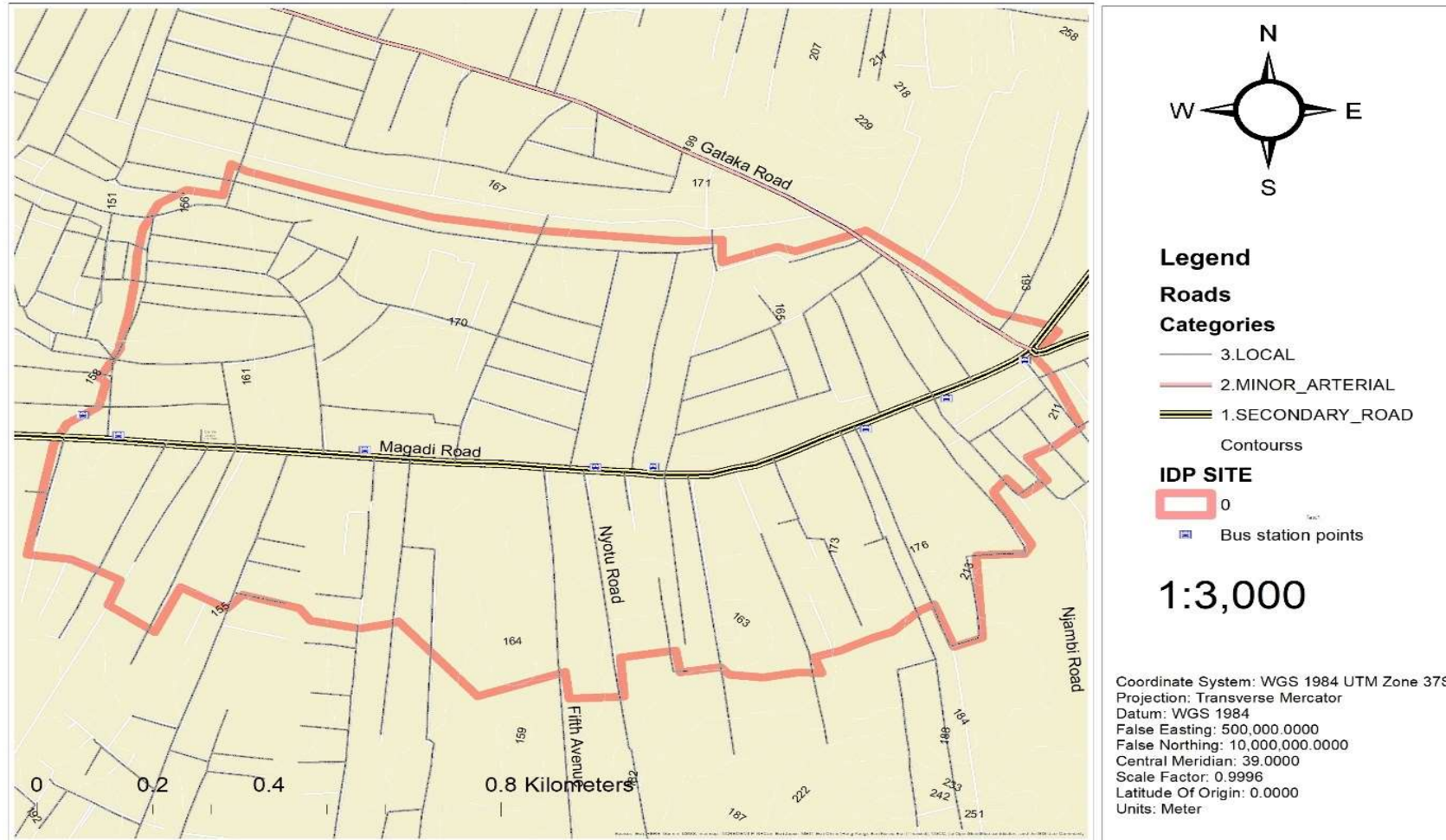
Magadi road, Ongata Rongai Town, Kajiado County

The study area is located in Ngong Municipality, Kajiado North Constituency within two wards Ongata Rongai ward and Nkaimurunya ward, Kajiado County, Kenya. The site area covers approximately a total of about 134Ha running from Magadi-Gataka road junction to Quarry Market.

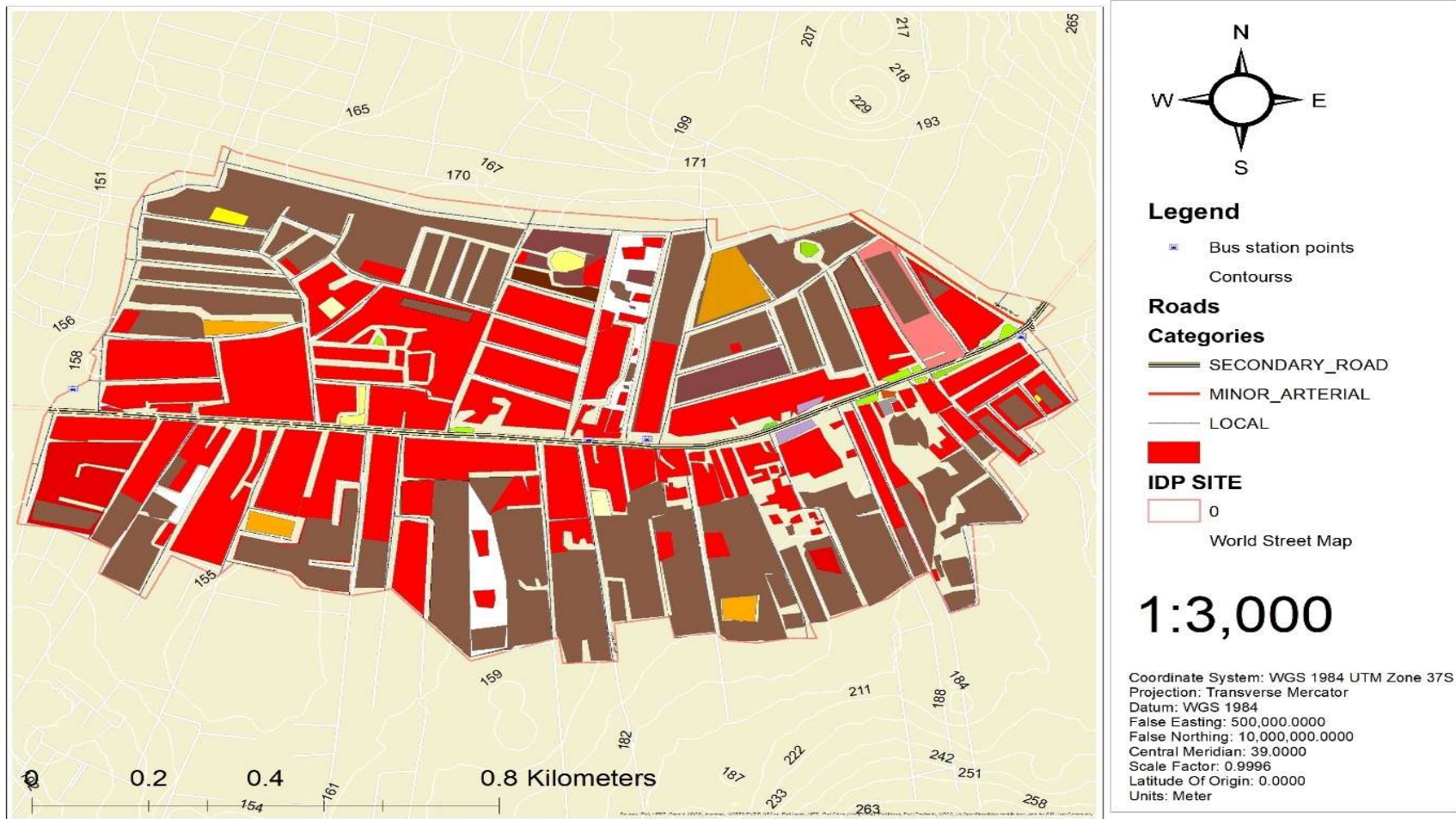
Map 1: Basemap

Map 1:Magadi Road Corridor Digitalized Basemap

Magadi Road Corridor Basemap



Magadi Road Corridor Basemap



1.2.5 PURPOSE/GOALS

The main purposes of the project are to enhance the efficiency and safety of transportation along the Magadi Road Corridor while promoting sustainable development and equitable growth. By addressing infrastructure deficiencies and traffic congestion, the recommendations aim to improve commuter experiences, reduce travel times, and stimulate economic productivity in urban, peri-urban, and rural areas served by the corridor. Furthermore, integrating land use planning and environmental conservation into development projects seeks to preserve natural resources, protect biodiversity, and mitigate the adverse impacts of urbanization on ecosystems and public health. Additionally, prioritizing social equity and community engagement aims to ensure that development initiatives benefit all residents, particularly those from marginalized communities, by providing affordable housing, access to services, and opportunities for socio-economic advancement.

Ultimately, the main purposes of this project are to transform the Magadi Road Corridor into a sustainable and inclusive transportation route that supports economic growth, preserves environmental integrity, and enhances the quality of life for all residents. By adopting a holistic approach that addresses infrastructure, land use, environmental, and social considerations, can unlock the corridor's full potential as a driver of prosperity and well-being in the region. Through collaborative efforts among government agencies, private sector stakeholders, and local communities, the recommendations aim to create a more resilient and vibrant corridor that meets the diverse needs of its users while safeguarding the natural and cultural heritage of the area.

1.2.6 OBJECTIVES

1. To review literature, principles, policies, legislations, development control guidelines and standards on sustainable linear development for enhancing liveability along road corridors.
2. To conduct contextual analysis of the Magadi road corridor planning area in Ongata Rongai Town.
3. To develop alternative plans, strategies, and policy options for the Magadi road corridor Linear Development and Management Plan in Ongata Rongai Town.
4. To assess the alternative plans and to formulate a preferred Magadi road corridor linear development and management Plan to be used in Ongata Rongai Town.
5. To formulate implementation, monitoring, and evaluation framework for Magadi road corridor Integrated Physical Land Use Plan in Ongata Rongai Town.

1.2.7 Limitations and Assumptions of the Project

While comprehensive in its approach, the Magadi Road Corridor Linear Development and Management Plan is subject to several limitations and assumptions that may impact its effectiveness and implementation. Firstly, the project's success relies heavily on the cooperation and coordination of various stakeholders, including government agencies, private developers, and local communities. However, differing priorities, competing interests, and bureaucratic hurdles may impede the seamless execution of the plan, leading to delays and compromises in its objectives.

The project operates under the assumption that the proposed infrastructure upgrades and transportation management strategies will adequately address the challenges posed by rapid urbanization and population growth along the corridor. However, unforeseen factors such as budget constraints, technological limitations, and external economic shocks may hinder the full realization of these interventions, limiting their impact on congestion, accessibility, and safety.

The plan assumes a certain level of environmental sustainability and resilience in its proposed land use and development strategies. However, the actual implementation of these measures may face obstacles related to land acquisition, regulatory compliance, and community resistance, particularly in areas where environmental conservation conflicts with economic development interests. As such, the project may need to adapt its approach to balance competing priorities and address unforeseen environmental challenges effectively.

Lastly, the effectiveness of the Magadi Road Corridor Integrated Physical and Land Use Plan is contingent upon accurate data, reliable forecasting, and robust monitoring and evaluation mechanisms. However, limitations in data availability, quality, and timeliness may constrain the project's ability to assess its progress accurately and adjust its strategies accordingly. Therefore, data collection, stakeholder engagement, and adaptive management approaches are essential to overcome these limitations and ensure the project's long-term success.

1.2.8 Scope of the planning area

The scope of the Magadi Road Corridor Linear Development and Management Plan in Ongata Rongai Town encompasses a multifaceted approach aimed at enhancing the sustainability and liveability of the corridor. The first objective involves an extensive review of existing literature, principles, policies, legislation, and development control guidelines about sustainable linear development. This review will inform subsequent planning processes by identifying best practices and guidelines for fostering liveability along the road corridor.

Following the literature review, the scope includes conducting a contextual analysis of the planning area encompassing the Magadi Road corridor in Ongata Rongai Town. This analysis will investigate into the socio-economic, environmental, and infrastructural characteristics of the area, providing insights into its current state and identifying key challenges and opportunities for development. The findings of this analysis will serve as the foundation for developing tailored strategies and policies aimed at addressing the specific needs and dynamics of the corridor.

Subsequently, alternative plans, strategies, and policy options will be formulated based on the identified challenges, opportunities, and best practices. These alternatives will undergo a thorough assessment to evaluate their feasibility, effectiveness, and alignment with the overarching goals of sustainability and liveability. The preferred plan will be selected based on these assessments, culminating in the formulation of the Magadi Road Corridor Integrated Physical and Land Use Plan for Ongata Rongai Town. Additionally, an implementation, monitoring, and evaluation framework be developed to ensure the effective execution of the chosen plan, enabling ongoing assessment and adaptation to changing circumstances over time.

1.3 Data Sourcing Approach and Methodology

To create development plan, the project methodology must be thorough to ensure that adequate and credible information is used in the planning process. Thus, this part details the data needs and requirements per objective, etc.

1.3.1 Elements of Analysis

The elements of analysis of this project are Legal and Policy and case studies on linear development of urban areas as well as the situational analysis of the project site.

1.3.2 Data Needs and Requirements

To solve the Magadi Road project problem, a comprehensive understanding of various data needs and requirements is essential. This includes traffic assess current and future road usage, road condition

surveys to determine necessary repairs and upgrades, and environmental impact assessments to ensure sustainable development. Socio-economic data is crucial to evaluate the project's impact on local communities, while geographic information system (GIS) data helps in mapping and planning. Additionally, financial data is needed to estimate costs and budget allocations, and legal and regulatory requirements must be reviewed to ensure compliance with local, regional, and national guidelines. Collecting and integrating these diverse data sets will facilitate informed decision-making and effective project implementation.

1.3.3. Data Sources

The area of interest was selected based on the demarcation of Independent Election and Boundary Commission wards (IEBC) in secondary sources. The data on country demarcation is available on the IEBC website. Consequently, Google Earth software, an open-source program that allows users to view the earth through high-resolution satellite images, was used to confirm and observe the characteristics of the study area. The data acquired was social, economic and environmental. These were collected using primary sources where questionnaires were administered to sample business people of Ongata Rongai and resource persons. The data was analysed using Kobo collect tool.

The spatial data acquired was geographical maps presenting the location features and boundaries of Ongata Rongai and the surrounding area using Geographical Information System (GIS) and Google Earth images. Population data of Ongata Rongai town was acquired from Census data by the Kenya National Bureau of Statistics (KNBS). Literature review articles and books were acquired both online and from the Nairobi university library both are secondary sources.

1.3.4 Methods of Data Collection, Analysis and Presentation

Methods of collecting data were classified in two ways as follows:

Primary Data

The primary data was collected through random sampling of business people along Magadi road and households. Moreover, a Key Informant Interview was conducted to urban management person and transport personnel in the Physical Planning sector of Kajiado County. Other data collected is for verification and ground truth by observation since the area being studied is large, simple random sampling was done for every class represented for land use and land cover change. This was done by using questionnaires and observation matrix. Photographs too were also taken to help in observing the current state of Magadi Road, Linear pattern and various land use and land cover activities which are current on the ground. These involve observing the current state of road infrastructure and both social and economic activities in Ongata Rongai town which were recorded through photographs

Secondary Data

A desktop review was conducted to conceptualize the concept linear urban pattern, undesirable growth of towns how the transport infrastructure has impacted their causes and how they are trying to control their impacts, given that Ongata Rongai growth is highly being influenced by its nearest to Nairobi City. Written documents such as journal articles, books, reports, and other documents acquired from electronic sources and the physical location were reviewed. Satellite images were considered significant data because they were used to generate Land Use and Land Cover maps.

Topographic maps were collected to generate basic layers, such as boundaries and roads, as well as to enhance satellite images and for geo-referencing purposes. Furthermore, socioeconomic data, such as population data was collected and used.

Data Analysis

Data collected was analyzed using various techniques depending on the nature of the data. Quantitative data was analyzed using Kobo Collect, SPSS, and Excel. The analysis included frequencies and descriptions. Qualitative data was analyzed using narration based on the various themes created. Spatial data such as maps were analyzed using GIS software. Observations and photos were analyzed using sketches, illustrations, and thematic areas. The analyzed data was presented using charts, tables, maps, and photographs

1.3.5 Limitation of the Methodological Approach

The key limitation encountered in the use of this methodological approach was the inadequacy of peer-reviewed studies wholly focusing on linear development as they combine it with other aspects of planning like compact cities and smart cities from a planning perspective.

1.4 Project Development Approaches

1.4.1 Site Assessment Approach

The assessment of the project site was conducted through a situational analysis of the site. This was preceded by a background study of the site that provided a brief history of the site and contextualized the site indicating its physical location at the national, county, sub-county, and neighbourhoods' levels. Elements of the site covered in the situational analysis included the demographic profile, physical and natural environment characteristics, existing physical and infrastructural developments on the site, abutting land use, transport and circulation analysis, utility and service analysis and the local policy and legal policies guiding spatial development on the site. These elements were appraised through a SWOT analysis to determine suitability of the site to the various proposed development option.

1.4.2 Methods of Evaluation of Development Option

The Method of evaluation that was used was the Goal Achievement Matrix (GAM). The goals achievement matrix outlines planned goals in clear detail and compares them to objectives as well as the required actions or measures to achieve the goals. Therefore, from the GAM, alternative development plan models were weighted and ranked. The model with the highest rank was then used as the preferred plan to facilitate the development and utilization of the Magadi Corridor.

1.4.3 Approaches to Planning, Design, and Space Organization

The planning, design, and space organization were functionality-based to ensure the efficiency of a space, ensuring that it meets the specific needs and requirements of its intended use. The approach will also be sustainability-oriented and aesthetically driven. The space, design, and space organization will be conducted contextually to ensure prioritization of the relationship between the space and its surrounding environment by ensuring that it is responsive to the site's history, culture, and natural features.

1.5 Structure and Organization of the Planning Development Report

This report is structured into six chapters.

Chapter One is the introductory chapter which commences with a background of the planning development project. This background starts by offering a summary of the planning research project

highlighting its project title, findings, and recommendations. The background then proceeds to underscore the research problem statement, justification, location and area coverage, goals and objectives, assumptions and limitations, and scope of the planning development project. The chapter then details the data sourcing and development approaches adopted in the project. The chapter culminates with the structure and organization of the planning development project.

Chapter Two of the project entails the review of literature on key concepts of the planning development project, policy and legal framework, urban planning and urban design guidelines adopted for the project, case studies, and ends in a conceptual design framework.

Chapter Three covers a detailed analysis of the project site offering a background of the study area, contextual location, demographic profile, physiographic characteristics, existing physical and infrastructural developments on the site, and local policies and laws governing the site. This detailed situational analysis is then used to conduct a site suitability analysis. The chapter ends with a synthesis of the findings of the site suitability analysis.

Chapter Four covers project planning and design. This chapter outlines the specifications for planning and design, introduces conceptual spatial plans for the site project, and concludes with a detailed design formulation of the preferred spatial plan.

Chapter Five details the implementation, monitoring, and evaluation strategies for the preferred spatial plan.

CHAPTER TWO:

2.0 REVIEW OF LITERATURE AND CONCEPTUAL /THEORETICAL FRAMEWORK

2.1 Overview

Linear urban forms have been known since the beginning of urban design history. The topic of linear and strip cities appeared in the second half of the 19th century and has very rich scientific literature. There is a quite large list of articles concerning linear cities. Architecture historian G.R. Collins, particularly interested in the literature and philosophy of the linear city, was a collector of literature on the subject and the author of many publications. According to him, the discussion about the essence of the linear city was started by Ch. Bouilhet in 1912, who supported the ideas of the linear development of cities existing along rivers and transport routes, while stressing that the rope plans of the cities would lead to “disappearance of cities as nuclei of human activities” [G. R. Collins 1959]. Another promoter of linear cities was Georges Benoit-Levy, who advocated the concept of the linear city in Madrid as one of the types of the famous Ebenezer Howard’s Garden city. These two types of city development as a linear and concentric system were the subject of comparisons and discussions [G. R. Collins, Oct. 1959]. Other valuable inputs into the subject and interesting overviews were done by [C.A. Doxiadis 1967], On Linear Cities, [T. Tufek-Memisevic, E. Stachura 2015], A linear city development under contemporary determinants and by the same authors [A. Tufek-Memisevic, E. Stachura, 2019], Linear megastructures. An eccentric pursuit in tackling urban sustainability challenges and by [A.V. Antyufeev, O. A. Antyufeeva 2019], Linear Cities: controversies, challenges and prospects. The research concerning the links between the “ideal” and linear cities was conducted by Z. Paszkowski, in his book *The Ideal City in the European Perspective and its Relations with Contemporary Urbanism*, where he devotes a whole chapter to the linear city idea [Z. Paszkowski 2011]. [T. TufekMemisevic, E. Stachura 2019]

THEORITICAL CONCEPT

“LA CIUDAD LINEAL” (LINEAR CITY MODEL)

The Spanish engineer Arturo Soria y Mata was the first to theorize a linear city concept when he presented his “La Ciudad Lineal” in 1882. In 1892–1894 a project was launched to build a 55-km-long linear city in the form of a circle around Madrid, and a five-kilometres-stretch of it was built (Furundzic & Furundzic 2012; Lynch 1981).

The concept included one central avenue lined with ribbons of buildings. The avenue would take care of transportation of people and goods on rail and road. The development would stretch through the countryside and would thereby encourage agricultural production along the linear city and raise living standards (Boileau 1959). The linear city can be realized on three levels. On a global level, linear cities could connect “Brussels to Beijing”, on a state level the linear city could unite each village with each other until reaching the capital city. On the third level a linear city could circumvent an existing city to form an independent parallel satellite city (Le 2011).

On a more practical level, Arturo Soria y Mata set up some principal design guidelines. The central avenue was the backbone of the city, and would cater for all traffic, but especially rail traffic in the form of trains or trams. Its width was set to be between 40 and 100 meters. The edge of the linear city was defined by two streets that run parallel to the central avenue and outside of these streets the countryside begins. Perpendicular streets would run from the central avenue at right angle thus forming blocks. Mostly single-family houses were planned on lots inside the blocks with a minimum lot size of 400 m².

Only 1/5 of the lot was to be built and 4/5 to be planted. Streets were to be lined with trees and for the delight of strollers' flowers, benches, fountains and kiosques should be placed along the streets. Squares could be placed along the streets to battle monotony (Le 2011).

The linear city shares many similarities with Ebenezer Howard's "Garden City" concept, presented 10 years after the linear city, in emphasizing fresh air, sunshine, vegetation and blending the town with the countryside, and fighting city congestion. However, the question of transportation was treated differently: Howard separated his new towns from the old town with empty transportation corridors, while Soria y Mata incorporated the transportation corridor into his town (Collins 1959).

Key features of the Linear City Model include:

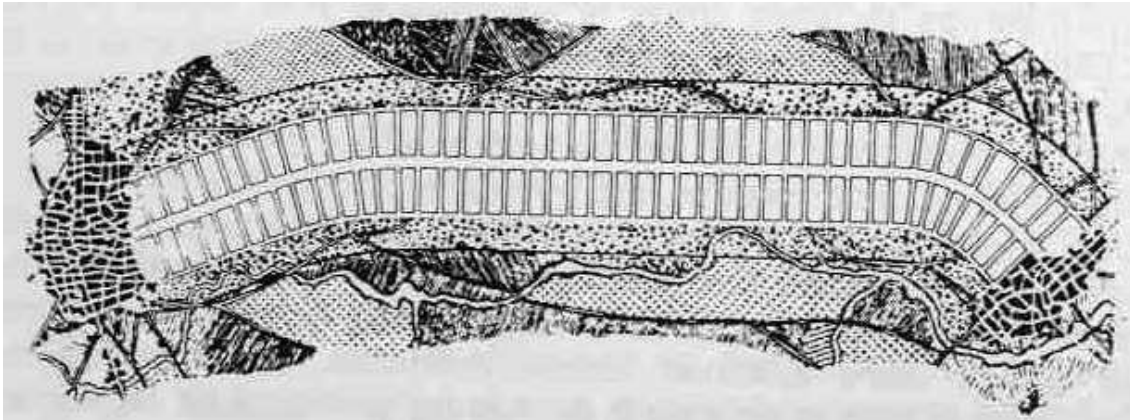


Figure 1: The Arturo Soria's first plan for a project in the neighbourhood Madrid

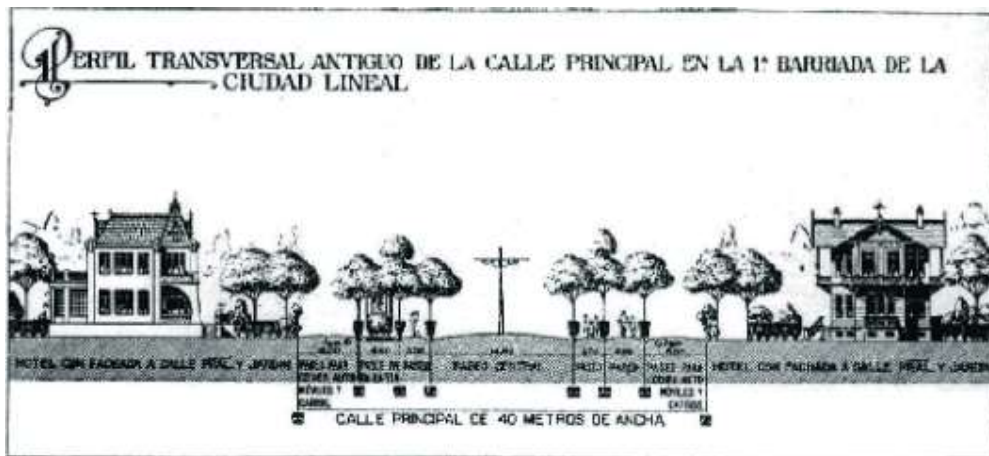


Figure 2: Side Elevation

Transportation Axis

The model assumes the presence of a central transportation route, such as a road or railway, along which development is concentrated. This transportation axis is crucial for facilitating movement and trade, making it a focal point for businesses and services.

Optimization of Accessibility

Businesses and services are strategically located along the linear axis to maximize accessibility and visibility. This arrangement allows for efficient transportation of goods and people along the route, contributing to economic efficiency.

Economic Rationality

The placement of businesses and residents is driven by economic rationality. Entities aim to position themselves in locations that offer the best economic advantages, considering factors such as proximity to markets, competitors, and transportation hubs.

Spatial Concentration

Development is concentrated along the linear axis, creating a distinctive spatial pattern. This concentration is based on the idea that businesses and services benefit from being close to each other and the primary transportation route.

Linear Nature of Development

Unlike other urban models that envision radial or concentric expansion, the Linear City Model suggests that development primarily occurs along a single line. This linear nature simplifies transportation logistics and creates a straightforward spatial structure.

The Linear City Model is often applied to situations where a prominent transportation route, like a major road or railway, influences the layout and growth of urban areas. It provides insights into how economic considerations and accessibility factors can shape the spatial organization of cities, making it relevant for understanding patterns of development along linear corridors such as Magadi Road in Ongata Rongai.

Application to Magadi Road Corridor Development

Central Transportation Axis: One of the key principles of the Linear City Model is the presence of a central transportation route, such as a road or railway, that serves as the backbone of development. In the case of Magadi Road, this concept aligns with the existing road infrastructure that connects Ongata Rongai to Nairobi. By focusing on development along this corridor, it optimizes transportation efficiency and accessibility for residents, businesses, and goods.

Optimization of Accessibility: Businesses and services along Magadi Road can be strategically located to maximize accessibility and visibility. This means that commercial and residential developments can capitalize on the proximity to the main transportation route, facilitating easier commuting and trade. This arrangement supports economic efficiency by reducing travel times and costs associated with logistics and commuting.

Economic Rationality: The Linear City Model emphasizes placing businesses and residents in locations that offer economic advantages, such as proximity to markets and transportation hubs. Along Magadi Road, developments can be planned to attract businesses that benefit from the corridor's visibility and accessibility, contributing to local economic growth and sustainability.

Spatial Concentration: Development along Magadi Road can be concentrated linearly, creating a distinct spatial pattern that optimizes land use and infrastructure planning. This concentration

minimizes urban sprawl and allows for efficient use of resources, while also fostering a cohesive urban fabric that integrates residential, commercial, and recreational spaces along the corridor.

Linear Nature of Development: Unlike radial or sprawling urban models, the linear nature of the development along Magadi Road simplifies transportation logistics and urban planning. It creates a straightforward spatial structure that enhances connectivity and community integration along the corridor.

Benefits for the Magadi Road Corridor Project

Efficient Land Use: By focusing development along Magadi Road, the Linear City Model promotes efficient land use, reducing the need for extensive infrastructure expansion and preserving green spaces outside the linear development zone.

Improved Transportation: Enhances transportation efficiency and accessibility for residents and businesses, thereby reducing congestion and improving overall mobility along the corridor.

Economic Growth: Attracts investment and promotes economic activities by providing a conducive environment for businesses to thrive due to strategic location and accessibility to markets.

Environmental Sustainability: Facilitates sustainable development practices by promoting compact urban forms, reducing carbon footprints associated with commuting and logistics.

Community Livability: Enhances quality of life by integrating residential areas with amenities such as parks, recreational spaces, and public facilities along the linear development, fostering a sense of community and well-being.

The Linear Megastructure Model

Overview

The Linear Megastructure Model is an urban planning concept that envisions a large, continuous urban form, typically stretched along a major transportation corridor by the famous architect Kenzo Tange. This model integrates various functions such as residential, commercial, and industrial activities into a singular, elongated structure. The concept is inspired by the idea of creating highly efficient, sustainable urban environments that maximize space usage and connectivity.

Key Features of the Linear Megastructure Model

The Linear Megastructure Model features a continuous, linear structure that extends along a major transportation route, such as a highway or railway. This form is designed to minimize land use while maximizing accessibility. Various urban functions, including housing, workspaces, shopping areas, and recreational facilities, are integrated into the megastructure, creating a self-contained urban environment where residents have access to all necessary amenities within a short distance. A central transportation axis runs through the megastructure, facilitating the efficient movement of people and goods. This axis typically includes high-capacity public transit systems such as trains or buses, as well as roads for private vehicles.

Sustainability is a core principle of the linear megastructure, which incorporates energy-efficient buildings, green roofs, and integrated renewable energy sources. The structure also promotes walking

and cycling, reducing reliance on motorized transport. Additionally, the megastructure can be built using modular construction techniques, allowing for phased development and easy adaptation to changing needs. This modularity enables rapid construction and flexibility in design. The model promotes high-density development, concentrating population and activities within the linear form, which reduces urban sprawl and preserves surrounding natural landscapes.

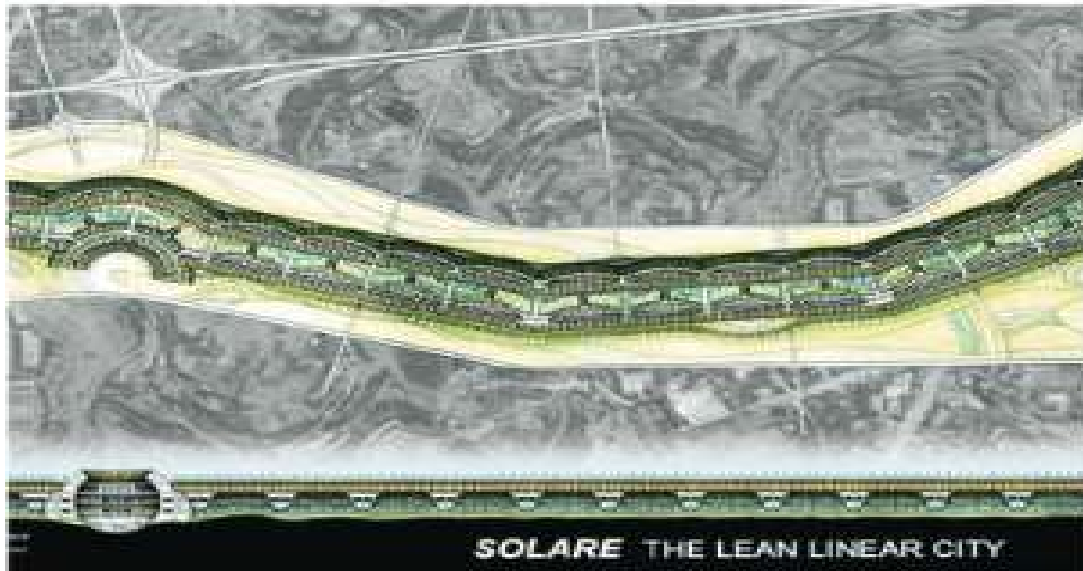


Figure 3: The Linear City

Application to the Magadi Road Corridor Development and Management Strategy Plan

For the Magadi Road Corridor, a linear megastructure can be developed to house various urban functions. This structure would run parallel to Magadi Road, providing a cohesive and organized urban form. The megastructure would integrate residential areas with commercial, industrial, and recreational spaces, creating a self-sufficient urban environment where residents would have access to all necessary services and amenities within the structure.

A central transportation corridor within the megastructure would include high-capacity public transit systems, such as light rail or bus rapid transit, ensuring efficient and sustainable transportation options for residents. The design of the megastructure would incorporate sustainable practices, such as energy-efficient buildings, green spaces, and renewable energy sources, creating an environmentally friendly urban environment. The construction of the megastructure could be modular, allowing for phased development. This approach would enable gradual growth and adaptation to changing urban needs over time.

Benefits for the Magadi Road Corridor Project

The Linear Megastructure Model maximizes the use of land along the Magadi Road Corridor, creating a dense, efficient urban form that reduces urban sprawl and preserves natural landscapes. The integration of a central transportation axis ensures that residents have easy access to public transit, reducing reliance on private vehicles and improving overall accessibility. The sustainable design of the megastructure promotes environmentally friendly practices, such as energy efficiency and reduced carbon emissions, contributing to a healthier urban environment.

The high-density, mixed-use nature of the megastructure supports economic efficiency by concentrating businesses, services, and residents in close proximity, fostering vibrant economic

activity. The modular construction approach allows for flexible and adaptable development, enabling the urban form to evolve in response to changing needs and technological advancements. By adopting the Linear Megastructure Model for the Magadi Road Corridor Development and Management Strategy Plan, the project can create a highly efficient, sustainable, and cohesive urban environment. This model leverages the benefits of high-density, mixed-use development and efficient transportation systems, making it a suitable approach for managing growth along the Magadi Road Corridor.

2.2 Definition of terms and concepts

It explains key concepts used in this development project.

1. **Linear City:** A linear city is a urban development concept where the city is built along a linear transportation axis, such as a road, railway, or river.
2. **Transportation Axis:** The central feature of a linear city, the transportation axis refers to the linear route along which development is concentrated.
3. **Optimization of Accessibility:** This term refers to the strategic placement of businesses, services, and residential areas along the transportation axis to maximize accessibility and visibility.
4. **Economic Rationality:** Economic rationality drives the location decisions of businesses and residents within a linear city.
5. **Spatial Concentration:** Spatial concentration refers to the dense clustering of development along the linear axis of the city.
6. **Linear Urban Pattern:** The linear urban pattern describes the spatial arrangement of development along a linear axis within a city or urban area.
7. **Transit-Oriented Development (TOD):** Transit-oriented development refers to a planning strategy that focuses on creating compact, mixed-use communities around transit nodes, such as train stations or bus stops.
8. **Infrastructure Corridors:** Infrastructure corridors are linear pathways that accommodate essential infrastructure elements, such as roads, railways, pipelines, or utility lines.
9. **Spatial Organization:** Spatial organization refers to the arrangement and distribution of land uses, buildings, and infrastructure within a city or urban area.
10. **Urban Economics:** Urban economics is a subfield of economics that examines the spatial distribution of economic activities within cities and urban areas.
11. **Socio-Economic Development:** Socio-economic development refers to the process of improving the economic, social, and environmental well-being of individuals and communities within a society.
12. **Mixed-Use Development:** Mixed-use development refers to the integration of diverse land uses, such as residential, commercial, and recreational activities, within the same area or building.
13. **Environmental Planning:** Environmental planning involves the integration of environmental considerations into the planning and decision-making processes related to land use, development, and natural resource management

14. Urban Form: Urban form refers to the physical shape, layout, and arrangement of buildings, streets, and open spaces within a city or urban area.
 15. Smart Growth: Smart growth is a planning approach that promotes compact, efficient, and sustainable urban development patterns.
 16. Community Engagement: Community engagement involves the active participation of residents, stakeholders, and community organizations in the planning and decision-making processes that affect their neighborhoods and communities
 17. Resilient Cities: Resilient cities are cities that have the capacity to withstand and recover from shocks and stresses, such as natural disasters, economic downturns, and social upheavals.
 18. New Urbanism: New Urbanism is an urban design movement that emerged in the late 20th century, advocating for the creation of walkable, mixed-use neighbourhoods with a focus on human-scale design, traditional architecture, and sustainable development principles.
-
19. Complete Streets: Complete Streets are streets designed to accommodate all modes of transportation, including pedestrians, cyclists, public transit users, and motorists, in a safe and accessible manner.

2.3 Policy and legal framework

OVERVIEW

This section details the legal basis for the formulation of Physical and land use plan and how this plan is linked to other policy, strategy, legislative and constitutional domains; These include The Constitution; Cities and Urban Areas Act 2011, The National Spatial Plan; The Big Four Agenda; The County Spatial Plan; The County Integrated Development Plan II(CIDP II); The New Urban Agenda; The Sustainable Development Goals(SDG's); The East African Community(EAC) Urban Development Strategy among others.

Urban Integrated Development Planning

Integrated development planning in the County takes two perspectives of county-wide 5-year Integrated Development Planning (CIDP) and urban-specific Integrated Development Planning (IDeP). Urban IDePs, 10-year Sectoral Plans, and Spatial Plans form the core inputs to the CIDP which is implemented through an annual program-based budget in a 3-year fiscal framework (MTEF).

The County Governments Act 2012, Section 108 (1) provides for integrated development planning for each county which shall have clear goals and objectives; an implementation plan with clear outcomes; provisions for M&E; and clear reporting mechanisms. Urban IDePs that favor 'action' and delivering infrastructural services within the urban metropolis are a sound building block to the County Integrated Development Planning. Rapid urbanization portends the challenge of infrastructure service provision in urban metropolis. Urban Integrated Development Planning provides a unified development framework that involves the full continuum of strategy analysis and planning, development of program-based budgets and ranked priorities, financing, and participatory implementation. The framework further provides for performance review through assessment of socio-economic changes in cumulative efficiency, effectiveness, and value for money.

Urban legislation branch provides legal and operational support for governments and cities to identify reforms and to adopt laws and legal frameworks that effectively regulate urbanization issues such as land use, urban planning, housing, infrastructure, taxation or safety. Urban legislation helps promote and develop governance models that are equitable, gender responsive and socially inclusive, advocating fair land use and access to basic services and infrastructure for all.

Constitutional Basis

The Constitution of Kenya is the supreme law of the country and provides the basis for planning. County Planning should take cognizance of the constitutional aspirations and principles.

- I. Article 60 on principles of land policy guides the formulation of land use and management policies
- II. Article 66 on regulation of use of any land, interest in or right over land in the interest of defence, public safety, public order, public morality, public health or land use planning provides powers to the National and County governments to regulate the use of land and property through spatial planning
- III. Article 67 (h) on Monitoring and oversight Provides powers to the National Land Commission to monitor and have oversight over land use planning
- IV. Fourth Schedule, Part I (21) on general principles of Land planning and the coordination of planning by the counties
- V. Fourth schedule, part 2 (8) provides for county planning and development The CoK restructured the governance and administrative structures of cities and urban areas commencing in 2013.
- VI. Article 184 of the Constitution provides the criteria for the governance and management of Urban Areas and Cities, including criteria for classifying areas as Urban Areas and Cities, Principles of governance and management of Urban Areas, participation of residents in the governance processes and mechanisms for identifying different categories of urban areas and cities including their governance.

County Government Act, 2012

(Section 102-115) - Provides principles of planning and development facilitation to guide county planning and developments

Urban Areas Cities Act, 2012

Section 40 of the Urban Areas and Cities Act 2011 provides for the contents of Integrated Urban Area Development plans with the highlights that include;

- i. Long term Board's vision for the urban area
- ii. An assessment of the existing level of development and log of basic services required
- iii. Framework for affirmative action
- iv. Development priorities and objectives
- v. Spatial development framework 37
- vi. Financial plan (Medium Term Expenditure Framework)
- vii. Key performance indicators and targets

Physical Planning Act Cap 286 – provides for Matters to be considered in spatial planning, a framework for development control, purpose and content of spatial plans. LINKAGES WITH OTHER PLAN AND POLICY DOCUMENTS

LINKAGES WITH OTHER PLAN AND POLICY DOCUMENTS

Magadi Road Corridor Development and Management Strategy Plan

1. Linkages with the County Government Act 2012

The County Government Act 2012 mandates the preparation of five-year integrated county development plans, urban area plans, spatial plans, and annual county budgets to ensure comprehensive planning and implementation. Under Section 102, county planning provides a platform for unified sector-wide planning, budgeting, financing, implementation, and performance review. The Act assigns the County Planning Unit the responsibility for coordinating integrated development planning. Additionally, it emphasizes the importance of engagement between government agencies, citizens, stakeholders, and interest groups, integrating economic, physical, social, environmental, and spatial planning. Therefore, the Magadi Road Corridor Development and Management Strategy Plan aligns with these requirements, ensuring it serves as a comprehensive blueprint for coordinated and inclusive development within the corridor.

2. Linkages with the Public Finance Management Act (PFMA), 2012

The Public Finance Management Act 2012 outlines the effective and efficient management of public resources, detailing the budget process for government agencies, which includes integrated development planning both in the long term and medium term. Article 126 specifically requires each county government to prepare an integrated development plan reflecting strategic priorities, responses to changes in the financial and economic environment, and detailed program delivery. The Magadi Road Corridor Development and Management Strategy Plan aligns with the PFM Act 2012, incorporating financial and economic priorities into its planning and budgeting processes. This ensures that the development projects within the corridor are financially sustainable and responsive to economic changes.

3. Linkages with the Urban Areas and Cities Act, 2011

The Urban Areas and Cities Act 2011 stipulates that boards of cities, municipalities, and towns must formulate and implement integrated development plans. Section 20 mandates this responsibility, making it a crucial aspect of urban management. Section 39 further mandates that a board or town committee must adopt a single, inclusive strategic plan within the first year of election. The Act emphasizes the need for five-year integrated development planning, ensuring these plans align annual budgets and guide all planning and decision-making processes. The Magadi Road Corridor Development and Management Strategy Plan adheres to these regulations, ensuring comprehensive urban management and sustainable development along the Magadi Road corridor.

4. Linkages with the Constitution of Kenya 2010

The Constitution of Kenya 2010 devolves the function of county planning and development to county governments, as stated in the Fourth Schedule. This requires counties to plan and budget for development programs within a specified timeframe. The preparation of the Magadi Road Corridor Development and Management Strategy Plan considers constitutional provisions, emphasizing the role of devolved units in planning for public amenities and urban infrastructure services. This ensures that the corridor's development plans align with constitutional mandates, promoting effective local governance and development.

5. Linkages with the Kenya Vision 2030

Kenya Vision 2030 is the national long-term development policy aimed at transforming Kenya into a newly industrializing, middle-income country by 2030. The Vision is anchored on three key pillars: economic, social, and political, implemented through five-year medium-term plans. The Magadi Road Corridor Development and Management Strategy Plan aligns with these pillars, ensuring that corridor development plans contribute to national goals. By integrating economic, social, and political objectives into development plans, the corridor can contribute to the broader vision of improving the quality of life for all citizens in a sustainable and secure environment.

6. Linkages with Sustainable Development Goals (SDGs)

The Sustainable Development Goals (SDGs) encompass a wide range of social and economic development issues, including poverty, health, education, climate change, and gender equality. The Magadi Road Corridor Development and Management Strategy Plan aligns its projects and programs to ensure they contribute to achieving these 17 global goals. By integrating SDGs into local planning, the Magadi Road Corridor can address various social and economic challenges, promote sustainability, and track progress toward global development targets.

7. Linkages with the County Spatial Plan

The County Spatial Plan includes a Spatial Development Strategy (SDS) with sub-strategies for urbanization, rural development, and housing. This strategy involves planning for existing urban areas, preparing physical development plans for major towns, compacting urban areas, and delineating urban fringes. The integration of these spatial plans ensures that corridor development is balanced and sustainable, promoting effective land use, infrastructure development, and urban-rural linkages.

8. Linkages with CIDP II

Counties are required by statute to prepare Integrated Development Plans (CIDPs) aligned with the electoral cycle. These plans focus on policy and budgetary aspects of output-based planning and reporting. The second round of CIDPs (CIDP II) aims to address the gaps identified in the first round, particularly regarding urban planning and management. By preparing and reviewing Urban CIDPs, the Magadi Road Corridor Development and Management Strategy Plan can enhance urban management, ensuring that development plans are comprehensive and effectively implemented.

9. Linkages with the New Urban Agenda (NUA)

The New Urban Agenda (NUA) aims to create inclusive and sustainable cities and human settlements that are safe, healthy, accessible, affordable, and resilient. It emphasizes the need for cities to fulfill their social functions, promote civic engagement, enhance social cohesion, achieve gender equality, and protect natural ecosystems. By aligning with the NUA, the Magadi Road Corridor Development and Management Strategy Plan promotes inclusive urban development, ensuring that all residents enjoy equal rights, opportunities, and a high quality of life.

10. Linkages with AU Agenda 2063

Aspiration 1 of AU Agenda 2063 aims for a prosperous Africa based on inclusive growth and sustainable development. It envisions high standards of living, quality of life, and modern infrastructure for all African people. By aligning with this agenda, the Magadi Road Corridor Development and Management Strategy Plan focuses on developing cities and human settlements with modern infrastructure, affordable housing, and access to basic necessities, contributing to the broader goal of sustainable development and improved living standards across the continent.

11. Linkages with East African Community Urban Development Strategy

The East African Community Urban Development Strategy aims to transform the region into a stable, competitive, and sustainable lower-middle-income area by 2021. It addresses regional development issues and priorities, emphasizing planned and controlled urbanization and optimizing strategic links. By aligning with this strategy, the Magadi Road Corridor Development and Management Strategy Plan promotes regional integration and development, ensuring that urbanization is managed effectively and contributes to regional growth and stability.

2.4 Urban Planning, Urban Design, and Site Planning Guidelines and Standards

2.4.1 Urban planning

Urban planning is a complex discipline aimed at arranging the development and organization of urban spaces to meet the needs of communities and foster sustainable growth. One of the seminal figures in this field is Sir Ebenezer Howard, whose work laid the foundation for modern urban planning concepts. In his book "Garden Cities of Tomorrow," originally published in 1902, Howard introduced the idea of garden cities as self-sustaining communities surrounded by greenbelts. His vision emphasized the integration of industry, agriculture, and residential areas within a balanced urban environment, promoting a high quality of life for inhabitants while preserving natural landscapes.

Howard's pioneering ideas continue to influence urban planners and policymakers worldwide, shaping strategies for managing urbanization and mitigating the challenges of rapid urban growth. His emphasis on decentralization, mixed land use, and the creation of green spaces reflects a holistic approach to urban development that prioritizes environmental sustainability and community well-being. By promoting compact, walkable neighborhoods with access to amenities and green infrastructure, Howard's garden city concept remains relevant in contemporary efforts to create livable, resilient cities.

Urban planning has evolved over the decades, incorporating new technologies, methodologies, and interdisciplinary approaches to address the complex and dynamic nature of urban systems. Today, urban planners collaborate with architects, engineers, economists, sociologists, and environmental scientists to develop innovative solutions to urban challenges. From transit-oriented development and smart growth initiatives to green building standards and participatory planning processes, contemporary urban planning seeks to create inclusive, equitable, and vibrant cities that enhance the quality of life for present and future generations.

2.4.2 Urban Design

Urban design is the process of designing and shaping cities, towns, and communities to make them functional, attractive, and sustainable for residents and visitors alike. Some of the key features are;

Population Density Distribution: This illustrates how population density is distributed across different areas within a city. It shows areas of high density (such as downtown cores or residential neighborhoods) and areas of low density (such as suburbs or industrial zones). Urban designers use this information to plan for adequate infrastructure, transportation, and public services based on population distribution.

Land Use Composition: This displays the breakdown of land use within the city, including residential, commercial, industrial, recreational, and green spaces. Urban designers aim for a balanced mix of land uses to create vibrant, livable communities. They consider zoning regulations, economic development goals, and environmental sustainability when planning land use.

Transportation Mode Share: This shows the percentage of trips taken by different modes of transportation, such as walking, cycling, public transit, and private vehicles. Urban designers prioritize sustainable transportation options to reduce traffic congestion, air pollution, and carbon emissions. They may advocate for infrastructure improvements, such as bike lanes, pedestrian-friendly streets, and efficient public transit systems, to promote alternative modes of transportation.

Quality of Life Indicators: measures various indicators of quality of life, such as access to parks and green spaces, air and water quality, housing affordability, cultural amenities, and community engagement. Urban designers strive to enhance quality of life for all residents by creating inclusive, accessible, and healthy urban environments. They work closely with community stakeholders to identify priorities and implement design strategies that improve overall well-being.

Urban design offers several features that contrast with the limitations of linear development, providing more holistic and adaptable solutions to urban challenges:

Integrated Approach: Unlike linear planning, which often focuses on isolated elements or projects, urban design takes a more integrated approach. It considers the interrelationships between various aspects of the urban environment, such as land use, transportation, public spaces, and infrastructure. By understanding how these elements interact, urban design can create more cohesive and sustainable urban landscapes.

Mixed Land Use: Linear planning tends to segregate land uses, resulting in sprawling, monotonous developments that rely heavily on automobile transportation. Urban design, on the other hand, promotes mixed land use, where residential, commercial, and recreational activities are integrated within the same neighborhood or district. This approach fosters walkability, reduces reliance on cars, and creates vibrant, diverse communities.

Human Scale: Linear planning often prioritizes the efficiency of movement, resulting in wide, impersonal thoroughfares and oversized infrastructure. Urban design emphasizes the human scale, prioritizing the needs and experiences of pedestrians and cyclists. Streets are designed to be narrow, shaded, and pedestrian-friendly, with amenities such as benches, trees, and public art, creating inviting and accessible public spaces.

Placemaking: Linear planning tends to overlook the unique character and identity of individual places, focusing instead on standardization and efficiency. Urban design employs placemaking strategies to celebrate and enhance the distinctive qualities of neighborhoods and communities. By incorporating local history, culture, and aesthetics into the design process, urban design creates memorable and meaningful places that foster a sense of belonging and pride among residents.

Flexibility and Adaptability: Linear planning often relies on rigid zoning regulations and master plans that may become obsolete or impractical over time. Urban design adopts a more flexible and adaptive approach, allowing for incremental change and evolution in response to shifting demographics, economic conditions, and environmental factors. By incorporating principles of adaptability and resilience, urban design ensures that cities can grow and thrive in a dynamic and uncertain world.

2.4.3 Site Planning Guidelines and Standards

These principles typically encompass aspects such as land use zoning, setbacks, building heights, road networks, utility provision, environmental considerations, and community amenities. These will guide linear development.

Zoning Regulations: Zoning regulations would designate areas suitable for linear development, such as transportation corridors or linear parks. These regulations would specify the permitted land uses, building densities, and setbacks along the linear development.

Setbacks and Building Heights: Guidelines would establish setbacks from the linear route to ensure safety, accessibility, and aesthetics. Building height restrictions may also be in place to maintain visual continuity and prevent the overshadowing of adjacent properties.

Road Networks and Access Points: Standards for road design and access points along the linear development would ensure smooth traffic flow, safe pedestrian access, and efficient connectivity with surrounding areas.

Utility Provision: Guidelines would address the provision of utilities such as water supply, sewage systems, electricity, and telecommunications infrastructure to support the linear development and adjacent properties.

Environmental Considerations: Regulations would include provisions for environmental impact assessments, mitigation measures, and sustainable design practices to minimize the ecological footprint of the linear development and preserve natural resources.

Community Amenities: Guidelines might require the inclusion of community amenities along the linear development, such as green spaces, pedestrian promenades, recreational facilities, and public transportation stops, to enhance the quality of life for residents and users.

2.5 Benchmarking and case studies -Global, Regional, National and Local

2.5.1 Case Study 1: THE LINE – THE SAUDI-ARABIAN LINEAR CITY CONCEPT AS THE PROTOTYPE OF FUTURE CITIES

The highlights of the Line City Project (The Line) are the following:

The Line will be an international city, inhabited by a number of more than 1 million people from all over the world. The location of the Line-City will cross and connect four distinctive landscapes and ecozones: Coastal, coastal desert, Mountain, and Upper Valley. The needed electrical energy will be totally produced from renewable sources (solar energy, wind energy and hydrogen-based power generation). This arrangement will ensure zero carbon emission resulting in clean and pollution-free environment of the Line-City urban area.



Figure 4: Saudi Linear City Elevation

According to the announcements distributed by Neom Company: "...the Line will be a revolution in urban living, which is developed as part of Neom, a planned \$500 billion cross-border city in the Tabuk Province of northwestern Saudi Arabia which will incorporate smart technologies. The site, near the Red Sea, will have borders of Egypt, Israel, and Jordan."

The Line will be the five-minute Habitat – spectacular due to the new definition of the role of pedestrian area. Walkability within the five-minute walk will define the settlement area, including essential daily services, schools, medical clinics, leisure facilities, as well as recreational areas with

green spaces. Thanks to that idea, no cars or car-planned streets will be needed. The Line will be composed of a 170 km chain of settlements (named: modules) hyper-connected by the high-speed underground train and/or hyperloop transportation. The settlements (modules) will consist of mixed-

use communities. They will be built around the nature with high respect to it, instead of over

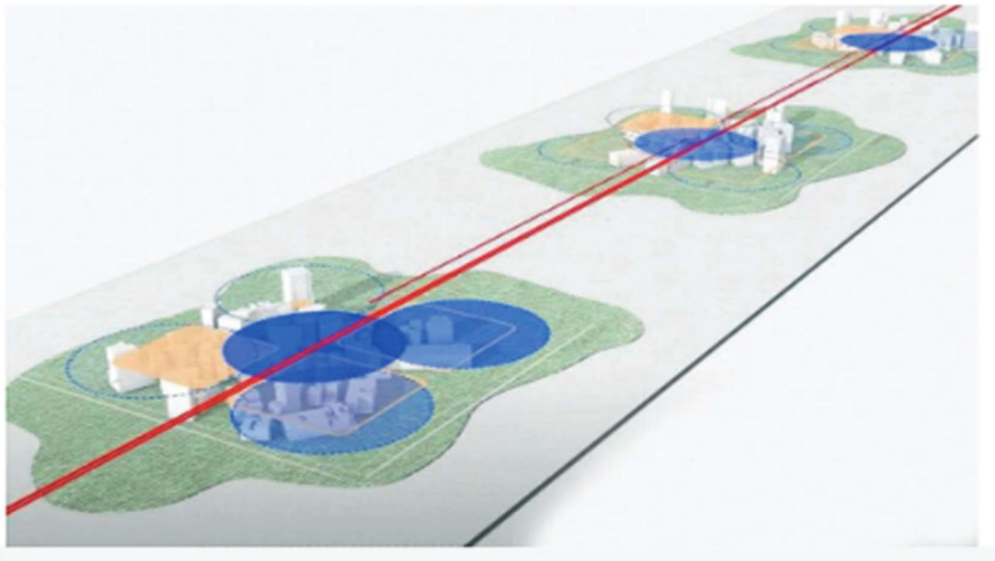


Figure 5: Overview of Saudi Linear Plan

it. The Line transportation system will be divided into three operational levels: The top level, called Pedestrian Layer, will be dedicated only to pedestrians. There will be no cars and streets, with green spaces instead. The second level, called Service Layer, will contain all daily service facilities such as schools, medical clinics, leisure facilities. At the lowest, third level, called Spine Level, there will be ultra-high-speed transit and autonomous vehicles “that will make travel easier and allow residents to reclaim time to spend on health and wellbeing.” It is expected no journey will be longer than 20 minutes.



Figure 6: Section of the Spine

The time plan schedules the beginning of the construction of The Line first quarter started in 2021 and is still underway.

This investment is said to be one of the most complex and challenging infrastructure projects in the world. It is also worth mentioning that it forms a part of extensive development works, being already underway at the Neom Project. Neom was announced in 2017 as part of Saudi Vision 2030 to develop public service sectors, the diversity of Saudi Arabia's economy and become less reliant on oil. The sustainability of the Line-City will be ensured due to facing directly the most pressing challenges for humanity today, such as "infrastructure, pollution, traffic, and human congestion", said Mohammed bin Sal- ARCHITECTURE ET ARTIBUS - vol.13 - 2(48)/2021 41man in his launching statement. The Line concept is intended to be a response to contemporary threats related to urbanization, implement the latest technological achievements in a model way and to create urban structures for safe and happy living, work and leisure.

Impact to the project

The LINE project in Saudi Arabia offers a revolutionary model for urban development that can significantly inform the Magadi Road Corridor Development and Management Strategy Plan. By emphasizing renewable energy, THE LINE sets a precedent for sustainability, using solar, wind, and hydrogen-based power generation to achieve zero carbon emissions. This focus on green energy can inspire the Magadi Road Corridor project to integrate similar technologies, reducing its environmental impact and promoting a healthier urban environment. Additionally, THE LINE's design prioritizes walkability and eliminates traditional car-centric infrastructure, creating pedestrian-friendly zones and extensive green spaces. Adopting this approach in Ongata Rongai could improve livability, reduce traffic congestion, and foster a more human-centric urban experience.

Moreover, THE LINE's innovative transportation solutions and modular urban design offer valuable lessons for the Magadi Road Corridor. The multi-layered transportation system, with designated levels for pedestrians, services, and high-speed transit, provides an efficient model for organizing urban functions. Implementing a robust public transportation network, such as bus rapid transit, can enhance connectivity and economic activity along the corridor. THE LINE's emphasis on integrating nature into urban development and building communities around natural landscapes aligns with sustainable planning principles, which can be applied to preserve natural habitats and create green buffers in the Magadi Road Corridor. By focusing on these aspects, the corridor can become a modern, sustainable, and economically vibrant urban area, attracting diverse industries and fostering innovation and growth.

2.5.2 Case study 2 VIHIGA MUNICIPALITY INTEGRATED DEVELOPMENT PLAN 2018-2022 Linear concept

Vihiga Township is one of the most urbanized parts of Vihiga County. It is unique in the sense that urban human settlements have developed on leapfrog basis along the major roads and particularly the AI Kisumu-Kakamega Road and C39 Chavakali-Kakamega road. Ribbon kind of development has taken place along the Kisumu-Kakamega Road. This works against an effective provision of infrastructure and service by the county government.

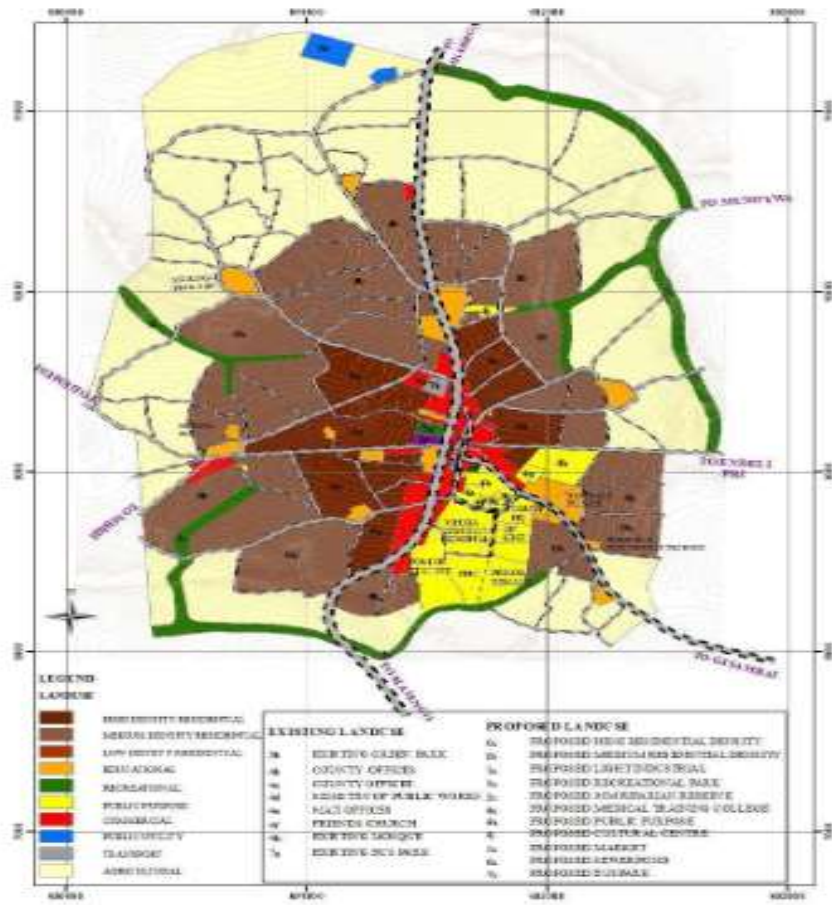
The township has four major urban centers including Mbale, Majengo, Chavakali and Mudete. Mbale is the major town and is linked to other satellite towns namely: Majengo and Chavakali along the Kisumu – Kakamega road, and Mudete on the Chavakali –Kapsabet road. The township has other

commercial nodes such as Manyatta, Chango, Kiritu, Kivagala, Bendera, Stand Kisa, Vihiga, and Lunyerere. There is a strong rural-urban linkage as people from the immediate interline bring agricultural produce to sell in town while these towns serve as distribution points for farm inputs and other commodities. It is important to highlight the historical development of the four major towns. The old Kisumu-Kakamega Road had several meanders (detours) due to the difficult landscape comprising of hills and valleys. Some market centers were bypassed by the new A1 road due to alignment and The Vihiga Municipality was experiencing rapid urbanization as more and more agricultural land was converted into urban and peri-urban land. More people are settling in urban areas with resultant pressure on infrastructure and services beyond the ability of the County Government to provide for. Urbanization has been taking place haphazardly without any approved Local Physical Development Plans. Further, most of the urban development activities are concentrated along the A1 Kisumu-Kakamega Road where they exhibit leap-frog type of development. Land within the planning area is predominantly under private tenure system and therefore the County Government had limited tax base. It was also extremely challenging for the County Government to control development on private land. The purpose of the plan, therefore, was to provide a spatial framework that will guide future development.

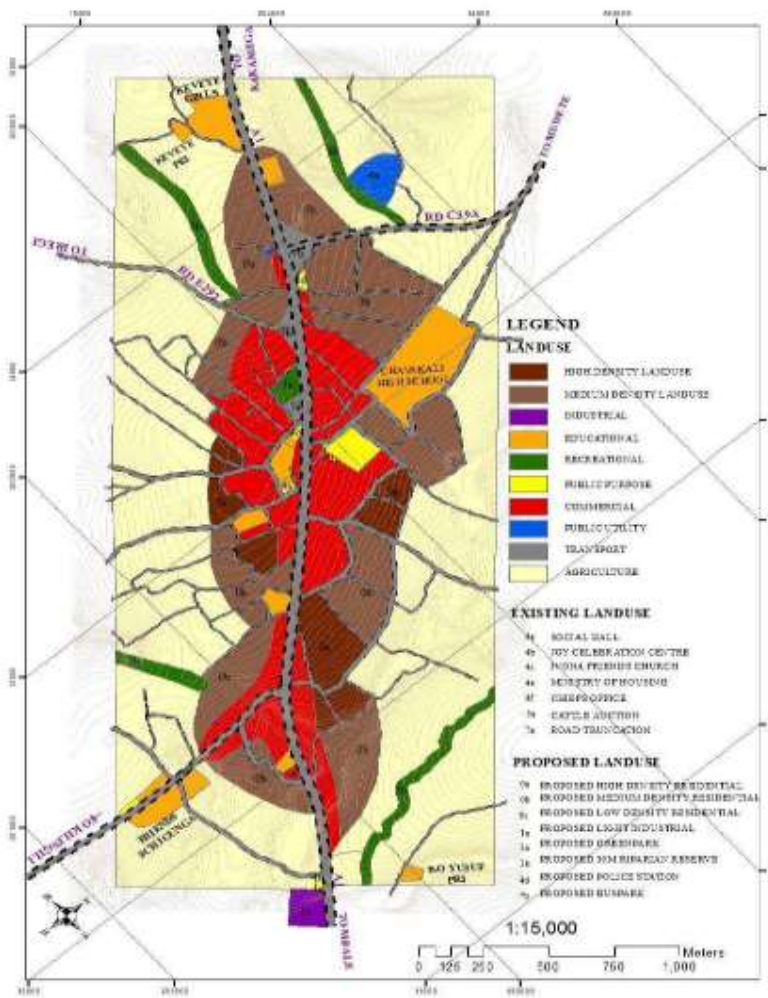
The Vihiga IDP outlines key plans to mitigate the problems associated with linear development. The plan focuses on creating a spatial framework that balances all land uses, thus promoting a more organized and sustainable urban growth pattern. By considering all land uses in its arrangement along the roads, the plan aims to rectify the haphazard urbanization and improve infrastructure provision and service delivery. It seeks to integrate the various commercial nodes and residential areas into a cohesive urban system, enhancing connectivity and accessibility. The plan also emphasizes the importance of preserving rural-urban linkages, ensuring that agricultural produce can efficiently reach urban markets while towns serve as distribution points for farm inputs and other commodities. This integrated approach aims to create a sustainable urban environment that can support the township's rapid growth and development.

The following are key plans which was used to solve the problem although is still underway.

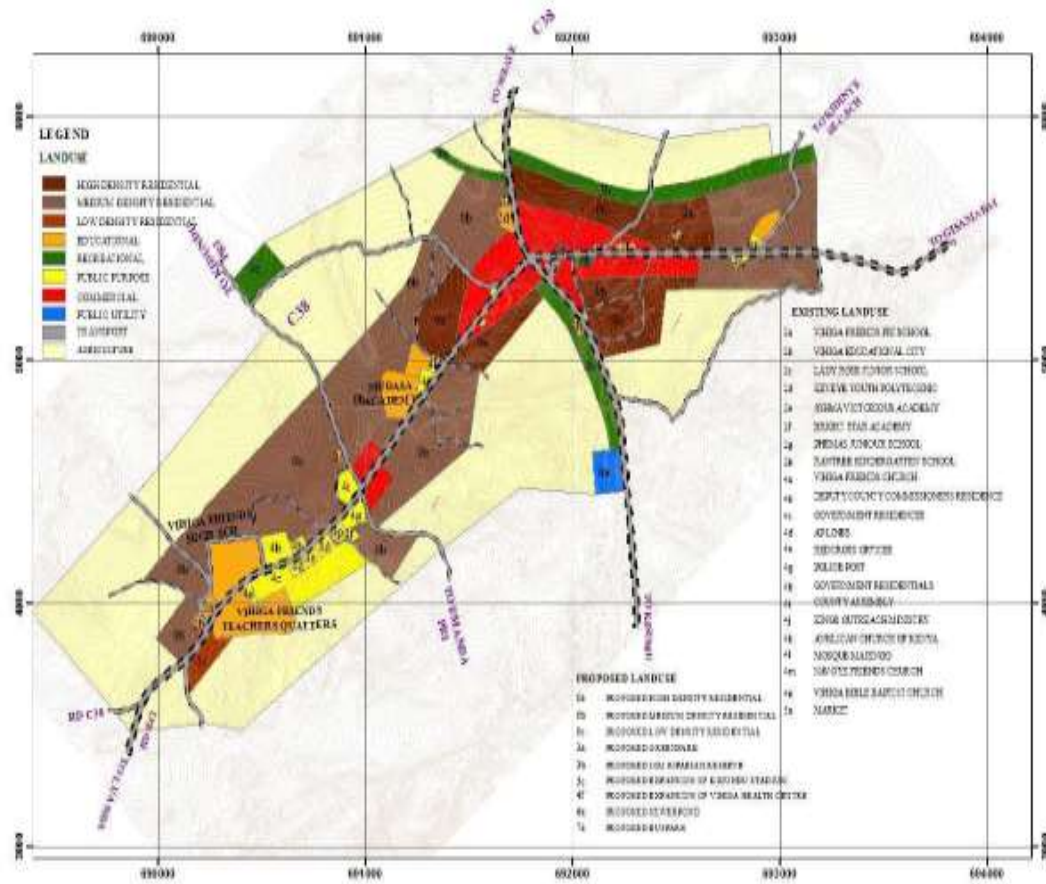
Map 1 : Mbale Area Action Plan



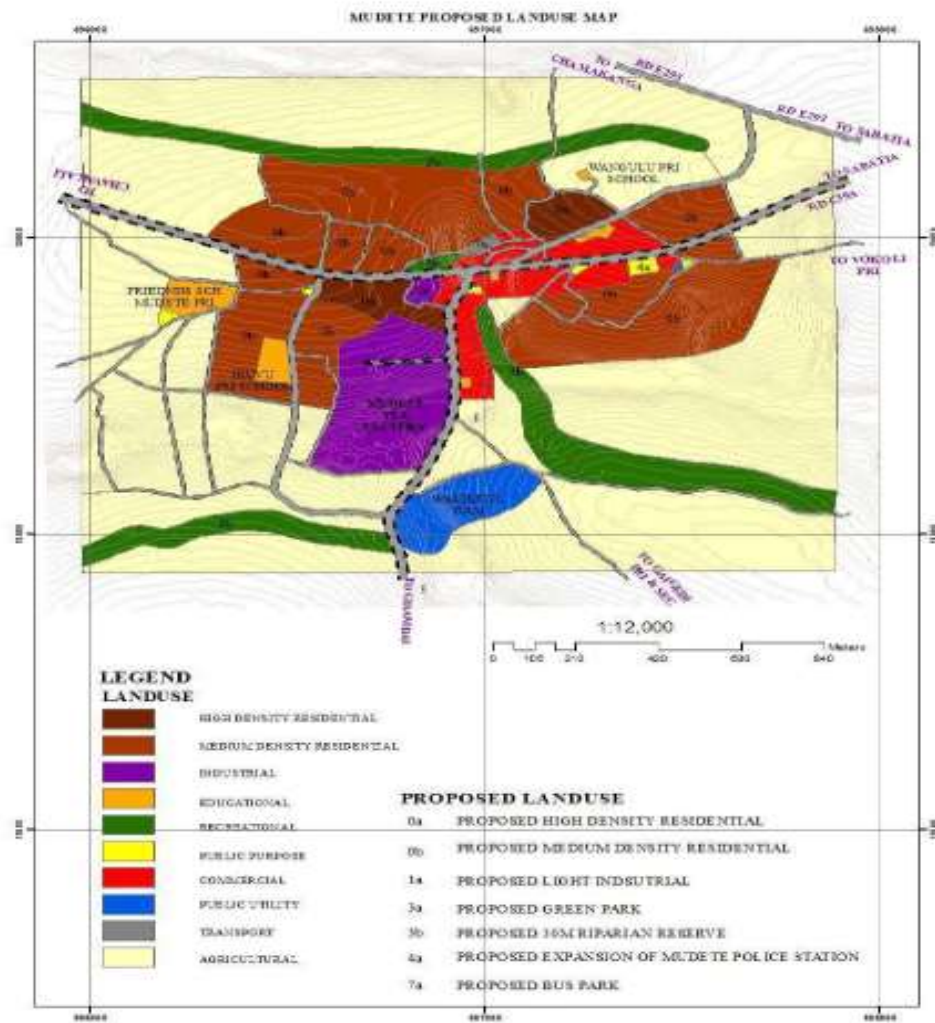
Map 2: Chavakali Area Action Plan



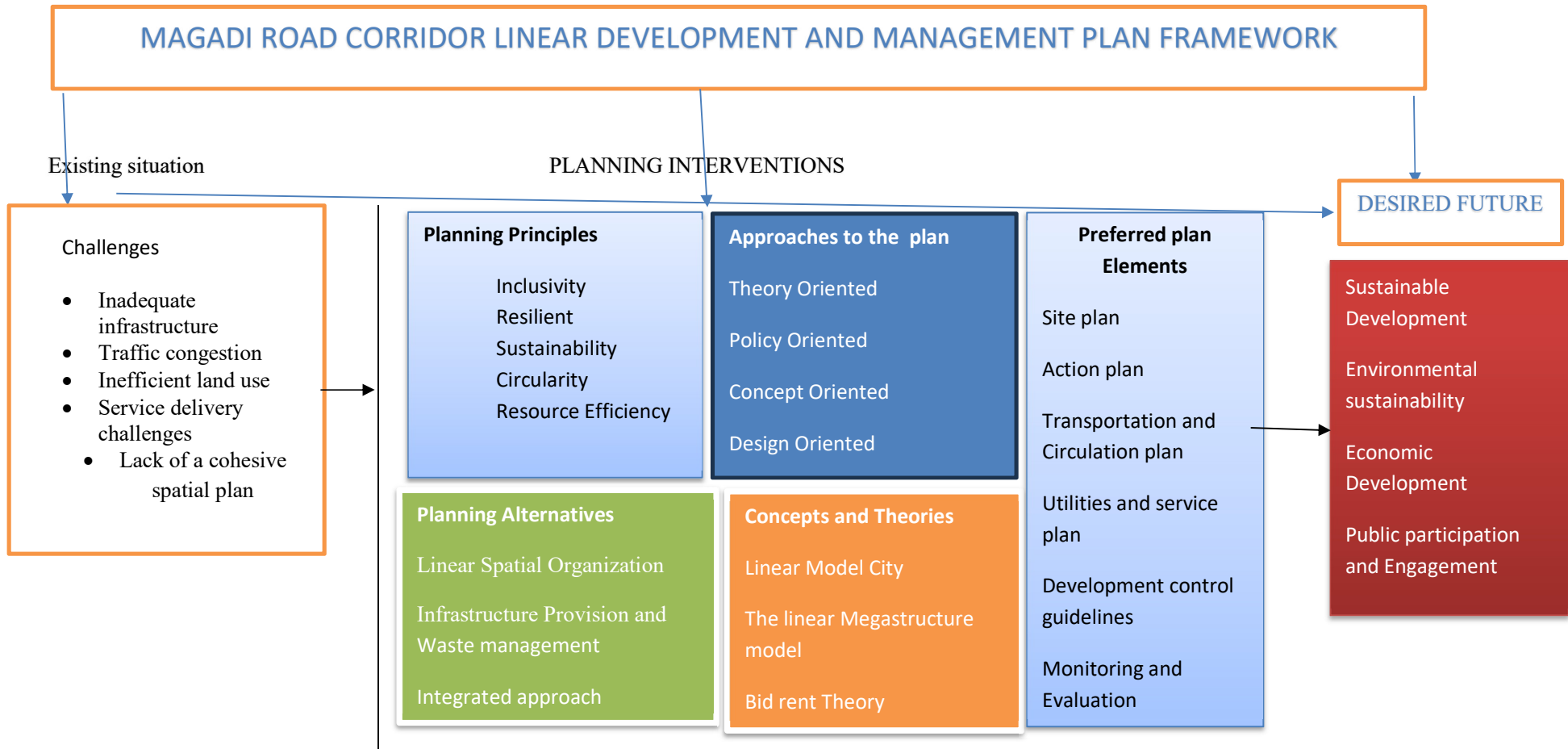
Map 3: Majengo/Vihiga Area Action Plan



Map 4: Mudete Area Action Plan



2.6 Design Conceptual framework



CHAPTER THREE:

3. SITUATION ANALYSIS OF THE PROJECT SITE

3.1 Context of the Project Area

3.1.1 Background of the Project Area

Magadi road, Ongata Rongai Town, Kajiado County

The study area is located in Ngong Municipality, Kajiado County, 17km south of Nairobi the capital city, and west of Ngong. Ongata Rongai town from Magadi road-Quary market is located at the following coordinates: 1.3939°S and 36.7442°E and lies 1731 meters above sea level. It is the most populous town in Kajiado County and the eleventh-largest urban center in Kenya with a population of 172,569 (KNBS, 2019). This is one of the fastest-growing towns in Kenya. The town is also part of the larger Nairobi Metropolitan Region where it serves as the dormitory town for Nairobi City.

The corridor traverses through diverse landscapes, including urban areas, peri-urban zones, and rural settlements. Ongata Rongai, situated along this corridor, has experienced significant population growth due to its proximity to Nairobi and relatively affordable housing options compared to the city centre. Magadi Road serves as a crucial artery for economic activities, connecting Nairobi with important industrial zones, agricultural areas, and trading centres. Ongata Rongai benefits from this economic connectivity, attracting businesses, residential developments, and services to cater to its growing population. The corridor facilitates the movement of people and goods between Nairobi and surrounding areas. It serves as a major commuter route for residents traveling to Nairobi for work, education, and other purposes. Additionally, it supports freight transportation, linking industries and markets along its path.

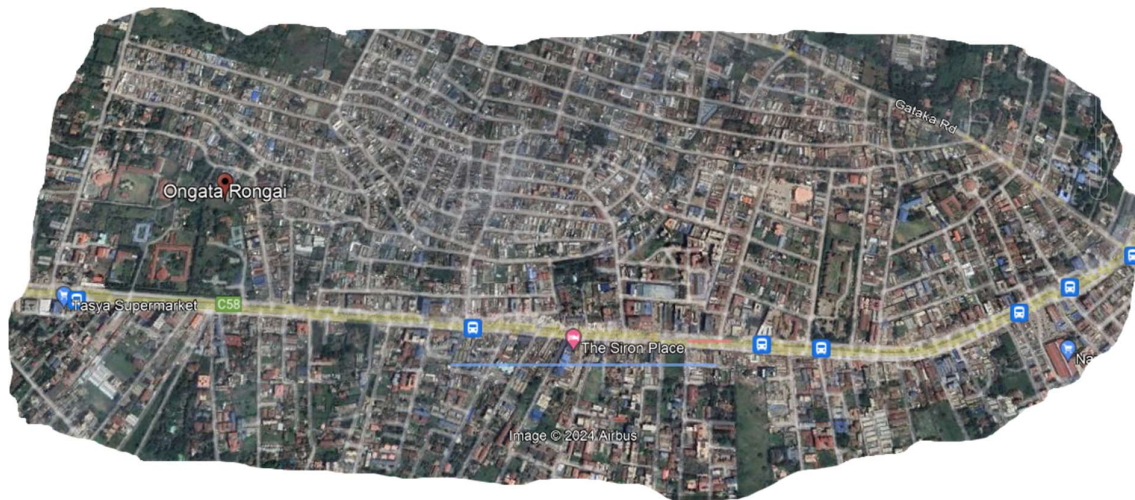


Figure 7: Google earth image showing Magadi road corridor

Historical Background

During Kenya precolonial period, the land was occupied by the Maasai community who were practicing pastoralism. The town developed from a cattle market, on the lower section, and a quarry site, on the upper side of the town and since 1950 the town has been rapidly growing to become the current township. Furthermore, due to Rongai's proximity to the capital city, intense development has been taking place since the 1990s due to the huge demand for housing by people working in the city. This has also resulted in the sprawling of the settlement beyond the border of Rongai town affecting the rural community who were formally nomadic pastoralists and forcing them to settle in addition to having alternative ways of managing cattle on small parcels thereby increasing land degradation. This has also reduced flora and fauna and increased human-wildlife conflict (County Government of Kajiado, 2019; Emily, 2000) The town is currently occupied by a population that is multi-ethnic due to immigration that has been taking place.

Through time the land tenure system has changed from community land and group ranches to individual ownership (Morara et al., 2014). There has been increased crop production and intensified livestock farming by the immigrants while around the peri-urban area of Rongai town, there has been an increased variety of land uses. The change inland use land cover has been due to intensified population pressure coupled with a lack of proper management of the town.

The history of Magadi Road traces back to the early days of Nairobi's development, where it initially served as a basic track connecting the burgeoning colonial administrative centre with surrounding regions. Named after Lake Magadi, which was a significant source of salt and a focal point for trade, the road facilitated the movement of goods, livestock, and people between Nairobi and the southwest. As Nairobi grew in the late 19th and early 20th centuries, so did the importance of Magadi Road, which gradually became a lifeline for the expanding city and its hinterlands.

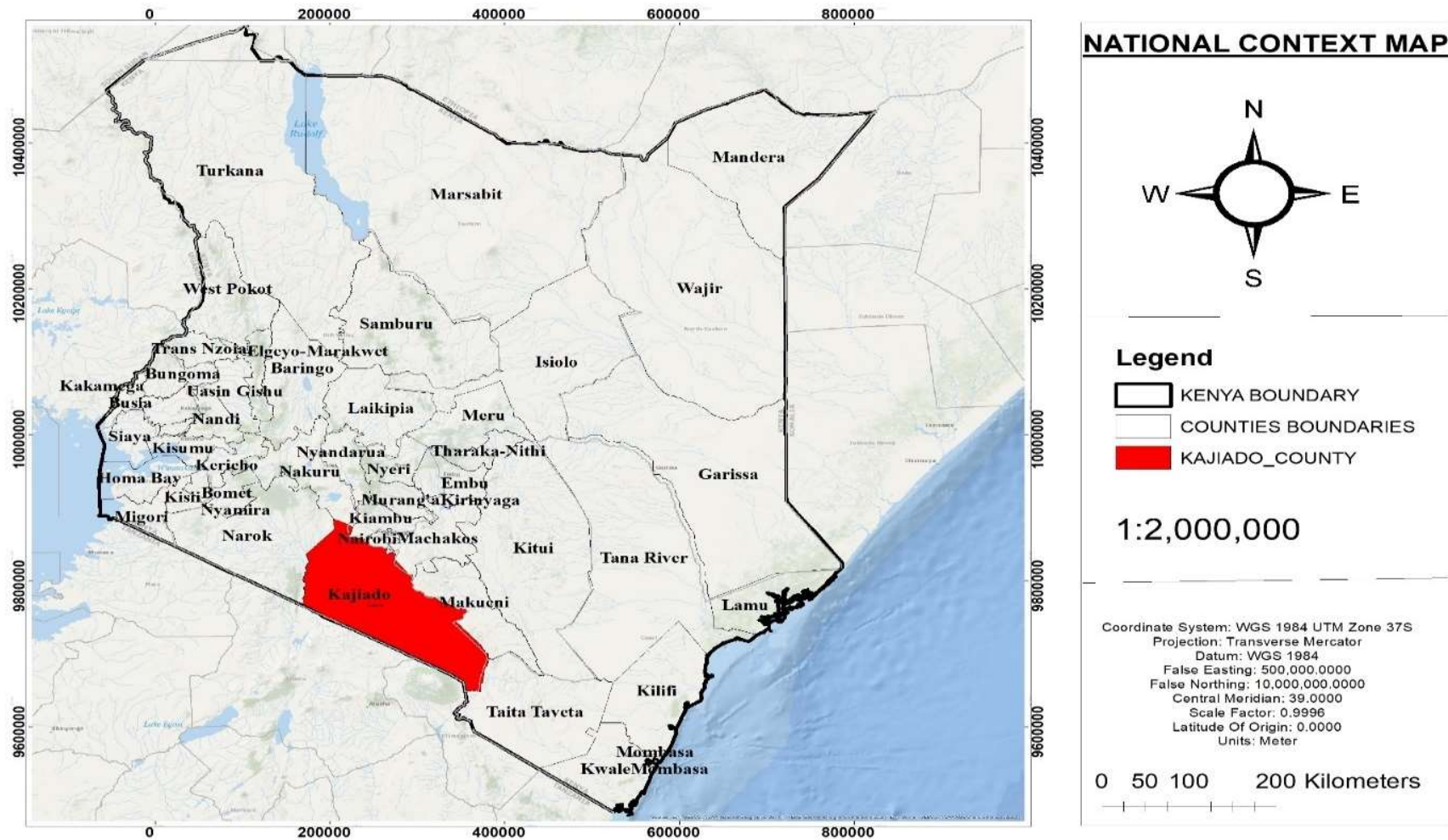
During the mid-20th century, Magadi Road underwent significant improvements as part of efforts to modernize Nairobi's transportation infrastructure. The road was widened, paved, and upgraded to accommodate the increasing volume of traffic, reflecting the city's rapid urbanization and industrialization. These enhancements transformed Magadi Road from a simple dirt track into a vital transportation artery, linking Nairobi with key satellite towns, agricultural areas, and industrial zones to the southwest.

Today, Magadi Road stands as a bustling thoroughfare lined with commercial establishments, residential neighbourhoods, and industrial facilities. It continues to play a crucial role in facilitating commuter traffic, freight movement, and economic activities between Nairobi and destinations such as Ongata Rongai, Kiserian, and Magadi Town. The road's history embodies the broader narrative of urbanization, economic growth, and infrastructure development in Kenya, illustrating the enduring significance of transportation in shaping the socio-economic landscape of the country.

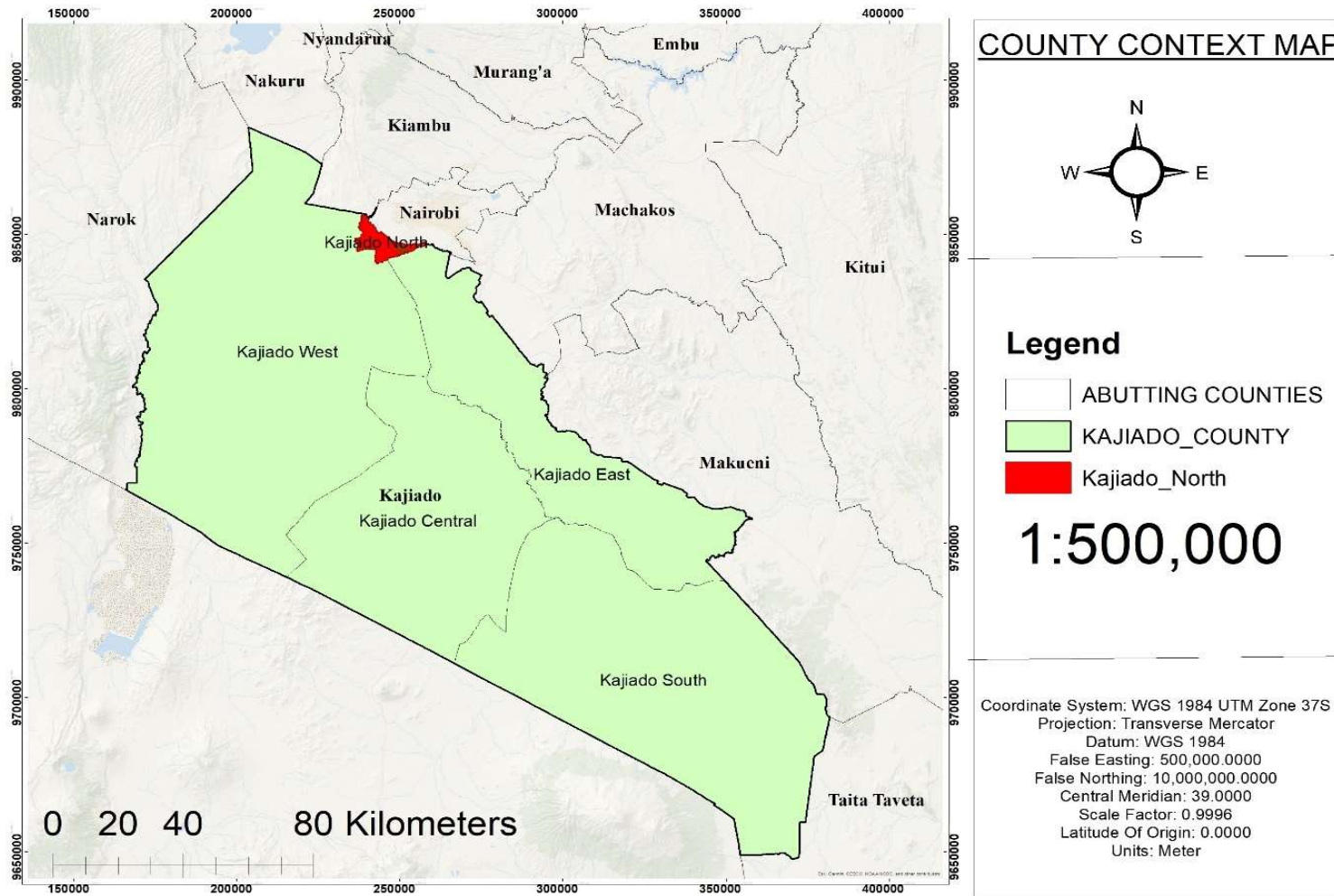
3.1.2 Contextual Site Location

Magadi road site in Ongata Rongai Town is situated in Kajiado North Constituency encompasses within two wards those are Nkaimurunya and Ongata Rongai ward, Kajiado County, Kenya. Ongata Rongai is approximately 17 kilometres from Nairobi's central business district (CBD), making it a popular residential area for people working in the city. The Contextual maps are arranged from National, County, Subcounty and the basemap respectively.

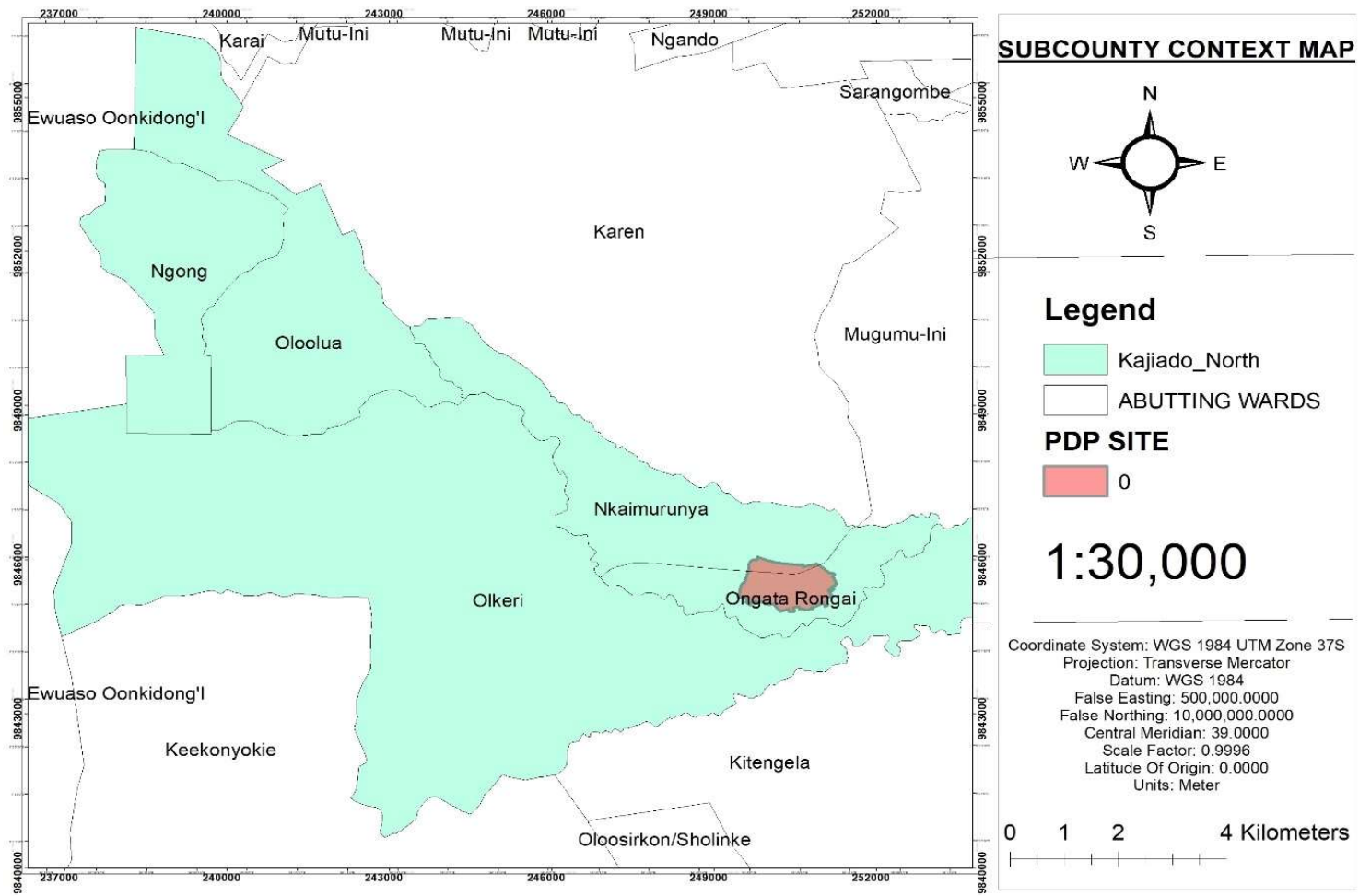
Map 2: National Context Map



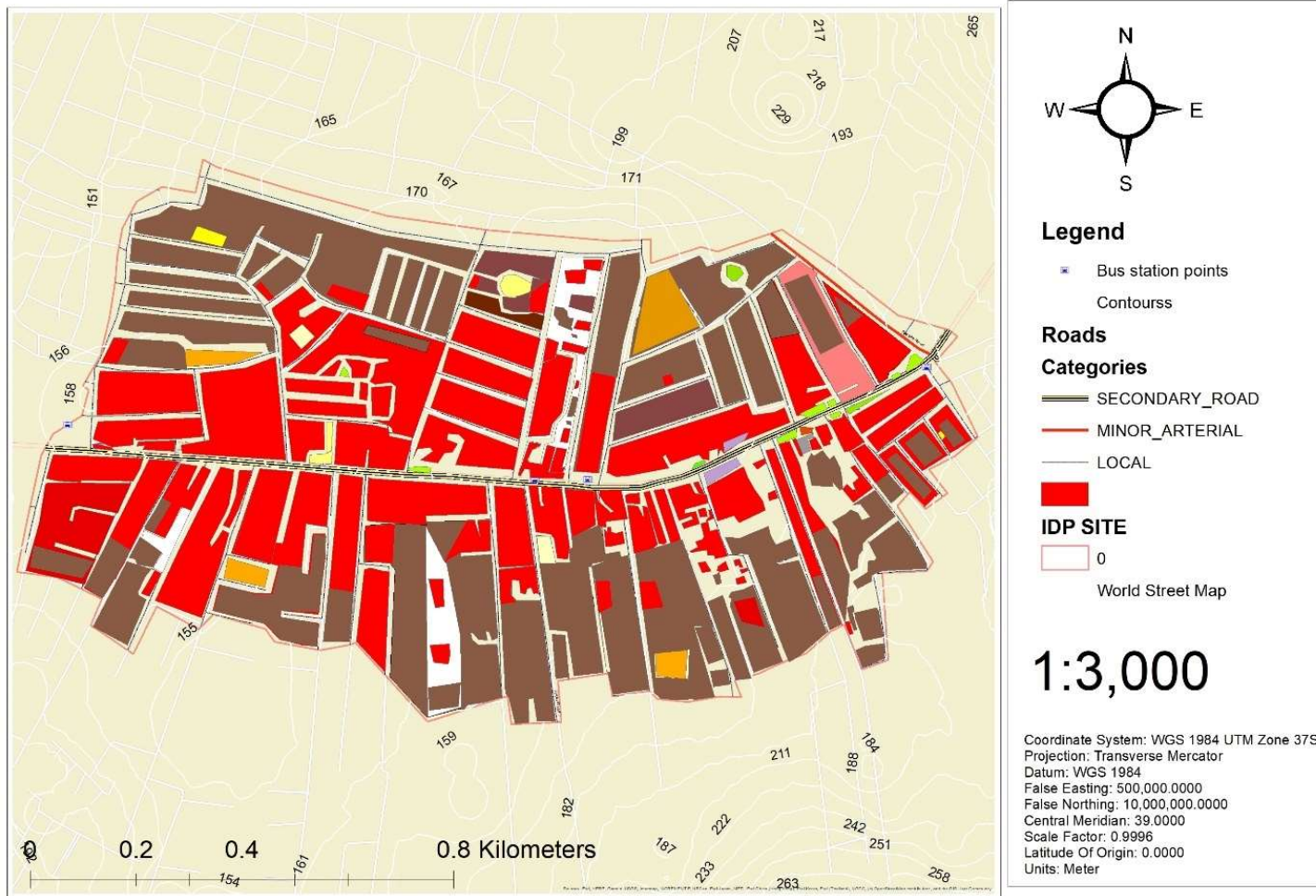
Map 3: County Context Map



Map 4: Subcounty Context Map



Magadi Road Corridor Basemap



3.2 Site Suitability Analysis

The process of assessing a site's conditions and capacity to satisfy the needs of the end user is known as site suitability analysis. It pertains to site planning. Site planning is the art of arranging the physical environment outside to facilitate human behavior. (Lynch, 1962).

3.2.1 Population and Demographic Projection

Over the past 10 years, Ongata Rongai, a satellite town on the edge of Kenya's capital, Nairobi, has experienced uncontrolled development and exponential population growth. Because of its appealing location close to the city, it's jumped from just under 40,000 residents in 2009 to a population of 172,569 in 2019. The most recent census data showed a high annual population growth rate of 16% but the population growth rate for the County is 5.5%.

Table 1: Population Projection

Year	2009	2019	2024	2029	2034	2039
Population Projection	40,178	172,569	226,018	292,505	378,657	490,190

3.2.2 Natural System Assessment

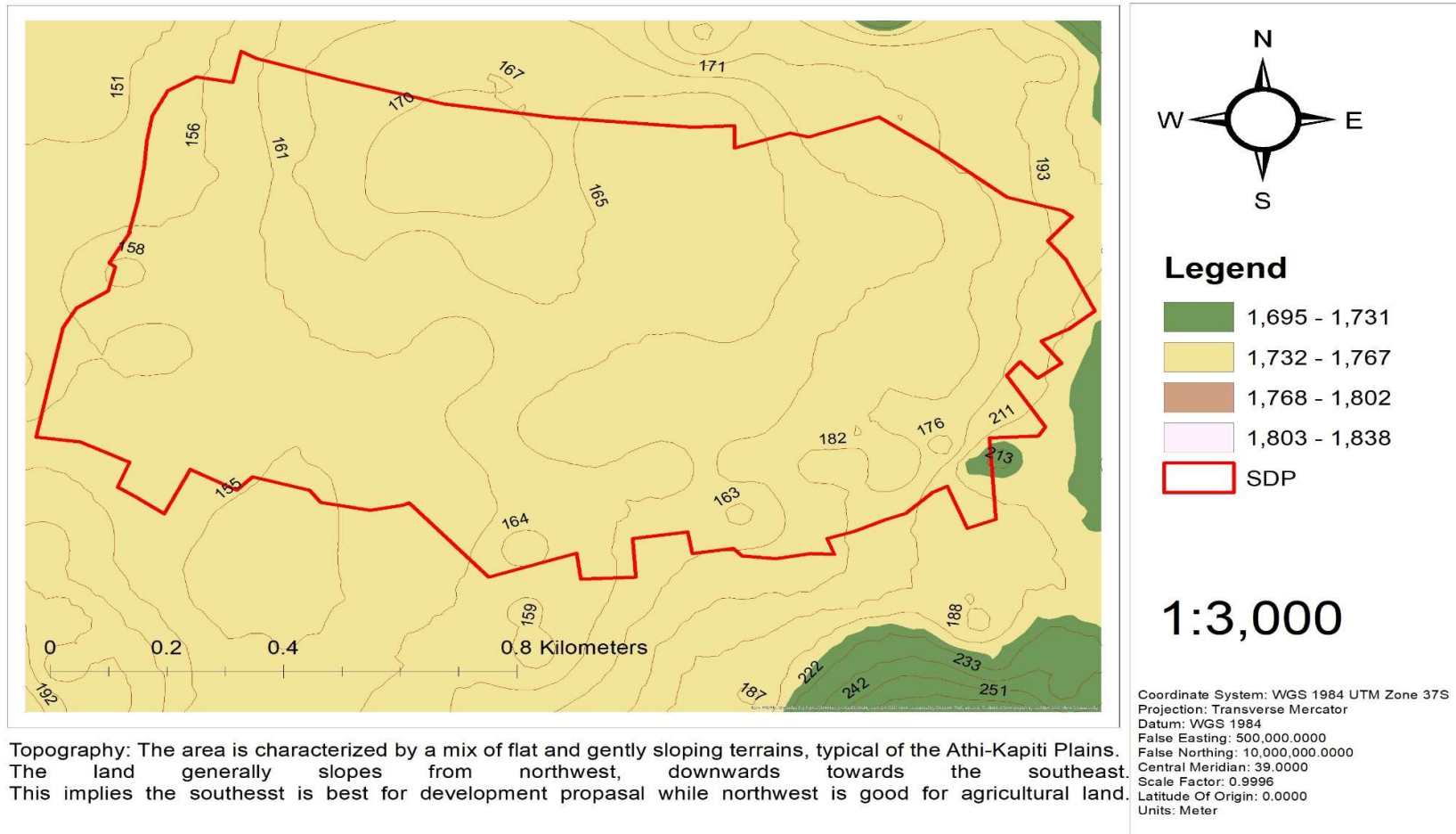
3.2.2.1 Physiography

Its deals with the understanding of natural features and processes of the Earth's surface. It encompasses the analysis of landforms, climates, vegetation, soils, and hydrology.

Topography and Slope

Map 5: Topography and Slope

Topography and Slope Map



Geology and Soils

Ongata Rongai exhibits geological and soil characteristics shaped by its position within the Great Rift Valley. The region's geology is predominantly influenced by volcanic activity, resulting in extensive deposits of volcanic rocks such as basalts, trachytes, and phonolites. These volcanic formations are interspersed with sedimentary layers, including lake sediments and alluvial deposits, reflecting the area's geological history of tectonic movements, volcanic eruptions, and sedimentation processes. The Rift Valley's tectonic activity has also contributed to the formation of faults and fractures, influencing the area's topography and groundwater distribution.

The soils in Ongata Rongai are largely derived from the weathering of volcanic rocks and alluvial deposits, resulting in a variety of soil types. Common soil types include black cotton soils (Vertosols), which are rich in clay and have high water retention capabilities, making them suitable for agriculture but challenging for construction due to their expansive nature. Other prevalent soils include loamy and sandy soils, which offer better drainage and are more suitable for building and infrastructure development. The soil fertility varies, with some areas being highly fertile due to the presence of volcanic ash, while others may require soil management practices to enhance agricultural productivity. Overall, the geological and soil characteristics of Ongata Rongai significantly influence land use, agricultural potential in some parts, and urban development in the region.

Hydrology and Drainage System

The hydrology and drainage system of Ongata Rongai, Magadi Road corridor, are integral aspects of the region's environmental and urban landscape. The area's hydrology is influenced by its position within the semi-arid environment of the Great Rift Valley. Rainfall in Ongata Rongai is seasonal and often sporadic, leading to variable water availability. The primary sources of surface water include seasonal rivers and streams that flow during the rainy season, such as the Mbagathi River. Groundwater resources also play a crucial role, with many residents relying on boreholes and wells for their water supply. The area's aquifers are recharged by rainfall, though the rate of recharge can be inconsistent, impacting water availability during dry periods.



Figure 8:Current Drainage system

Image: drainage system along Magadi road

The drainage system in Ongata Rongai is shaped by the natural topography and the underlying geological structures. The terrain, characterized by gentle slopes and occasional steep escarpments, directs surface runoff towards the seasonal rivers and streams. However, rapid urbanization has put pressure on the natural drainage system, often leading to issues such as flooding and erosion, especially during heavy rainfall events. The expansion of impervious surfaces like roads and buildings reduces the land's natural ability to absorb water, exacerbating these problems.

3.2.2.2 Climate Condition

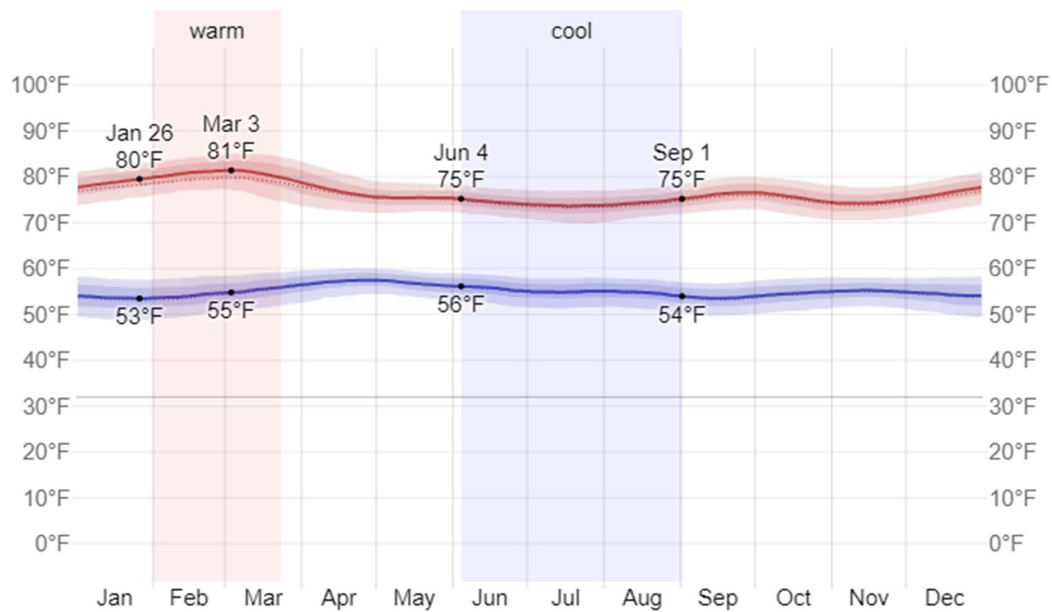
Average Temperature

The temperature of Ongata Rongai Town varies with season. The warm season lasts for 1.7 months, from February 1 to March 23, with an average daily high temperature above 80°F. The hottest month of the year in Rongai is March, with an average high of 80°F and low of 56°F.

The cool season lasts for 2.9 months, from June 4 to September 1, with an average daily high temperature below 75°F. The coldest month of the year in Rongai is July, with an average low of 55°F and high of 74°F.

Graph: Average High and Low Temperature in the Summer in Rongai

Graph 1: Average Temperature



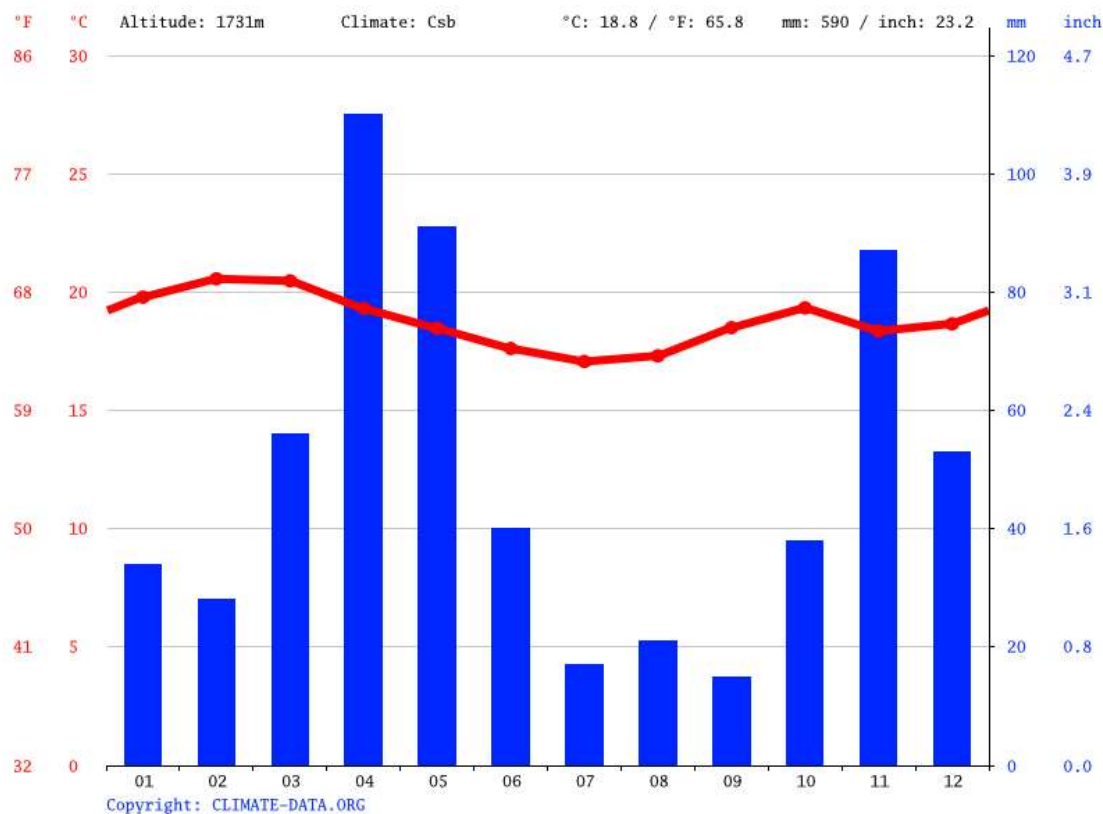
Source: weatherspark.com

Rainfall

The driest month is September, with 15 mm of rain. The highest amount of precipitation occurs during the month of April, with an average quantity reaching up to 110 mm.

Rainfall and temperature graph for Ongata Rongai

Graph 2: Average Rainfall



Humidity

The perceived humidity level in Rongai, as measured by the percentage of time in which the humidity comfort level is *muggy*, *oppressive*, or *miserable*, does not vary significantly over the course of the year, remaining a virtually constant 0% throughout.

Graph 3: Humidity



3.2.3 Existing Physical Development on Site

Magadi road corridor Ongata Rongai has experienced significant physical development over the years, driven by its proximity to Nairobi and increasing population.

1. Residential Development

Residential Estates and Housing: Gated Communities: Various gated communities and housing estates such as Rimpa Estate, Tusker Estate, and Nkoroi offer a mix of middle to high-end residential options. **Apartment Complexes:** Numerous apartment buildings cater to the growing population, providing affordable housing options. **Single-Family Homes:** Many areas feature single-family homes, ranging from modest dwellings to larger, more luxurious houses.

2. Commercial Development

Shopping Centers and Malls; Maasai Mall: A major shopping center with retail stores, supermarkets, and eateries. **Tuskys and Naivas Ongata Rongai:** Another significant shopping destination providing a variety of goods and services. **Ongata Mall:** Features shops, restaurants, and entertainment facilities.

Markets and Retail Outlets: Local Markets: Traditional markets such as the Ongata Rongai and Quarry Market offer fresh agricultural produce, clothing, and other goods. **Retail Shops:** Numerous small retail outlets and businesses line the main roads and residential areas.

3. Institutional Development

Educational Institutions such as Multimedia University: A major higher education institution offering a range of academic programs. **Ongata Academy:** A well-known private school providing primary and secondary education. **Public Schools:** Several public primary and secondary schools serve the local community.

Healthcare Facilities:

Ongata Rongai Health Centre is a key public healthcare facility. **Private Hospitals and Clinics:** Numerous private healthcare providers offer specialized medical services, including Nairobi Women's Hospital and Gertrude's Children's Hospital branch.

4. Infrastructure Development

Road Network: Magadi Road: The primary arterial road connecting Ongata Rongai to Nairobi and other areas. **Internal Roads:** Various internal roads connect different parts of the town, though some are in need of upgrades and maintenance. **Public Transportation:** Matatus and Buses: The main modes of public transport, provide connectivity within the town and to Nairobi. **Boda-Bodas:** Widely used for short-distance travel within the town.

5. Recreational and Social Amenities

Entertainment and Leisure: Cinemas and Theaters: Facilities like the one in Maasai Mall offer entertainment options. **Restaurants and Cafes:** A variety of dining establishments serve different cuisines and cater to diverse tastes.

7. Industrial and Light Manufacturing

Small Industries: Workshops and Light Manufacturing Units: Scattered across the town, providing employment and services. Construction Material Suppliers: Numerous businesses supply building materials, reflecting ongoing development.

Map 6: Existing Physical Development

Existing Physical Development on the Site



1. School
2. Drainage System
3. Magadi Road
4. Residential buildings
5. Commercial Buildings
6. Hospital

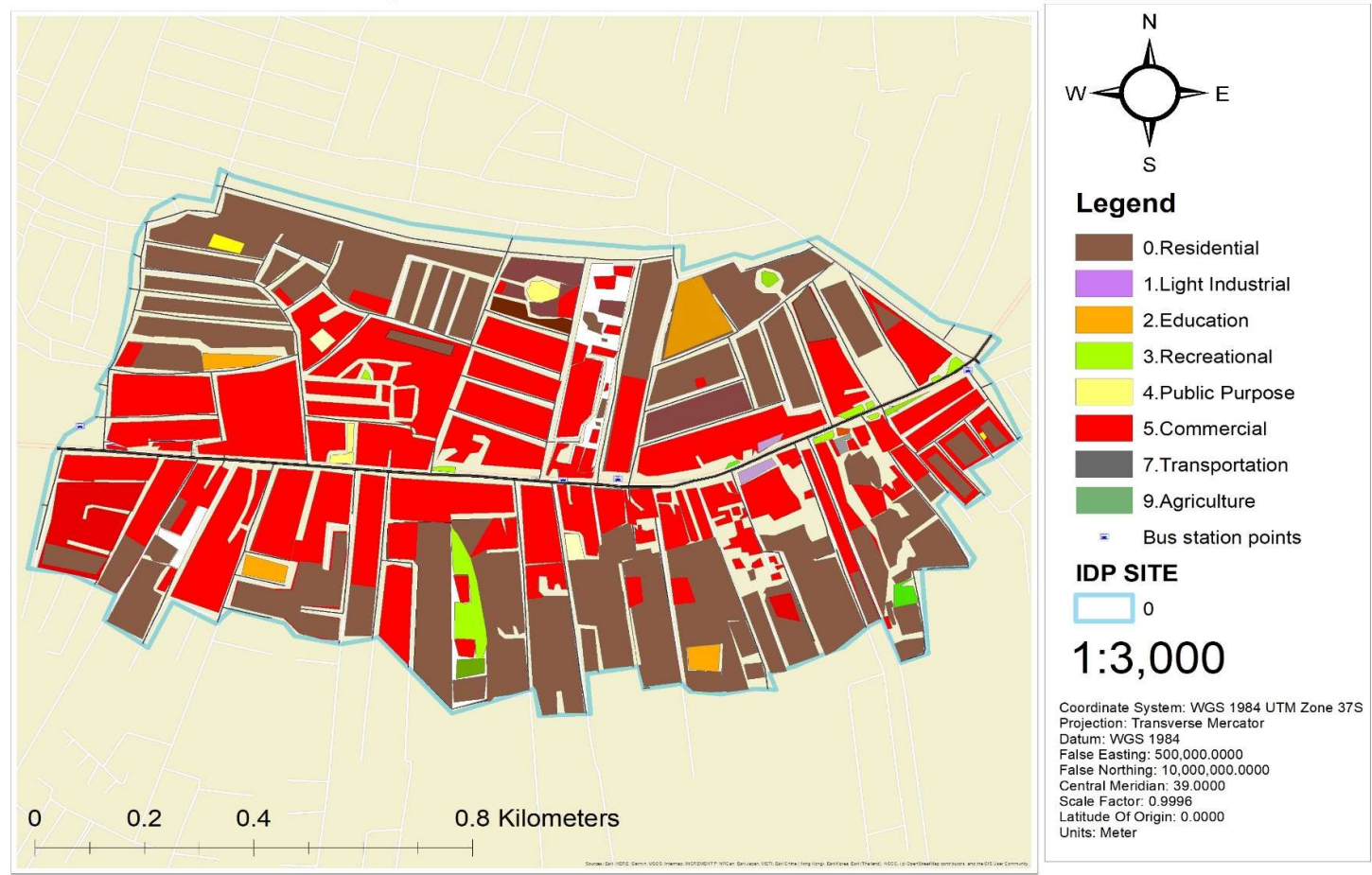


3.2.4 Abutting Land Uses

The land uses in the site range from different positions. Residential: Areas designated for housing, including gated communities, apartments, and single-family homes. Commercial: Zones for businesses and trade activities, encompassing shopping centers, markets, and retail outlets. Industrial and Light Manufacturing: Spaces allocated for manufacturing operations, workshops, and production facilities. Institutional: Land reserved for educational institutions, healthcare facilities, and other public services. Recreational and Open Space: Areas set aside for parks, sports facilities, and leisure activities. Agricultural: Land used for farming, crop cultivation, and livestock rearing. Mixed-Use Development: Zones incorporating a combination of residential, commercial, and recreational functions within a single area. Utility and Infrastructure: Land dedicated to essential services like water supply, electricity, and waste management. Transportation and Road Network: Spaces utilized for roads, highways, public transportation, and associated infrastructure.

Map 7: Land Use

Magadi Road Corridor Land Use Map



3.2.5 Site Views and Nodes

Ongata Rongai Town offers a variety of site views and nodes that define its urban landscape and contribute to its identity. These elements play crucial roles in the town's spatial organization, functionality, and aesthetic appeal.

Site Views

Ongata Rongai provides a mix of natural and built environment views. To the west, residents and visitors can enjoy panoramic views of the Ngong Hills, which are a prominent natural landmark offering scenic beauty and recreational opportunities. The eastern views often showcase the expansive plains of the Great Rift Valley, providing a sense of openness and connection to the broader landscape. Within the town, views are shaped by the built environment, including residential areas, commercial centers, and institutional buildings. Iconic structures such as churches, schools, and markets serve as visual anchors in the townscape. The contrast between the natural hills and the urban development creates a dynamic visual experience for those traveling through or living in Ongata Rongai.

Nodes

Several key nodes within Ongata Rongai serve as focal points for activity and interaction. The town center along Magadi Road is the primary commercial hub, bustling with shops, markets, and eateries. This area is characterized by high pedestrian and vehicular traffic and acts as a central meeting point for the community. Another significant node is the intersection near Maasai Mall, which functions as a commercial and social gathering spot with a mix of retail stores, restaurants, and entertainment facilities. Additionally, key institutional nodes, such as educational institutions (e.g., Multimedia University) and healthcare facilities, draw specific groups of people, contributing to the town's vibrancy. These nodes not only facilitate economic activities but also enhance social cohesion by providing spaces for communal interaction.



Figure 9: Site View and Nodes along the Magadi Road

3.2.6 Transport and Circulation Analysis

The transportation network is characterized by a mix of roadways, public transport systems, and informal transit options that cater to the needs of its residents and facilitate movement to and from Nairobi and other neighboring regions.

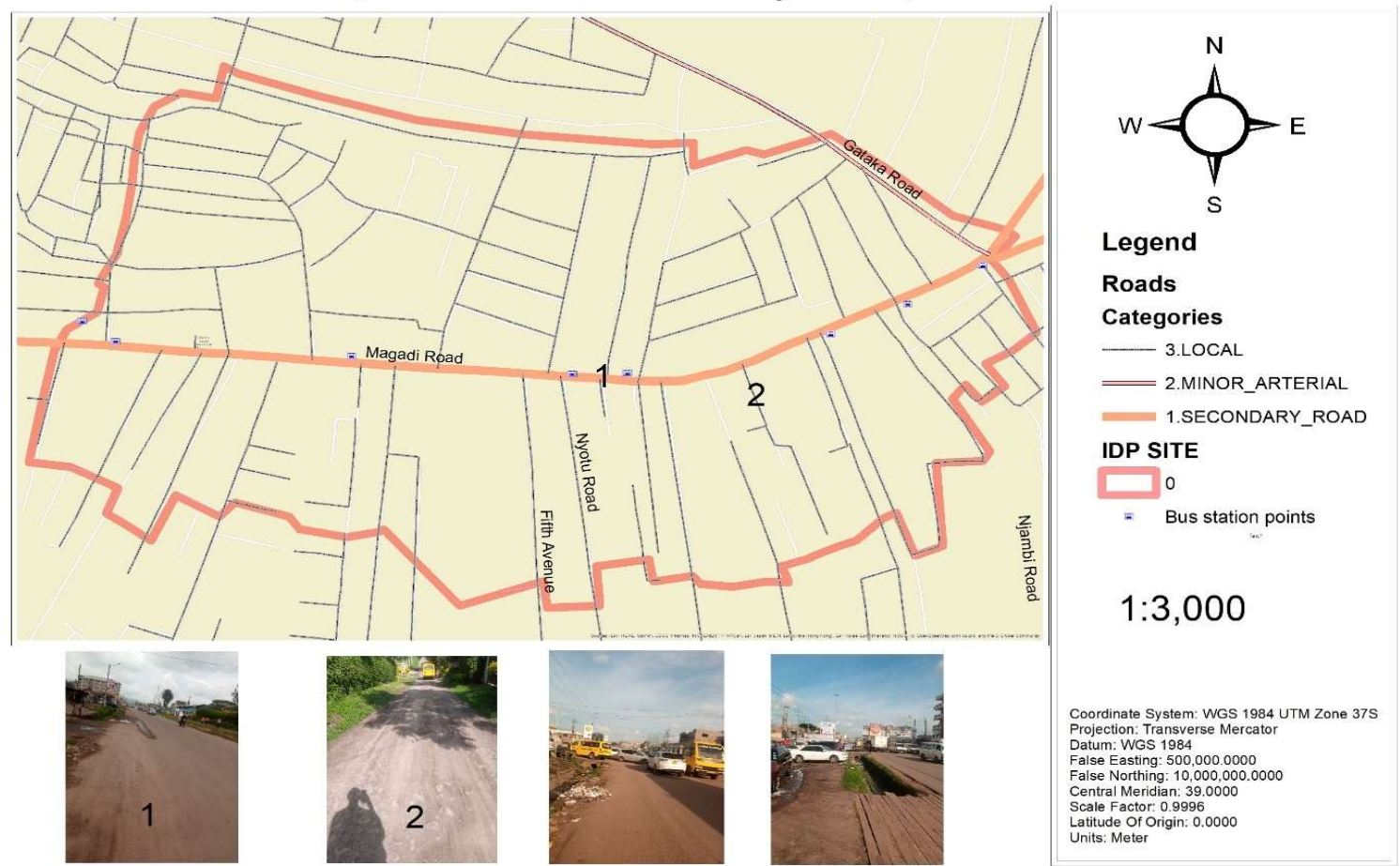
Internally, Ongata Rongai relies heavily on a network of primary and secondary roads that connect various neighborhoods, commercial centers, and key institutions. Magadi Road class **C58** is the main arterial road running through the town, serving as the backbone for local and through traffic. This road supports significant vehicular movement, including private cars, matatus (shared minibusses), and boda-bodas (motorcycle taxis), which are the primary modes of transport within the town. Traffic congestion is a common issue, especially during peak hours, due to the high volume of vehicles and the rapid urbanization that outpaces infrastructure development. Pedestrian circulation is also notable but not designed, with sidewalks and informal pathways used extensively by residents for short-distance travel and access to local amenities.

Some of the major roads connecting to Magadi Road, Ongata Rongai include;

1. Lang'ata Road: Lang'ata Road intersects with Magadi Road in Nairobi's southern suburbs.
2. Southern Bypass: The Southern Bypass is a major highway that runs parallel to Magadi Road, connecting Nairobi's southern suburbs with other parts of the city and beyond.
3. Ngong Road: Ngong Road intersects with Magadi Road near Karen and Ngong Town, forming an important connection between Nairobi and the Rift Valley region.
4. Kiserian-Isinya Road: This road branches off from Magadi Road in Ongata Rongai and extends towards Kiserian and Isinya towns.
5. Mbagathi Way: Mbagathi Way also intersects with Magadi Road. This road serves as another important connector within Nairobi, facilitating east-west movement across the southern part of the city and linking various residential and commercial areas.

Map 8: Existing Transport and Circulation Analysis

Transport and Circulation Analysis Map



3.2.7 Utility and Services Analysis

3.2.7.1 Water Supply

Ongata Rongai faces challenges with water supply due to its rapid population growth and limited infrastructure. The primary sources of water include boreholes, wells, and rainwater harvesting systems. Many residents rely on private water vendors for their daily needs, which can be costly and inconsistent. Efforts are ongoing to improve water supply infrastructure, including extending Nairobi's water pipeline network to serve the town better and implementing community water projects.

3.2.7.2 Sanitation and Waste Management

Sanitation services in Ongata Rongai are a mix of private and public provisions. Most households use septic tanks and pit latrines, as the sewerage system is not fully developed. This has led to environmental and health issues, especially during the rainy season when contamination risks increase. Solid waste management is handled by both private companies and the local government, with regular collection services for household and commercial waste. However, illegal dumping and inadequate waste disposal remain challenges that require more effective management strategies and public awareness campaigns.



Figure 10: Open and poorly managed Drainage

3.2.7.3 Power and Lighting

Ongata Rongai is connected to the national electricity grid managed by Kenya Power. The electricity supply is relatively stable, though power outages can occur, especially during adverse weather conditions. Many households and businesses have adopted alternative energy sources such as solar panels to supplement the grid supply and ensure a more reliable energy supply. The adoption of renewable energy sources is encouraged to reduce dependency on the national grid and promote sustainability.

3.2.8 Environmental Issues

Waste Management and Pollution

One of the major environmental issues in Ongata Rongai is ineffective waste management. The town's rapid population growth has outpaced the development of adequate waste disposal systems. As a result, illegal dumping of solid waste is common, leading to unsightly piles of garbage in various parts of the town. This not only mars the town's aesthetics but also poses significant health risks, as decomposing waste can attract vermin and spread diseases. Moreover, the inadequate collection and disposal of waste contribute to water and soil pollution, further degrading the local environment.

Water Supply and Quality

Water scarcity and poor water quality are pressing concerns in Ongata Rongai. The town relies heavily on boreholes, wells, and water vendors, which are often inconsistent and insufficient to meet the demand. During dry seasons, water shortages become acute, affecting both domestic and commercial activities. Additionally, the quality of water from these sources can be questionable, as there is a risk of contamination from improperly disposed waste and sewage. Contaminated water sources pose serious health risks, including waterborne diseases, which are exacerbated by inadequate sanitation infrastructure.

Deforestation and Loss of Green Spaces

The expansion of residential and commercial developments has led to significant deforestation and the loss of green spaces in Ongata Rongai. Trees and natural vegetation are often cleared to make way for new buildings and infrastructure, leading to habitat destruction and a decrease in biodiversity. The loss of green spaces also reduces the natural absorption of rainwater, increasing the risk of flooding during heavy rains. Moreover, the reduction in vegetation cover contributes to the urban heat island effect, making the town warmer and less comfortable, especially during hot weather.

Air Pollution

Air pollution is another environmental challenge in Ongata Rongai, primarily due to emissions from vehicles and industrial activities. The high volume of traffic on Magadi Road, combined with the use of older, poorly maintained vehicles, contributes to elevated levels of exhaust fumes and particulate matter in the air. Additionally, industrial activities, such as small-scale manufacturing and workshops, can emit pollutants that degrade air quality. Poor air quality has significant health implications, particularly for respiratory conditions, and affects the overall livability of the town.

Inadequate Drainage and Flooding

Ongata Rongai's drainage infrastructure is often inadequate to handle heavy rainfall, leading to frequent flooding. The town's rapid urbanization has increased the amount of impervious surfaces, such as roads and buildings, which prevents rainwater from naturally soaking into the ground. Consequently, during heavy rains, water accumulates quickly, overwhelming the existing drainage systems. Flooding not only damages property and infrastructure but also exacerbates sanitation issues by spreading waste and contaminants, posing further health risks to residents.

Soil Erosion

Soil erosion is a significant environmental issue in Ongata Rongai, particularly in areas where vegetation has been cleared for development. Without adequate ground cover, soil is easily washed away by rain, leading to the degradation of land and the siltation of waterways. Erosion reduces soil fertility, affecting any remaining agricultural activities and the stability of foundations for buildings and infrastructure. The loss of topsoil also contributes to further environmental degradation and the loss of natural landscapes.

Human-wildlife conflict

Wildlife-human conflict is a significant environmental issue in Ongata Rongai, particularly due to its proximity to Nairobi National Park and other natural habitats. As urban development encroaches on wildlife habitats, interactions between humans and wildlife become more frequent, leading to various conflicts and challenges.

Environmental Challenges Map



3.2.9 Legal, Institution, and Financial Issues- Local

1. Legal Issues

The planning development project is guided by legal and policy frameworks as discussed in chapter two of this report such as the Constitution of Kenya, Urban Areas and Cities Act, Water Act, Physical Planning Handbook, Vision 2030, Environmental Management and Coordination Act (EMCA), Physical Land Use and Planning Act (PLUPA), National Land Use Policy among others. These laws and policies do not have adequate provisions for the integration of land use and planning in Kenya, therefore, posing a challenge in the preparation of this planning development project. Additionally, the lack of blue economy planning provision standards and guidelines in Kenya has resorted to the use of other countries' guidelines which may not necessarily be applicable in Kenya.

2. Institutional Issues

Kajiado County government (KCG) is in charge of all development, management, and service delivery in Ongata Rongai. Other stakeholders and institutions, such as private sector developers' investors, CBOs, NGOs, the County Governments will have to be involved due to the project's complexity. Major institutions involved in the sustainable planning for Magadi Road Corridor include;

1. Ministry of Transport, Infrastructure, Housing, Urban Development and Public work which is in charge of development and maintainers of roads
2. NCC- the department of physical planning is in charge of providing development control guidelines and carrying out enforcement. This plays a key role in assisting the implementation of this project.
3. NLC- this is the body in charge of managing public land on behalf of the national government. The commission is a key factor in guiding land acquisition for the provision of public services.
4. KeNHA, KURA, KeRRA, KRB which are under roads.
5. Ministry of Environment and Forestry

3. Financial Issues

Since finances are required for the implementation of the project, several actors are to be involved who include:

- 1) Government Institutions.
- 2) Private investors.
- 3) NGOs.
- 4) Private Sector Institutions.
- 5) International Institutions.

3.2.10 SWOT Analysis of Site Condition and Evaluation of Site Suitability

SWOT analysis is a strategic planning tool used to identify and evaluate the strengths, weaknesses, opportunities, and threats of an area. This analysis will inform strategic decision-making and help in developing strategies to capitalize on its strengths and opportunities while mitigating its weaknesses and threats.

Table: SWOT Analysis

STRENGTH	• Proximity to Nairobi: Close to Nairobi's CBD, providing easy access for commuters. • Major Arterial Road: Magadi Road is a key transport route, facilitating movement and trade. • Affordable Housing: Offers more affordable housing options compared to Nairobi city center. • Educational Institutions: Presence of schools and universities attracts a younger demographic. • Commercial Growth: Increasing number of businesses, shopping centers, and markets. • Healthcare Facilities: Availability of hospitals and clinics. • Public Transport: Well-served by matatus and buses, enhancing connectivity. • Urban-Rural Balance: Offers a blend of urban amenities and access to nature. • Economic Opportunities: Growth in real estate and retail sectors provides employment opportunities. • Community Services: Presence of social amenities like banks, supermarkets, and recreational facilities.
WEAKNESSES	• Traffic Congestion: High traffic volumes on Magadi Road, especially during peak hours. • Infrastructure Strain: Existing infrastructure often struggles to keep up with rapid population growth. • Water Supply Issues: Inconsistent water supply in some areas. • Waste Management: Insufficient waste disposal and recycling facilities. • Security Concerns: Rising crime rates in some parts of the town. • Public Transport Reliability: Irregular public transport schedules can cause inconvenience. • Pollution: Increasing air and noise pollution from traffic. • Road Maintenance: Poor road conditions and frequent need for repairs. • Health Services Overload: Local healthcare facilities often face overcrowding due to increasing population. • Housing Quality: Some housing developments are of substandard quality.
OPPORTUNITIES	• Infrastructure Development: Potential for significant improvements in roads, water, and sanitation. • Real Estate Investment: Growing demand for residential and commercial properties.

	• Tourism Development: Proximity to Nairobi National Park can boost eco-tourism. • Business Expansion: Opportunities for new businesses and retail outlets. • Education Sector Growth: Expansion of existing institutions and establishment of new ones. • Public Transport Enhancements: Potential for improved public transport systems, such as BRT. • Green Spaces: Development of parks and recreational areas. • Technological Advancements: Adoption of smart city technologies to manage urban growth. • Community Projects: Increased community engagement in development projects. • Sustainable Practices: Implementation of sustainable development practices and green building technologies.
THREATS	• Overpopulation: Rapid population growth strain resources and infrastructure. • Environmental Degradation: Urban sprawl leading to loss of green spaces and wildlife habitats which causes Human-Wild Animal conflicts. • Economic Disparities: Growing economic divide between different segments of the population. • Land Disputes: Potential for conflicts over land ownership and usage. • Infrastructure Overload: Existing infrastructure are overwhelmed by rapid development. • Crime Rates: Potential increase in crime rate causes insecurity. • Health Epidemics: Risk of health epidemics due to dense population and inadequate healthcare. • Climate Change: Effects of climate change such as drought is an impact to the area. • Political Instability: Changes in political climate affecting development policies. • Economic Downturns: Vulnerability to economic fluctuations that could affect investment and development.

Map 9: Challenges on the site



Constrains Map

1. Poor waste management along Magadi road
2. Traffic Jam due to many private vehicles
3. Lack of traffic sign to control the movement
4. Poor drainage system along the road
5. Potholes along Magadi road
6. Informal business invading road reserves



3.3 Synthesis

3.3.1 Proposed Project Characteristics

The proposed development project for Ongata Rongai, including the upgrade of Magadi Road, creation of a green belt, implementation of a Bus Rapid Transit (BRT) system, development of well-connected corridors, and establishment of formal markets, aims to enhance infrastructure, promote sustainable urban growth, and improve the quality of life for residents.

1. a) Upgrade of Magadi Road to Four Lanes with a Green Belt Buffer

Expanding Magadi Road to four lanes will significantly increase its capacity, reducing traffic congestion and improving the flow of vehicles, making commutes faster and more efficient. The upgraded road will include enhanced safety features such as proper signage, pedestrian crossings, and barriers, improving safety for both motorists and pedestrians. A green belt along the road will serve as a buffer zone to prevent urban sprawl and encroachment by informal businesses, preserving natural habitats and enhancing biodiversity. The green belt will also beautify the area, creating a visually appealing environment with lush greenery and well-maintained landscapes. Vegetation in the green belt will help improve air quality by absorbing pollutants and providing oxygen, contributing to a healthier environment for residents.

The green belt will protect natural areas from urban encroachment, ensuring sustainable urban growth and conservation of local ecosystems. The presence of green spaces will help mitigate the urban heat island effect, enhance stormwater management, and increase the town's resilience to climate change impacts.

b) Implementation of a Bus Rapid Transit (BRT) System

Characteristics:

The BRT system will provide a high-capacity, efficient public transportation option, reducing reliance on private car. BRT buses will operate in dedicated lanes, ensuring fast and reliable service with minimal delays. Stations will be designed to be easily accessible to all users, including people with disabilities. By promoting public transport use, the BRT system will help reduce vehicle emissions, contributing to better air quality and environmental sustainability.

3. Community-Led Waste Management for Ongata Rongai Town

The community-led waste management system for Ongata Rongai Town is characterized by its decentralized, grassroots approach that actively involves residents in managing waste. It features neighborhood collection points managed by volunteers, local recycling hubs for sorting and selling recyclables, and communal and household composting initiatives. Education and awareness campaigns are integral, with workshops and school programs promoting waste reduction and recycling. Participation is incentivized through reward programs and recognition awards, fostering community engagement and competition. Collaborations with NGOs, private sector partners, and local government provide essential support and resources. Land acquisition for necessary infrastructure is handled transparently, with fair compensation to ensure community buy-in and minimal disruption. This approach emphasizes sustainability, community ownership, and resilience.

4. Development of Well-Connected Corridors in Ongata Rongai Town

Characteristics:

The development of well-connected corridors will facilitate better movement within the town, making it easier for residents to access different parts of Ongata Rongai. These corridors will integrate various modes of transport, including walking, cycling, and public transit, promoting sustainable urban mobility. Improved connectivity will stimulate local businesses by providing better access for customers and suppliers, fostering economic growth. The corridors will be aligned with urban planning principles to ensure orderly development and efficient land use.

5. Establishment of Formal Markets in Quarry Open Market and Other Locations

Characteristics:

Developing formal markets will provide organized spaces for vendors, reducing the chaotic nature of informal markets and improving the shopping experience. The markets will be equipped with essential infrastructure such as proper sanitation facilities, water supply, electricity, and waste management systems. The markets will create economic opportunities for local vendors and entrepreneurs, supporting small businesses and boosting the local economy. The markets will serve as community hubs where residents can buy and sell goods, fostering social interaction and community cohesion.

3.3.2 Project Development Impact Analysis

Table 2: Project Impact Analysis

Project	Advantages	Disadvantages
1. Upgrade of Magadi Road to Four Lanes with a Green Belt Buffer and Implementation of a Bus Rapid Transit (BRT) System	Reduced Traffic Congestion. Improved Safety. The green belt will improve air quality by absorbing pollutants, providing oxygen, and creating a pleasant, green environment. Biodiversity Conservation. Aesthetic Improvement. Dedicated lanes for BRT buses will ensure fast, reliable public transport, reducing delays and travel times.	Construction Disruption: The road expansion project might cause temporary inconvenience, including traffic diversions and noise. The project might be expensive, requiring significant financial investment for construction and maintenance. Potential Displacement.
2. Community-Led Waste Management for Ongata Rongai Town	Increased Community Engagement and Ownership Cost-effective and Sustainable. Enhanced Public Awareness and Education.	Potential for Inconsistent Implementation Limited Resources and Capacity Coordination and Management Challenges
3. Development of Well-Connected Corridors in Ongata Rongai Town	Improved Mobility: Better connectivity within the town will make it easier for residents to move around, accessing	Land Use Changes: Development might require changes in land use, potentially displacing current land uses or residents.

	various parts of the town efficiently. Economic Growth: Improved access will stimulate local businesses by attracting more customers and facilitating supply chains. Sustainable Urban Mobility: Integration of different modes of transport, including walking and cycling, will promote sustainable and healthy transportation options.	Initial Disruption: Construction activities might temporarily disrupt local activities and businesses. Cost: Significant investment is required for planning, construction, and maintaining the corridors.
4. Establishment of Formal Markets in Quarry Open Market	Organized Commerce: Formal markets will provide organized, designated spaces for vendors, reducing the chaos of informal markets. Improved Facilities: The provision of essential infrastructure such as sanitation, parking space water, and waste management will enhance the shopping experience and hygiene.	Displacement of Informal Vendors: Informal vendors might be displaced, leading to potential loss of livelihood if they cannot secure space in the formal markets. Cost: Developing and maintaining formal markets requires significant financial investment.

CHAPTER FOUR:

4.0 PROJECT PLANNING AND DESIGN

4.1 Specification for planning and designs

Planning design specifications are a set of detailed instructions and requirements that outline the parameters and goals for a particular project. These specifications are essential to ensure that the project is completed successfully, within budget and meets the desired outcomes

4.1.1 Guiding planning and design principles

1. a) Upgrade of Magadi Road to Four Lanes with a Green Belt Buffer

Planning Principles:

Traffic Efficiency and Safety: Expanding Magadi Road to four lanes addresses the principle of enhancing mobility by increasing road capacity, which reduces traffic congestion and travel time. Incorporating safety features such as proper signage, pedestrian crossings, and barriers ensures a safer environment for both motorists and pedestrians.

Environmental Sustainability: The green belt buffer aligns with sustainable urban development principles by preserving natural habitats, enhancing biodiversity, and mitigating the urban heat island effect. It also contributes to air quality improvement and stormwater management.

Design Considerations:

Road Expansion: The four-lane design includes two lanes in each direction, with additional safety measures like designated pedestrian crossings, bicycle lanes, and barriers.

Green Belt: A landscaped green belt with native vegetation acts as a buffer zone. This belt is designed to be visually appealing, with a mix of trees, shrubs, and grass to enhance biodiversity and air quality.

Aesthetics and Functionality: The green belt also serves an aesthetic purpose, creating a visually pleasing environment and offering recreational spaces for residents.

1 b) Implementation of a Bus Rapid Transit (BRT) System

Planning Principles:

Public Transportation Efficiency: BRT systems focus on providing a high-capacity, reliable public transport option, which reduces the reliance on private cars and decreases traffic congestion.

Accessibility and Inclusivity: Stations are designed to be accessible to all users, including those with disabilities, promoting inclusivity and equal access to public transportation.

Design Considerations:

Dedicated Lanes: BRT buses operate in dedicated lanes to avoid delays caused by mixed traffic, ensuring a fast and reliable service.

Stations and Stops: BRT stations are strategically located and designed with user accessibility in mind, featuring ramps, clear signage, and seating areas.

Environmental Impact: By reducing the number of private vehicles on the road, the BRT system helps lower vehicle emissions, contributing to better air quality and environmental sustainability.

2. Development of Well-Connected Corridors in Ongata Rongai Town

Planning Principles:

Connectivity and Integration: The principle of creating well-connected corridors emphasizes the integration of various transport modes, including walking, cycling, and public transit, to promote seamless urban mobility.

Economic Stimulus: Improved connectivity supports local businesses by facilitating easier access for customers and suppliers, thus fostering economic growth.

Design Considerations:

Urban Planning Alignment: The corridors align with comprehensive urban planning principles, ensuring orderly development and efficient land use.

3

4. Establishment of Formal Markets in Quarry Open Market and Other Locations

Planning Principles:

Economic Opportunity: Developing formal markets provides organized spaces for vendors, supporting small businesses and boosting the local economy.

Community Cohesion: Markets serve as community hubs, fostering social interaction and community cohesion.

Design Considerations:

Infrastructure: Markets are equipped with essential infrastructure such as sanitation facilities, water supply, electricity, and waste management systems to ensure a clean and functional environment.

Organization: The design includes clearly defined stalls and pathways to reduce chaos and improve the shopping experience.

4.1.2 Planning and Development Regulation

Various regulations and standards in Kenya have informed the planning development regulations below. These standards and guidelines outline the ideal zoning laws and land uses that will result in implementing the Magadi road corridor development plan.

Table 3: Planning Development Regulation

AREA OF CONCERN	REGULATION
Roads	For the proposed upgrade of Magadi Road in Ongata Rongai: Standard Vehicle Lanes: Each lane should be between 3.25 to 3.5 meters wide to accommodate general traffic. BRT Lanes: Dedicated BRT lanes should be between 3.5 to 3.75 meters wide to ensure efficient bus transit. Bicycle Lanes: If included, should be 1.5 to 2.0 meters wide for safe cycling.

	Pedestrian Paths: Sidewalks should be 1.5 to 2.5 meters wide to ensure pedestrian safety and comfort. Setback from road to building is about 6 meters These are per the Kenya Roads Act (2007): Which governs the development and maintenance of roads in Kenya, ensuring that roads meet specific standards for safety and capacity. Kenya National Highways Authority (KeNHA) and Kenya Urban Roads Authority (KURA) Guidelines: Provide specific standards for the design and construction of roads, including lane widths, road markings, and signage. National Transport and Safety Authority (NTSA) Regulations: Include safety standards for road design to minimize accidents and improve traffic flow
Green belt	Physical and Land Use Planning Act, 2019: Promotes the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Urban Planning Guidelines: Provides guidelines for the allocation of land for green spaces in urban development plans. The belt size will depend on the available land
Urban corridors and buildings	Physical and Land Use Planning Act, 2019: This act provides guidelines for urban development, including the creation of well-connected transport and pedestrian corridors. Urban Areas and Cities Act, 2011: Emphasizes the importance of integrated urban planning and development, ensuring efficient land use and connectivity. Corridor Design Considerations: Size Between Plots in Urban Planning Plot Size Regulations: Planning Standards: The size of plots and the spacing between them in urban areas like Ongata Rongai are guided by local zoning and planning regulations set by the county government and urban planning authorities. Minimum Plot Sizes: Urban residential areas typically have minimum plot size requirements that can range from 200 square meters to 500 square meters or more, depending on the zoning category (e.g., high-density, medium-density, or low-density residential areas). Building Setbacks and Spacing: Front Setback: The minimum distance between the plot boundary and the building facade, usually ranging from 3 to 6 meters, depending on local regulations. Side Setback: The minimum distance between the side plot boundary and the building can be around 1.5 to 3 meters. This spacing helps ensure privacy, ventilation, and light access. Rear Setback: The distance between the rear plot boundary and the building, typically around 3 to 6 meters. Plot Coverage and Floor Area Ratio (FAR): Plot Coverage: Regulations often specify the maximum percentage of a plot that can be covered by buildings, typically ranging from 50% to 75% for residential plots.

	Floor Area Ratio (FAR): Limits the total floor area of buildings relative to the plot size, promoting balanced development. For example, an FAR of 1.5 means the total floor area of all buildings on the plot cannot exceed 1.5 times the plot size.
Markets	Public Health Act, 2012: Sanitation: Requires markets to have adequate sanitation facilities, including toilets, handwashing stations, and waste disposal systems to maintain hygiene and public health. Cleanliness: Mandates regular cleaning and maintenance of market areas to prevent the spread of diseases. Layout and Organization: Stalls and Booths: Designated stalls and booths should be organized systematically to facilitate easy navigation for shoppers and efficient use of space by vendors. Zoning within the Market: Different zones can be designated for different types of goods (e.g., fresh produce, dry goods, clothing) to enhance organization and ease of shopping. Essential Infrastructure: Sanitation Facilities: Provide adequate and accessible toilets, handwashing stations, and waste management systems to ensure hygiene. Water Supply: Ensure a reliable water supply for cleaning, drinking, and other needs. Electricity: Provide sufficient lighting and electrical outlets to support vendors' needs and enhance safety during evening hours. Accessibility and Safety: Paths and Walkways: Design wide, well-marked paths and walkways to ensure smooth movement of shoppers and vendors, including provisions for people with disabilities. Fire Safety: Install fire extinguishers, emergency exits, and conduct regular fire safety drills to prepare for emergencies. Security: Implement measures such as security personnel, CCTV cameras, and proper lighting to ensure the safety of vendors and shoppers.

4.1.3 Site Accommodation Schedule

A schedule of accommodations is a detailed list of the accommodations and provisions required by the final user of a construction project. A schedule of accommodations is created to help determine the minimum space requirements for the building and, consequently, the site space requirements to achieve a specific design as suggested by the project brief. It assists with cost projections as well. The project site for the preferred plan is approximately 134 Ha. There are many different land uses on the property. Each of these uses aims to advance the creation of a sustainable linear development of the Magadi corridor.

Table 4: Site Accommodation Schedule

Land Use	Typology	Facilities Provisions	Area (Ha)
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Residential	High-density housing, low-density housing	Apartments, single-family homes, duplexes, community centers, playgrounds, parking areas	53.6
Commercial	Retail, office spaces	Shopping centers, office buildings, restaurants, banks, parking areas, public restrooms	26.8
Public Purpose	Government buildings, community centers	Municipal offices, community halls, public libraries, cultural centers, emergency services	8.0
Industrial	Light industrial, warehouses	Factories, distribution centers, warehouses, service facilities, loading docks	4.0
Education	Schools, colleges	Primary and secondary schools, vocational training centers, college campuses, libraries, sports facilities	8.0
Recreational/Open Space/Conservation Areas	Parks, nature reserves, sports complexes	Public parks, nature trails, sports fields, playgrounds, picnic areas, conservation zones	9.4
Transportation Infrastructure	Roads, public transit hubs	Roadways, bus stops, BRT lanes, parking facilities, pedestrian pathways, bike lanes	10.7
Public Utilities/Facilities	Utility services, waste management	Water treatment plants, electricity substations, waste collection and recycling centers	6.7
Agriculture	Urban farming, community gardens	Farmlands, greenhouses, community gardens, farmers markets	6.7

4.1.4 Land Use Budget

The future land requirements of a given population and their activities are calculated using a land use budget, a model for optimizing land use. It entails estimating how much space will be needed to adapt to anticipated future change and allocating land to specific uses. The budget for land use for this project lists the amount of land needed for each type of land use to support a sustainable Magadi road corridor development. Given that the total land is 134 ha.

Table 5: Land Use Budget

Land Use Type	Area in Ha	Percentage (%)
Residential	53.6	40
Commercial	26.8	20
Public Purpose	8.0	6
Industrial	4.0	3
Education	8.0	6
Recreational/Open Space/Conservation Areas	9.4	7
Transportation Infrastructure	10.7	8
Public Utilities/Facilities	6.7	5
Agriculture	6.7	5
Total	134	100

4.1.5 Land Use Structuring Elements

The land use structuring elements for the proposed development projects in Ongata Rongai involve strategic zoning and planning to ensure a balanced and sustainable urban environment. Transportation infrastructure, including the upgraded Magadi Road and a dedicated BRT system, will ensure efficient mobility, while green belts and open spaces will improve environmental quality and provide recreational areas. Sustainability measures such as resource management and climate resilience will be integral to all developments, ensuring long-term viability and enhanced quality of life for residents.

4.1.6 Functional Organization of Site Land Use

The proposed development projects in Ongata Rongai, including the upgrade of Magadi Road, creation of a green belt, implementation of a Bus Rapid Transit (BRT) system, development of well-connected corridors, and establishment of formal markets, will use a strategic approach to the functional organization of site land use. This approach ensures that residential, commercial, industrial, and public areas are appropriately zoned and interconnected. Residential areas should be organized based on density, ensuring accessibility to essential amenities like schools and healthcare facilities. Commercial zones should be concentrated in central business districts and distributed neighborhood shops to facilitate easy access. Industrial areas need to be located on the town's periphery, with buffer zones to minimize pollution and noise impacts on residential and commercial zones. Public and semi-public areas, including parks, recreational spaces, and public service buildings, should be integrated throughout the town to enhance livability and community cohesion.

Transportation and mobility infrastructure is critical, with the upgraded Magadi Road and BRT system ensuring efficient and safe movement. Well-connected corridors that integrate walking, cycling, and public transit will reduce reliance on private vehicles and improve overall mobility. Green belts and open spaces along major roads and within the town will enhance environmental quality, provide recreational opportunities, and act as buffers to prevent urban sprawl. Sustainability considerations, such as efficient resource management and climate resilience, will be incorporated into all aspects of development to ensure long-term viability and quality-of-life improvements for Ongata Rongai residents. This comprehensive, phased, and stakeholder-inclusive approach will create a balanced, vibrant, and sustainable urban environment.

4.2 Conceptual Spatial Plans

Conceptual spatial plans offer different approaches to organizing land use and infrastructure in urban development.

Approaches

A conceptual-oriented approach

A conceptual-oriented approach for sustainable linear development in Magadi road corridor from Magadi – Gataka junction to Quarry Market, Ongata Rongai focuses on utilizing major transportation corridors, particularly Magadi Road, as the backbone for urban growth. This approach aims to develop high-density residential, commercial, and industrial zones along the road and the Bus Rapid Transit (BRT) system, creating a continuous and efficient urban spine. Key features include integrating mixed-use developments that combine residential, commercial, and recreational facilities to reduce travel distances and promote walkability. Green belts and buffer zones will be strategically placed along the corridors to prevent urban sprawl, enhance biodiversity, and provide aesthetic and environmental benefits. These green spaces will also help mitigate the urban heat island effect, manage stormwater, and improve air quality, contributing to a healthier living environment.

The linear development will be supported by a well-planned transportation network that includes pedestrian pathways, cycling lanes, and feeder roads connecting the main corridor to surrounding neighborhoods. This network will ensure seamless mobility and accessibility, encouraging public transport use and reducing reliance on private vehicles. Formal markets, public amenities, and essential services will be distributed along the corridor, providing convenient access to residents and stimulating economic activities. By concentrating development along the main transport routes, the linear development approach minimizes land consumption, preserves natural landscapes, and promotes sustainable urban growth. This method aligns with principles of smart growth, emphasizing efficient infrastructure use, environmental conservation, and the creation of vibrant, connected communities.

4.2.1 Alternatives

Map 10: Problems to be solved by the alternatives



Alternative 1

a) Upgrade of Magadi Road to Four Lanes with a Green Belt Buffer

The first alternative for the project in Ongata Rongai involves upgrading Magadi Road to four lanes with the addition of a green belt buffer. This development will significantly enhance the road's capacity, reducing traffic congestion and improving the overall flow of vehicles, which will make commutes faster and more efficient. The widened road will include modern safety features such as proper signage, pedestrian crossings, and barriers, ensuring safety for both motorists and pedestrians. The green belt will serve as a critical buffer zone, preventing urban sprawl and encroachment by informal businesses, thereby preserving natural habitats and enhancing biodiversity. This green belt will not only beautify the area with lush greenery and well-maintained landscapes but also help

improve air quality by absorbing pollutants and providing oxygen, contributing to a healthier environment for residents.

In addition to environmental benefits, the green belt will enhance the town's resilience to climate change impacts by mitigating the urban heat island effect and improving stormwater management. It will act as a recreational space for residents, offering areas for walking, cycling, and other outdoor activities, thus promoting a healthier lifestyle. The buffer will also help in sound dampening, reducing noise pollution from the busy road. Overall, this alternative aligns with sustainable urban development principles by balancing infrastructural improvements with environmental conservation and community well-being, ensuring that Ongata Rongai can grow efficiently without compromising its natural resources and quality of life. The land for the upgrade is already available, there will be only strict regulations to evacuate those settlements on the road reserve.

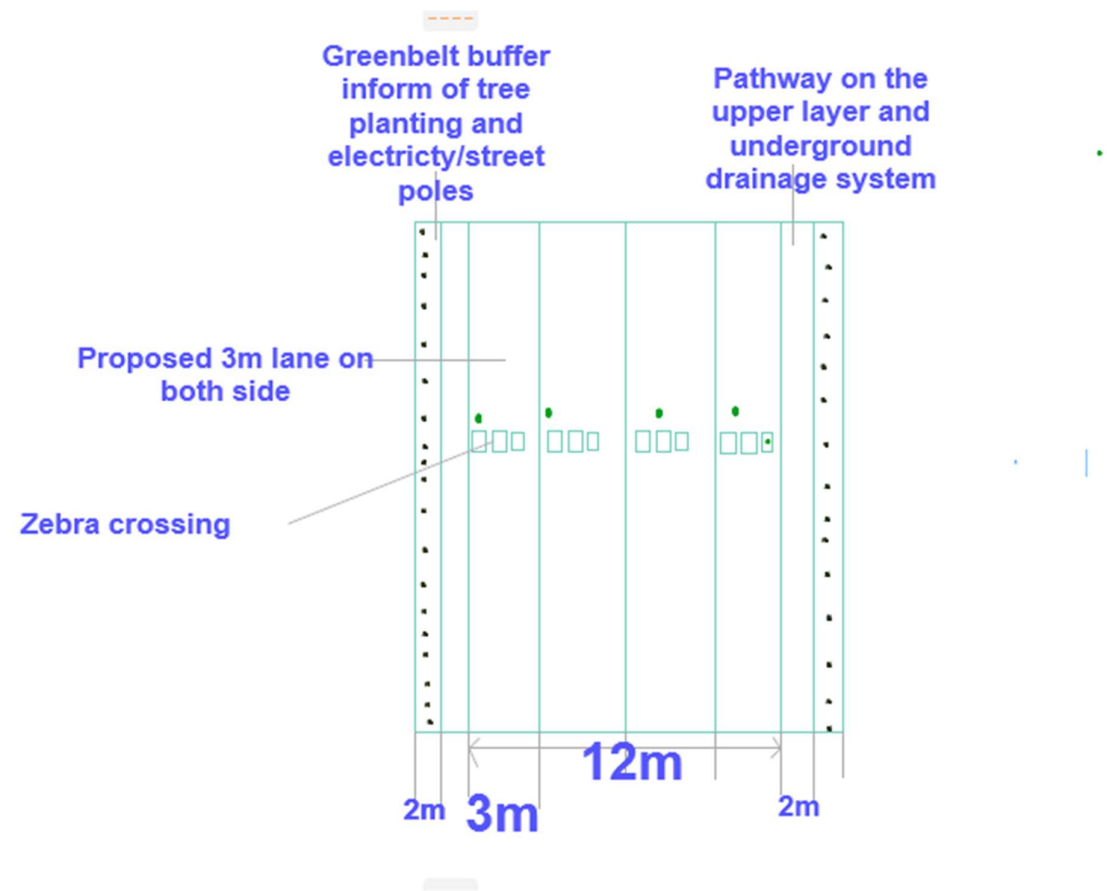


Figure 11: Proposed Magadi Road Upgrade drawing

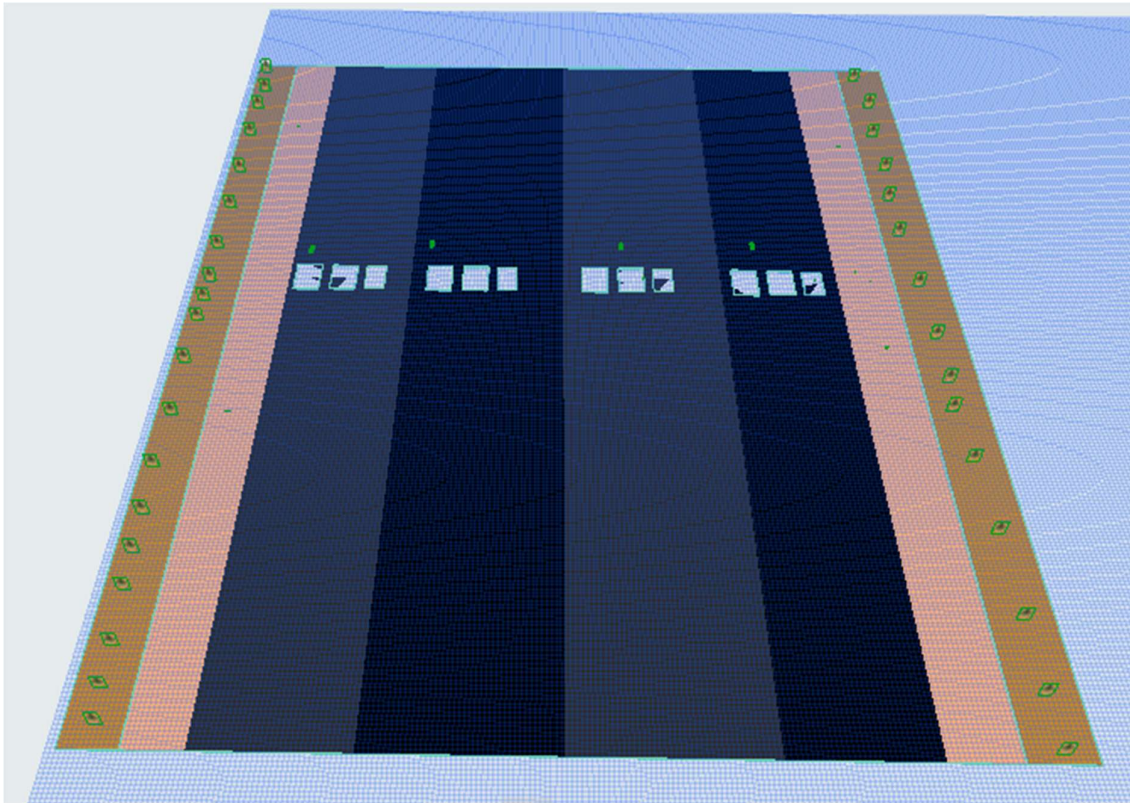


Figure 12: 3D drawing

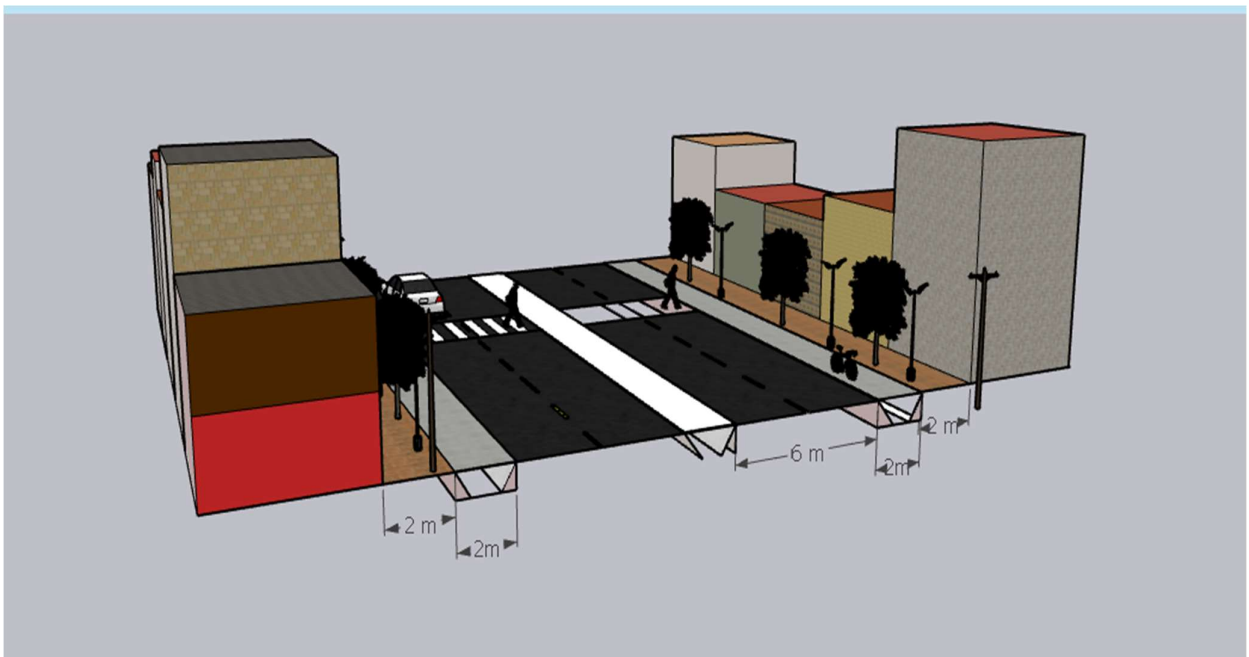
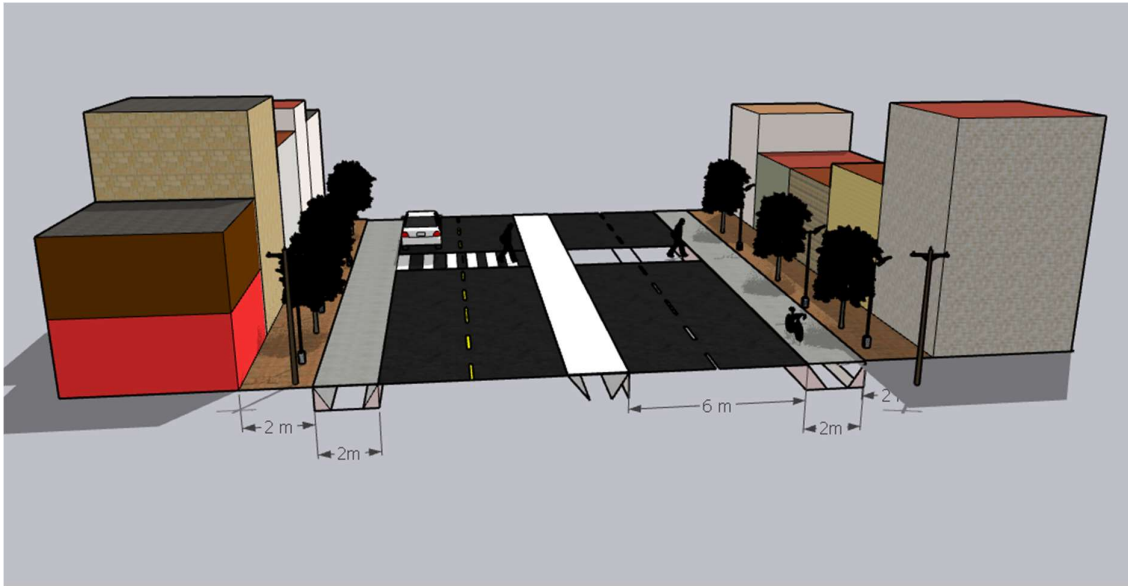


Figure 13:3D Model



b) Implementation of a Bus Rapid Transit (BRT) System with a dedicated lane

The alternative for the Ongata Rongai development project focuses on implementing a Bus Rapid Transit (BRT) system with dedicated lanes. This BRT system will provide a high-capacity, efficient public transportation option, significantly reducing reliance on private cars. By operating in dedicated lanes, BRT buses will offer fast and reliable service with minimal delays, ensuring consistent travel times for commuters. Stations will be designed to be accessible to all users, including those with disabilities, thus promoting inclusivity and ease of use. The BRT system will also feature modern amenities such as real-time information displays, comfortable waiting areas, and secure ticketing systems to enhance the overall user experience.

Beyond improving transit efficiency, the BRT system will have substantial environmental and social benefits. By encouraging the use of public transportation, it will help decrease vehicle emissions, thereby improving air quality and contributing to environmental sustainability. The reduction in traffic congestion will lead to lower noise pollution and improved urban living conditions. Economically, the BRT system will stimulate local businesses by increasing foot traffic and making it easier for customers and employees to access commercial areas. Socially, it will provide affordable and reliable transportation, enhancing mobility for all residents, particularly those who do not own private vehicles. This alternative aligns with sustainable urban development principles, fostering a more connected, accessible, and environmentally friendly city.

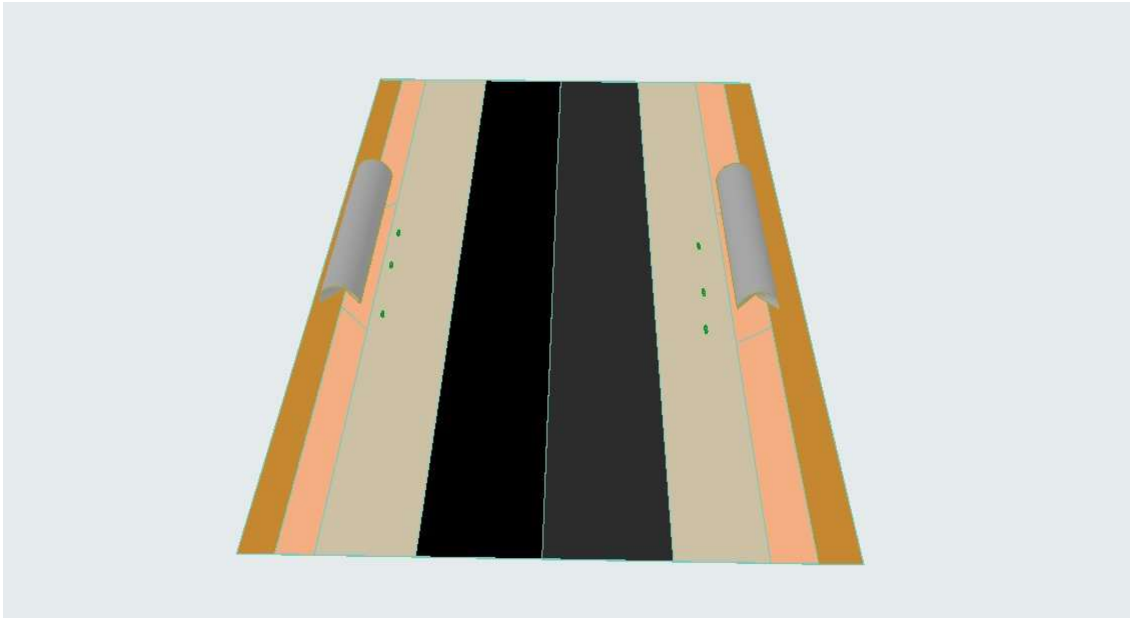
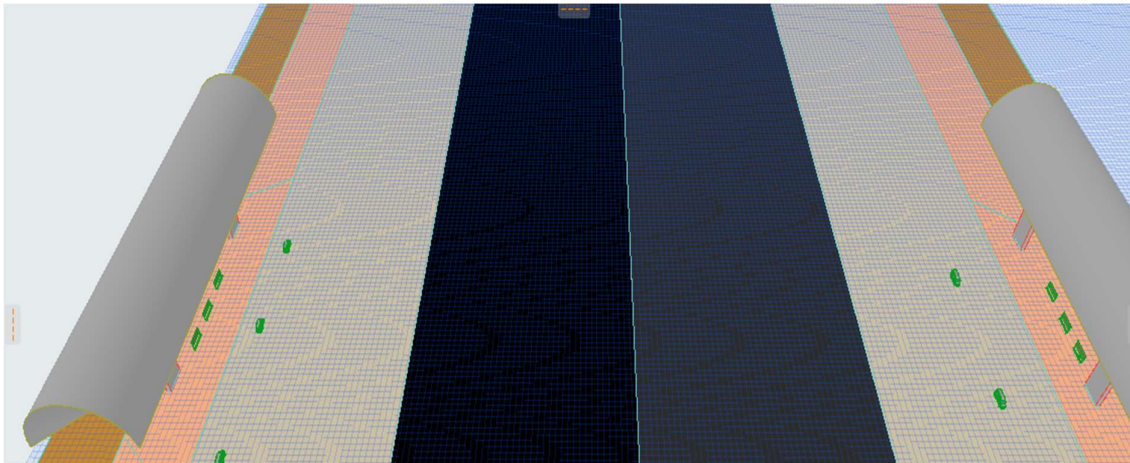


Figure 14: 3D BRT stage



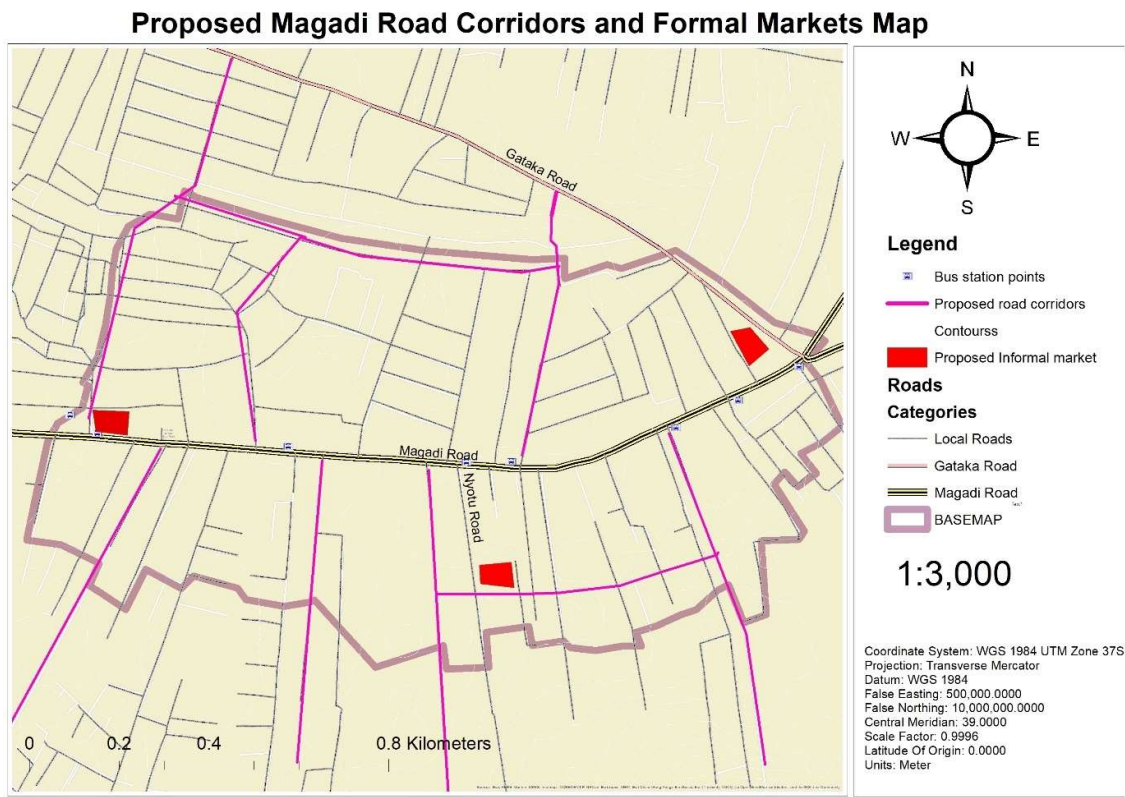
Alternative 2

The second alternative for the Ongata Rongai development project focuses on creating well-connected corridors throughout the town, complemented by the establishment of formal markets. This approach aims to facilitate efficient movement and accessibility by developing a network of interconnected roads and pathways that seamlessly link residential, commercial, industrial, and public spaces. These corridors will integrate various modes of transport, including walking, cycling, and public transit, to promote sustainable urban mobility. By improving connectivity, this plan will reduce travel times, ease traffic congestion, and provide residents with convenient access to essential services and amenities. Additionally, the corridors will be designed with safety features such as pedestrian crossings, bike lanes, and proper lighting to ensure safe and comfortable travel for all users.

Land acquisition for these corridors and markets will involve a transparent process to ensure fairness and minimal disruption. The government will conduct thorough land assessments and engage with landowners to negotiate fair compensation based on market value, considering factors such as the land's current use and potential future value. Compensation may include monetary payments,

relocation assistance, or offering alternative land parcels. Public consultations and community involvement will be prioritized to address concerns and ensure the affected parties are adequately informed and supported throughout the process.

The inclusion of formal markets along these well-connected corridors will further enhance the town's functionality and livability. These markets will provide organized, accessible spaces for vendors, reducing the chaos of informal markets and improving the overall shopping experience. Equipped with essential infrastructure such as proper sanitation facilities, water supply, electricity, and waste management systems, the markets will support small businesses and local entrepreneurs, fostering economic growth and community development. By serving as community hubs, these markets will also promote social interaction and cohesion among residents. This integrated approach ensures that the development not only improves physical infrastructure but also strengthens the economic and social fabric of Ongata Rongai, contributing to a more vibrant, sustainable, and resilient urban environment.



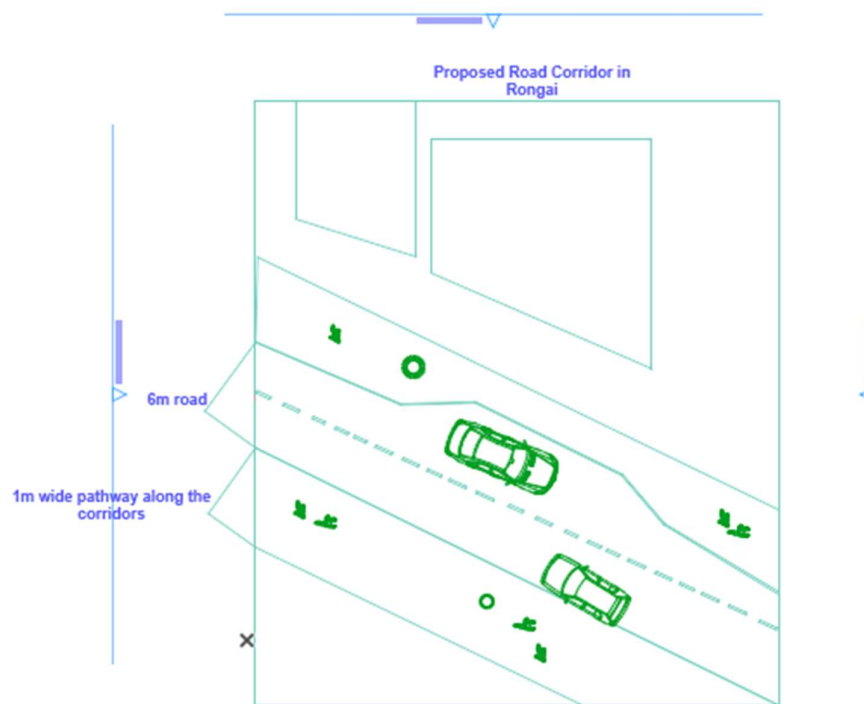


Figure 15:3D drawing of corridor road

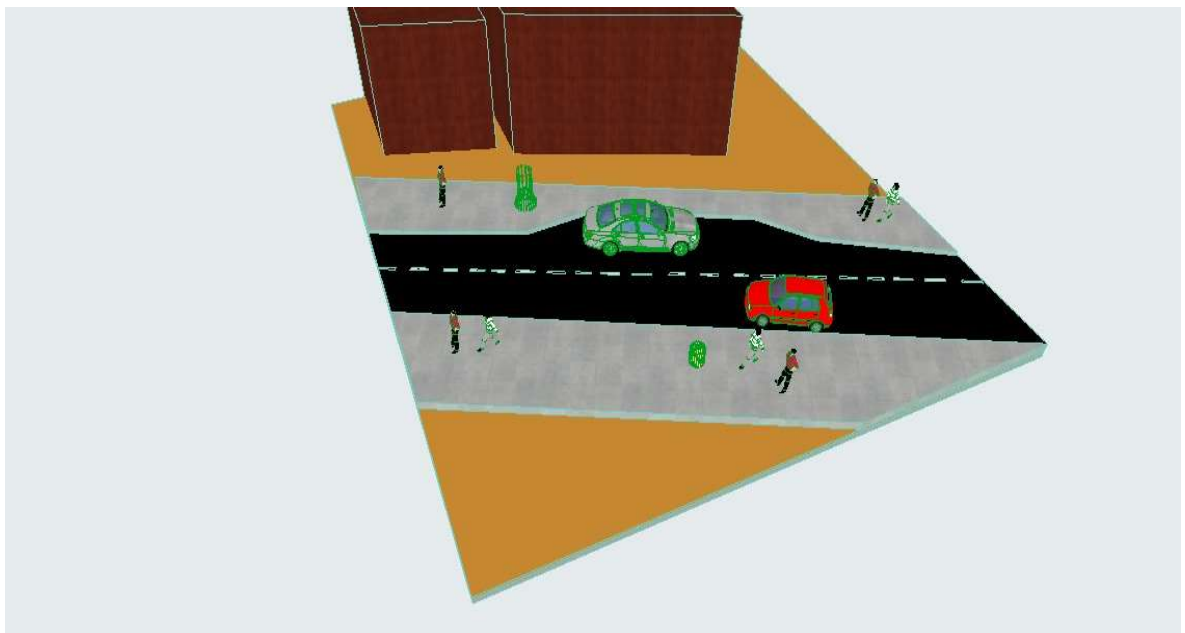


Figure 16: 3D Model of road corridor

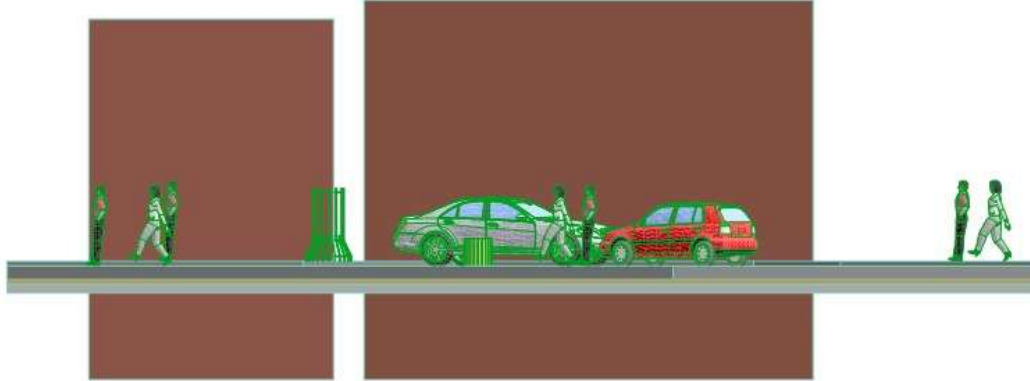


Figure 17: 3D Elevation of the road corridor

Alternative 3: Convectional and Community-Led Waste Management for Ongata Rongai Town

The third alternative for waste management in Ongata Rongai Town involves a decentralized, community-led approach. This strategy emphasizes the active involvement of local communities to manage waste at the neighborhood level, focusing on waste reduction, recycling, and composting initiatives. By empowering communities, this approach aims to foster a sense of ownership, increase participation, and improve overall waste management effectiveness.

Key Components of the Community-Led Waste Management System:

Neighborhood Collection Points:

Local Collection Centers: Create accessible waste collection points in each neighborhood with separate bins for organic waste, recyclables, and non-recyclables. **Community Volunteers:** Form local volunteer groups to manage these collection points, ensuring proper sorting and cleanliness.

Local Recycling Programs:

Community Recycling Hubs: Establish small recycling hubs where recyclables can be sorted, cleaned, and prepared for sale to recycling companies, managed by local cooperatives or small enterprises. This will be done in all small urban centers around the town. **Upcycling Projects:** Encourage community-led upcycling projects to create useful items from waste materials, generating income and reducing waste.

Neighborhood Composting:

Community Composting Sites: Develop communal composting sites for processing household organic waste into compost, maintained by trained community members. **Household Composting Kits:** Provide composting kits and training to households, promoting home composting of organic waste.

Education and Awareness:

Workshops and Training: Conduct workshops and training sessions on waste reduction, sorting, recycling, and composting, targeting residents of all ages. **School Programs:** Integrate waste management education into school curricula, encouraging student participation and awareness.

Participation Incentives:

Reward Programs: Implement incentives for households and neighborhoods that actively participate in waste reduction and recycling, such as utility bill discounts or local service vouchers.

Awards: Establish annual awards for outstanding community groups and individuals in waste management, fostering competition and recognizing contributions. This can be done by the Area MCAs.

Land Acquisition and Compensation: For neighborhood collection centers and composting sites, land acquisition will involve identifying suitable public spaces or negotiating with private landowners for small parcels. Fair compensation will be provided through monetary payments or community benefits like improved infrastructure, with transparent processes and community consultations to ensure local support and minimize conflicts.



Figure 19: Neighbourhood dustbin



Figure 18: Separation Point

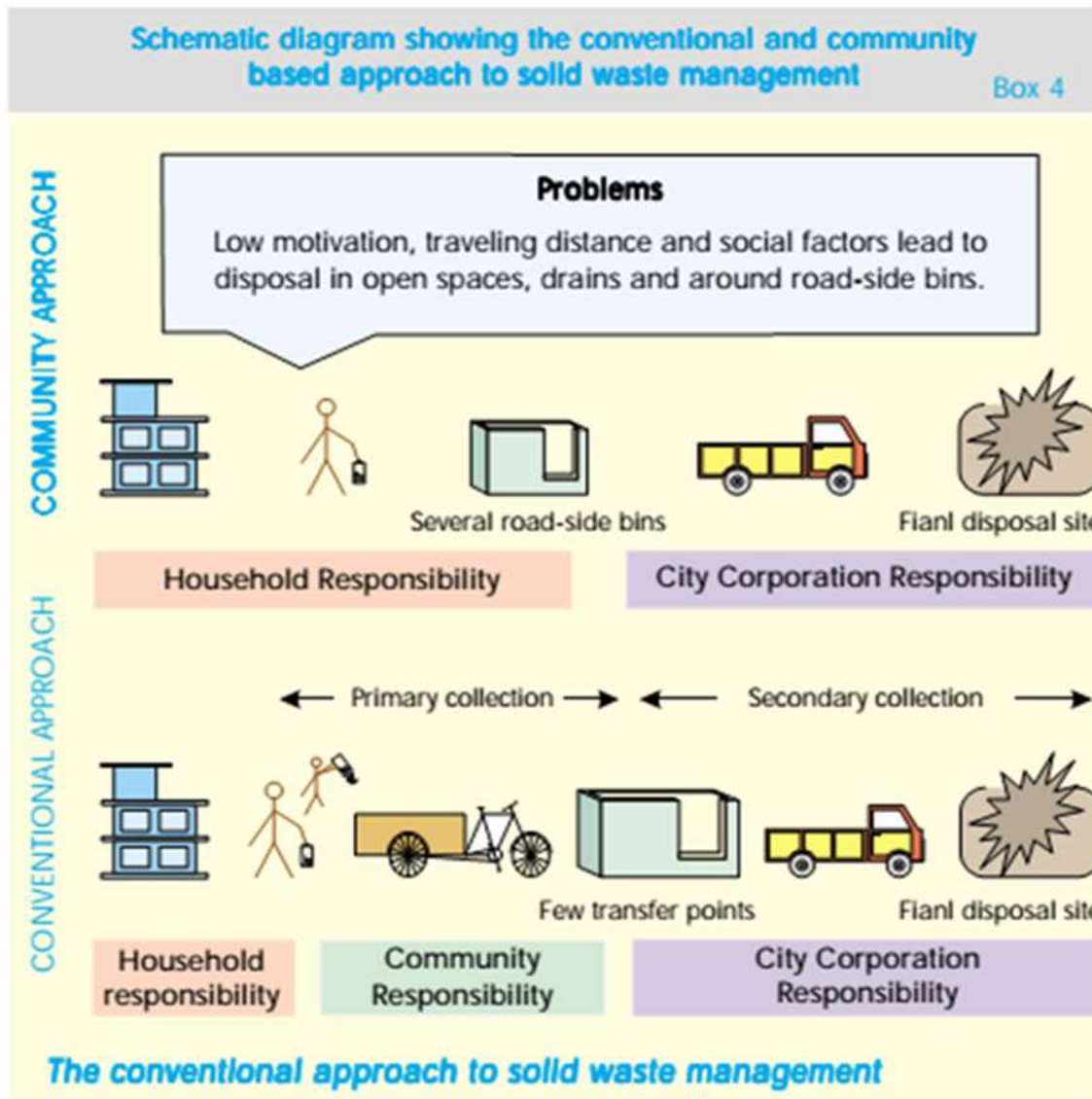


Figure 20: Waste collection drawing

Copy right Yusuf Rigasa Abdullahi environmentalist

Alternative 4

Integrated approach

The integrated approach for the Magadi road corridor, Ongata Rongai development project combines upgrading Magadi Road to four lanes with a green belt, implementing a Bus Rapid Transit (BRT) system with dedicated lanes, Convectional and Community-Led Waste Management, developing well-connected corridors, and establishing formal markets. This comprehensive strategy addresses multiple urban challenges by enhancing road capacity and safety, promoting efficient public transportation, and

reducing traffic congestion. The green belt will improve air quality and prevent urban sprawl, while the BRT system will ensure fast, reliable public transit. Well-connected corridors will support various transport modes, stimulating local businesses and facilitating economic growth. Formal markets along these corridors will provide organized vendor spaces and essential infrastructure, fostering small businesses and community cohesion. This holistic approach ensures sustainable, balanced, and resilient urban development, improving residents' quality of life and preserving the environment.

4.2.2 Evaluation of the Alternatives

Evaluating the four alternatives for the Magadi road corridor within Ongata Rongai development project involves assessing their effectiveness in addressing key urban challenges, their impact on residents' quality of life, economic benefits, environmental sustainability, and feasibility.

Table 6: Goal Achievement Matrix

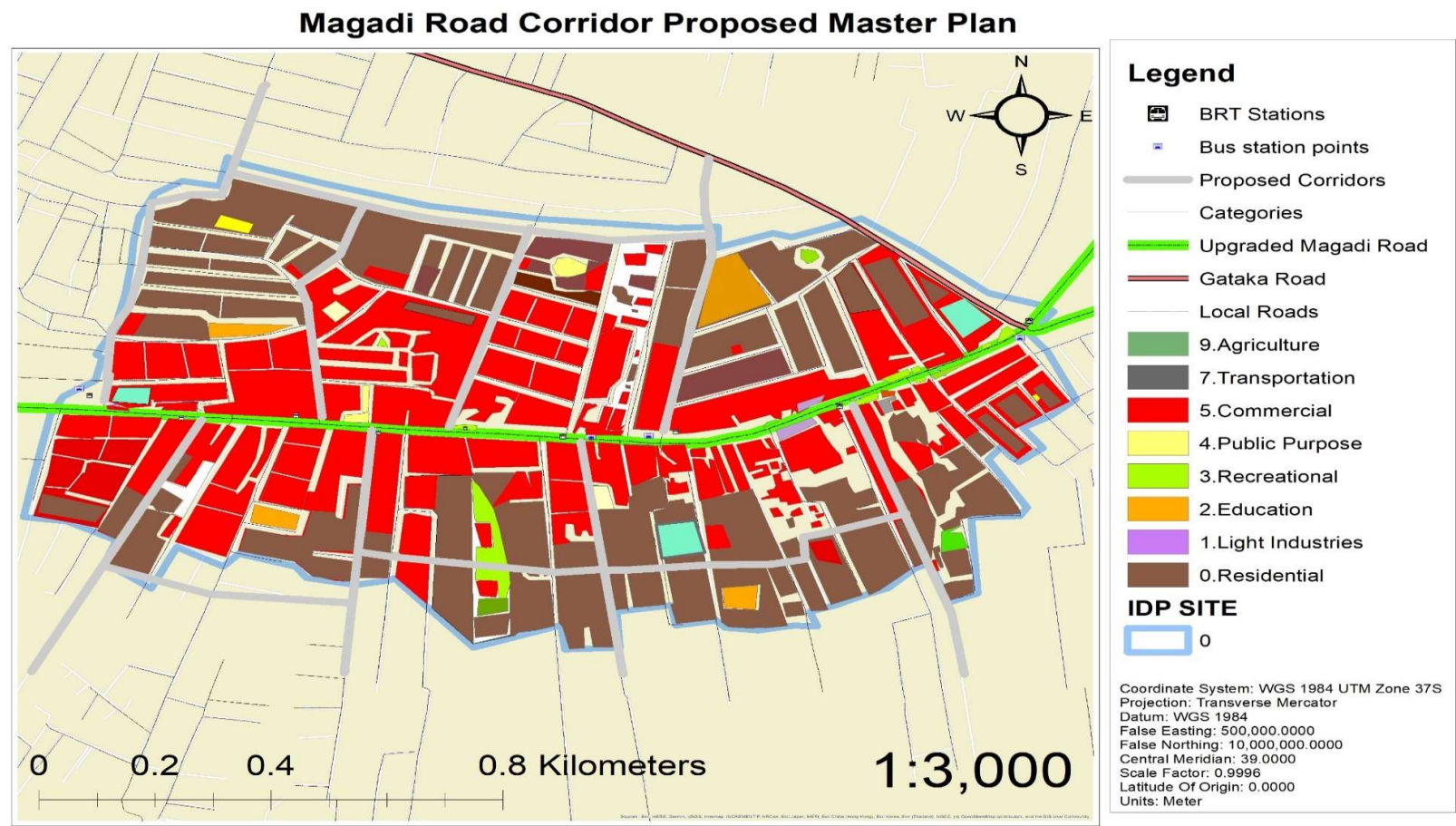
Objectives	Assigned Weight	Alternative 1	Sub-Total	Alternative 2	Sub-Total	Alternative 3	Sub-Total	Integrated Development	Sub-Total
Model 4									
1	0.25	2	0.5	2	0.5	3	0.5	3	0.75
2	0.25	2	0.5	3	0.75	3	0.75	3	0.75
3	0.25	1	0.25	2	0.5	3	0.75	3	0.75
4	0.25	1	0.25	3	0.75	3	0.75	3	0.75
TOTAL	1		1.5		2.5		2.75		3
RANK		FORTH		SECOND		THIRD			FIRST

(Likert scale: 1=low, 2= moderate, 3= Good)

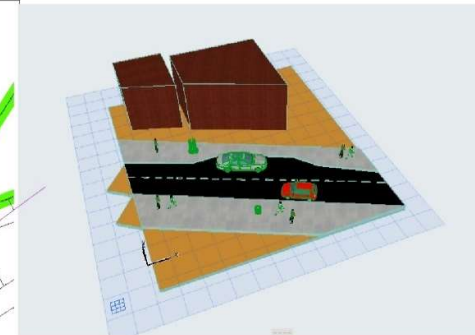
4.3 Detailed Design Formulation

4.3.1 Preferred Site Plan

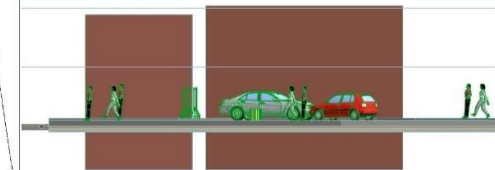
Map 11: Mater plan



Magadi Road Corridor Proposed Master Plan

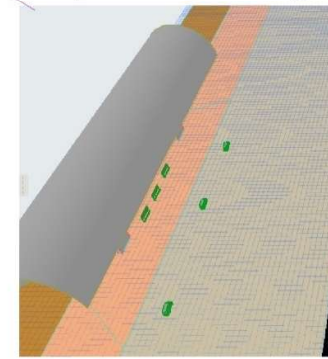
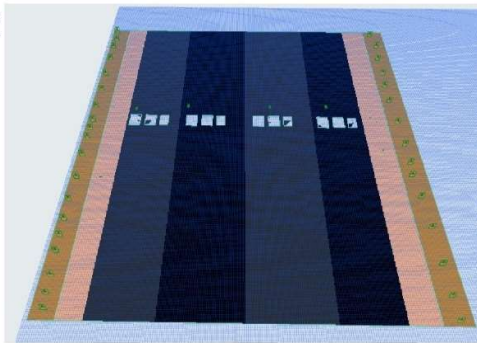
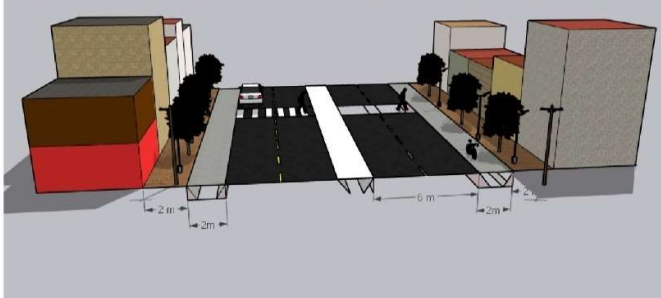


Proposed Corridor road



Proposed BRT Stage

Proposed Magadi road upgrade and green belt

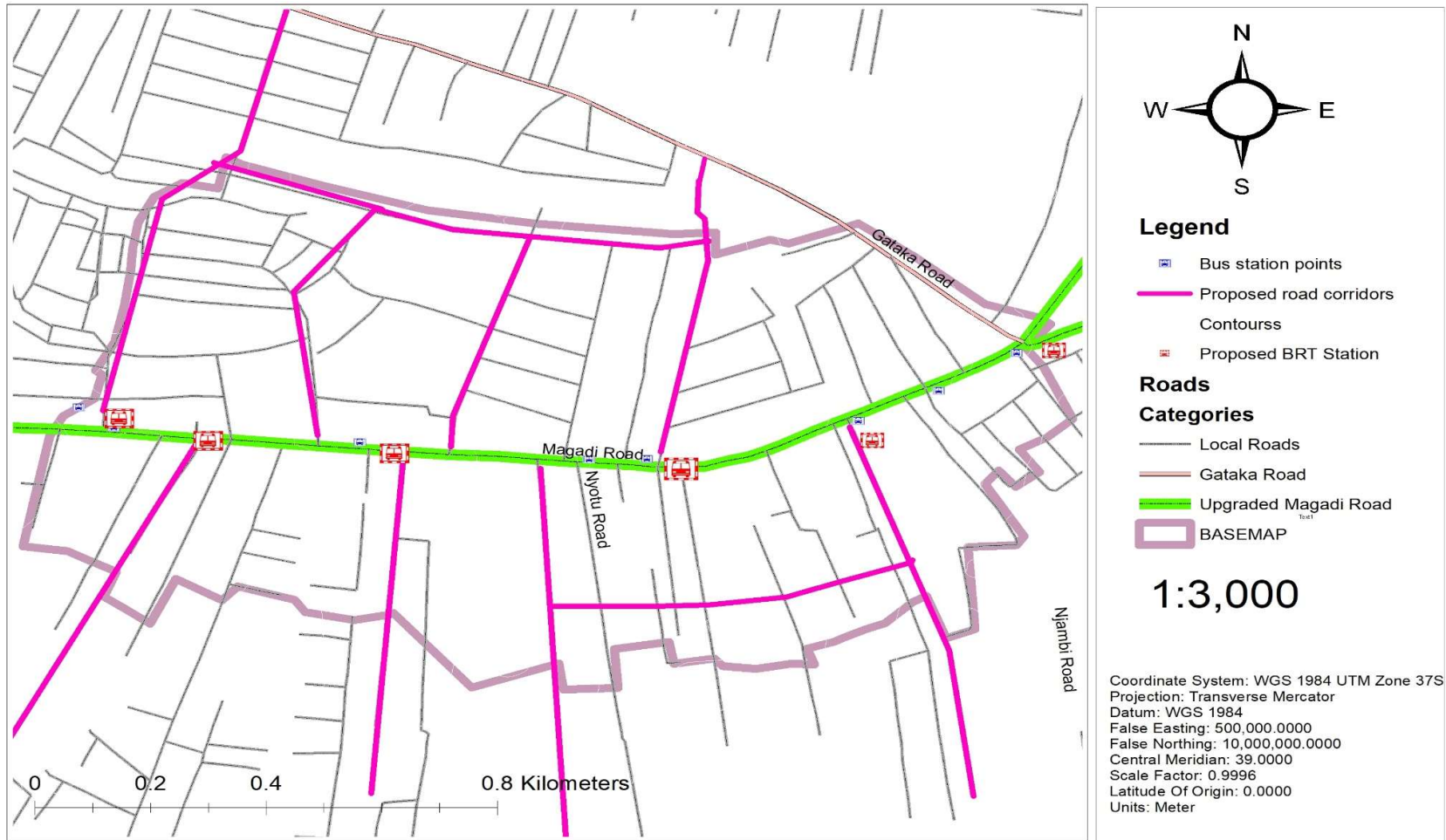


ACTION AREA PLAN



4.3.2 Transport and Circulation Plan

Map 12: Transport and circulation Plan



4.3.3 Utilities and Services

The Magadi Road project's utilities and services provision plan focuses on delivering essential infrastructure to support the road's functionality and the surrounding communities. This includes the installation of water supply and drainage systems to ensure efficient water management and prevent flooding. The road will be equipped with street lighting to enhance safety and visibility during nighttime. Additionally, electricity lines will be laid out to support both the road infrastructure and the adjacent areas, promoting economic activities and improving living standards. Telecommunications infrastructure, including internet and mobile networks, will be expanded along the road to enhance connectivity for residents and businesses. These utilities will not only support the road itself but also contribute to the overall development and modernization of the region.

Moreover, the project will integrate public amenities such as bus stops and rest areas to facilitate public transport and provide convenience for travelers. Proper signage and road markings will be implemented to improve traffic management and safety. Waste management services will be set up, including strategically placed bins and regular collection schedules to maintain cleanliness. Emergency services, such as ambulance and fire stations, will be strategically located to ensure rapid response in case of accidents or emergencies. By incorporating these utilities and services, the Magadi Road project aims to create a well-equipped, safe, and sustainable transportation corridor that supports the needs of the community and stimulates economic growth.

4.3.4 Building Prototypes



Figure 21: Market building

4.3.5 Site Environmental Management Plan

Structured environmental management plan for the Magadi Road Corridor Development and Management Strategy Plan in Ongata Rongai Town:

Table 7: Environment Management Plan

Component	Objective	Actions	Responsible Entity	Indicators	Notes
Air Quality Management	Improve air quality	- Implement green belt buffer along road	Environmental Agency	Air quality levels, green cover increase	Regular monitoring required
		- Promote use of BRT system	Transport Authority	Public transit usage rates, vehicle emissions	Public awareness campaigns needed
Noise Control	Reduce noise pollution	- Install noise barriers near residential areas Use of Electric cars	Urban Planning Dept.	Noise level measurements	Focus on high-traffic areas
		- Enforce regulations on noise levels from construction sites	Environmental Agency	Compliance reports	Regular inspections
Water Management	Ensure sustainable water use	- Implement rainwater harvesting systems	Infrastructure Dept.	Number of systems installed, water savings	Community participation essential
		- Construct drainage systems to manage stormwater	Civil Engineering Dept.	Instances of flooding, water quality tests	Prioritize flood-prone areas
Soil Conservation	Prevent soil erosion and degradation	- Establish vegetation cover along corridors	Environmental Agency	Soil erosion rates, vegetation cover increase	Use native plant species
		- Implement soil conservation techniques in construction	Construction Companies	Compliance with soil management practices	Regular training for workers
Waste Management	Promote sustainable waste management practices	- Set up waste collection and recycling points	Municipal Authority	Waste collection efficiency, recycling rates	Regular public education

		- Enforce waste segregation at source	Environmental Agency	Compliance reports, reduced landfill waste	Regular monitoring and enforcement
Energy Efficiency	Enhance energy efficiency in urban development	- Encourage use of energy-efficient lighting and appliances	Urban Planning Dept.	Energy consumption rates, number of efficient installations	Incentives for adoption
		- Promote renewable energy sources	Energy Authority	Renewable energy usage, installations	Include in building codes
Green Spaces	Increase and maintain urban green spaces	- Develop parks and recreational areas	Parks and Recreation Dept.	Area of green spaces per capita, user satisfaction	Community involvement in maintenance
		- Protect existing natural habitats	Environmental Agency	Biodiversity indices, habitat preservation	Focus on critical areas
Sustainable Transport	Promote sustainable transport options	- Develop and maintain BRT system	Transport Authority	Public transit ridership, reduction in private vehicle use	Integrated with other transport modes
		- Create pedestrian and cycling paths	Urban Planning Dept.	Length of paths, usage rates	Safety features and accessibility focus

This environmental management plan outlines the key components, objectives, actions, responsible entities, timelines, indicators, and notes necessary to ensure the Magadi Road Corridor project is environmentally sustainable and addresses the various aspects of urban development in Ongata Rongai.

4.3.6 Development Control Guidelines

The table outlines the development control guidelines for the Magadi Road Corridor project in Ongata Rongai Town, aligned with Kenya's development control guidelines:

Table 8: Development Control

Component	Guideline	Details	Responsible Entity	Compliance Measure
Land Use	Zoning Regulations	Adhere to designated zoning areas (residential, commercial, industrial)	Municipal Planning Authority	Zoning maps and permits
	Mixed-Use Development	Encourage mixed-use development along the corridor	Urban Planning Dept.	Approved development plans
Building Standards	Building Heights	Maximum building heights as per zoning area	Building Control Authority	Height measurements and compliance checks
	Setback Requirements	Minimum setbacks from roads and property boundaries	Municipal Planning Authority	Site inspections and approval processes
	Plot Coverage	Maximum allowable plot coverage	Building Control Authority	Plot coverage ratio calculations
Infrastructure	Road Design Standards	Roads to comply with national design standards (width, materials)	Transport Authority	Road construction audits
	Utility Provision	Ensure access to water, electricity, and sanitation	Utility Companies	Utility connection approvals
Environmental Management	Green Space Requirements	Allocate a minimum percentage of area to green spaces	Environmental Agency	Green space plans and site inspections
	Waste Management Systems	Implement waste segregation and recycling systems	Municipal Authority	Waste management audits
	Stormwater Management	Construct proper drainage systems to manage runoff	Civil Engineering Dept.	Drainage system inspections
Public Amenities	Public Facilities	Include public facilities such as parks, schools, and health centers	Municipal Planning Authority	Facilities plans and community feedback

	Accessibility	Ensure accessibility for disabled individuals in all public buildings	Building Control Authority	Accessibility audits and certifications
Transportation	Public Transit Integration	Integrate BRT system and other public transport with road design	Transport Authority	Public transit usage data
	Non-Motorized Transport	Develop pedestrian and cycling paths along the corridor	Urban Planning Dept.	Path lengths and user surveys
Safety	Fire Safety	Implement fire safety standards in all buildings	Building Control Authority	Fire safety inspections and certifications
	Traffic Safety	Install proper signage, pedestrian crossings, and speed controls	Traffic Police Dept.	Traffic safety audits and incident reports
Economic Development	Support for Small Businesses	Designate areas for formal markets and support small businesses	Economic Development Dept.	Business licenses and market inspections
	Incentives for Green Businesses	Provide incentives for businesses adopting sustainable practices	Environmental Agency	Number of green business certifications
Community Involvement	Public Participation	Engage community in planning and decision-making processes	Municipal Planning Authority	Public consultation records and feedback
	Community Monitoring Groups	Establish and support local monitoring groups	Environmental NGOs	Group activity reports and outcomes

4.3.7 Impacts of Preferred Alternative

Table: Impact of Preferred Alternative

Table 9: Impact of preferred plan

Impacts of preferred alternative	Benefits
Environmental Impacts	The green belt along Magadi Road improves air quality, absorbs pollutants, and enhances biodiversity. - Reduction in greenhouse gas emissions and mitigation of urban heat island effect through the BRT system. - Green spaces and corridors contribute to carbon sequestration, reducing overall carbon footprint.
Social impacts	BRT system provides affordable, accessible, and reliable transportation for residents.

	• Well-connected corridors improve mobility and access to essential services. • Formal markets create organized spaces for vendors, fostering social interaction and community cohesion. • Improved public health and hygiene standards through the provision of essential amenities in markets.
Economic Impacts	• Enhanced connectivity and transportation infrastructure attract businesses and stimulate local commerce. • Formal markets support small businesses and entrepreneurs, creating economic opportunities and jobs. • Increased foot traffic and accessibility benefit existing businesses and encourage new enterprises. Reduced travel costs and time increase productivity and economic efficiency.
Urban development impacts	• Upgraded road and BRT systems alleviate traffic congestion, improving town navigability. • Well-connected corridors promote smart land use and integrate various urban functions seamlessly. • Green belt prevents urban sprawl, ensuring orderly and sustainable development. • Holistic planning leads to a more organized, efficient, and livable urban environment.
Overall impacts	• Balanced and sustainable development trajectory for Ongata Rongai. • Maximizes benefits across environmental, social, economic, and urban development sectors. • Enhances community resilience and vibrancy while preserving natural resources and improving quality of life. • Promotes long-term sustainability and ensures the well-being of residents and the environment.

CHAPTER FIVE:

5.0 IMPLEMENTATION MONITORING AND EVALUATION

5.1 Project Log-frame / Strategy Matrix

Table: Project Strategy Matrix

Table 10: Project Strategy Matrix

Goals	Activities	Outcomes	Actors	Time
1. Inventory of Existing Site Development and Conditions	- Conduct site surveys and assessments	- Identification of areas for improvement	County Government(Planner)	6 months
	- Analyze current traffic patterns and road conditions including land uses along the major route	- Assessment of existing road capacity and land uses	Engineering Consultants Physical and Land Use Planner	3 months
	- Evaluate environmental impact and biodiversity	- Identification of green belt opportunities	Environmental Agencies	4 months
	- Assess existing public transport infrastructure	- Analysis of public transportation needs	Transport Authority	2 months
			Community Groups	
2. Upgrade of Magadi Road to Four Lanes with a Green Belt Buffer	- Design and planning of road expansion	- Increased road capacity	County Government	1 year
	- Implementation of modern safety features	- Reduced traffic congestion	Engineering Consultants	2 years
	- Establishment of green belt buffer	- Enhanced safety for motorists and pedestrians	Transport Authority	1.5 years

		- Improved air quality	Construction Contractors	2 years
		- Biodiversity conservation	Environmental Agencies	1.5 years
			Community Groups	
3. Implementation of a Bus Rapid Transit (BRT) System with Dedicated Lanes	- Feasibility study and route planning	- Efficient public transportation	County Government	1.5 years
	- Construction of dedicated BRT lanes	- Reliable and inclusive transit option	BRT Operators	2 years
	- Installation of BRT system amenities	- Reduced emissions	Transport Authority	1.5 years
		- Improved urban mobility	Construction Contractors	2 years
		- Economic stimulation	BRT Operators	1.5 years
			Service Providers	
4. Development of Well-Connected Corridors with Formal Markets and Waste management	- Design and construction of interconnected corridors	- Enhanced connectivity	Urban Planning Dept.	2 years
	- Establishment of formal markets with waste management	- Reduced travel times	Market Authorities	1.5 years
	- Integration of safety features and amenities	- Stimulated local economy	Small Business Owners	1 year
		- Improved shopping experience	Local Government	
		- Safe and accessible public spaces	Community Groups	

5.2 Implementation Schedule

Table 11: Implementation Schedule

ITEM	ACTIVITY	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
1	Inventory of existing site development and conditions	✓	✓	✓	✓							
2	Design and planning of road expansion	✓	✓	✓	✓	✓	✓	✓	✓			
3	Implementation of modern safety features					✓	✓	✓	✓	✓	✓	✓
4	Establishment of green belt buffer					✓	✓	✓	✓	✓	✓	✓
5	Feasibility study and route planning for BRT system	✓	✓	✓	✓	✓	✓					
6	Construction of dedicated BRT lanes							✓	✓	✓	✓	✓
7	Installation of BRT system amenities									✓	✓	✓
8	Design and construction of interconnected corridors	✓	✓	✓	✓	✓	✓	✓	✓			
9	Establishment of formal markets							✓	✓	✓	✓	✓
10	Integration of safety features and amenities					✓	✓	✓	✓	✓	✓	
11	Coordination of all components	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
12	Monitoring and adjustment of implementation									✓	✓	✓
13	Progressive maintenance									✓	✓	✓
14	Evaluation and feedback									✓	✓	✓

Key:

- Q1: First Quarter
- Q2: Second Quarter
- Q3: Third Quarter
- Q4: Fourth Quarter

This table provides a structured timeline for the project's implementation, covering a period of ten years from 2024 to 2034.

5.3 Project Phasing Framework

Preferred Alternative Phasing Framework for Magadi Road Corridor Project

Table 12: Project Phasing Framework

Phase	Stage	Activities Conducted	Time Period
Short Term	Preliminary Stage	- Proposed integrated development plan identification	2-3 years
		- Preparation of alternative plans and designs	
		- Preferred plan selection	
		- Public Notification of Project	
		- Conducting an EIA	
		- Project Phasing and Costing	
		- Project submission for approval	
Medium Term	Design Stage	- Design plan for Magadi Road expansion	3-5 years
		- Design proposal for the Market	
		- Design proposal for public transport facilities	
		- Proposal for road infrastructure expansion	
		- Provision of greenery along the roads	
Long Term	Construction Stage	- Construction of the proposed transport infrastructure	5-10 years
		- Rehabilitation of existing markets	
		- Tree planting along the Magadi Road	
		- Expansion of Magadi Road to four lanes	
		- Construction of dedicated Bus Rapid Transit (BRT) lanes	
		- Construction of the Solar power plant	
		- Construction of the wastewater recycling plant	
		- Expansion of access roads to neighborhoods	
Post-Design and Construction Stage	Post-design and Construction Stage	- Development control enforcement	Ongoing
		- Rehabilitation and maintenance of infrastructure	
		- Site management	
		- Regulation and management of the project	
		- Public sensitization on the sustainability of the development	

5.4 Project Costing and Resource Requirements

Table: Project Costing and Resource Requirements for Magadi Road Corridor Project

Table 13: Project Cost

Activity	Actions	Resource Requirement	Resource Mobilization	Cost
Site preparation and assessment	- Feasibility of site, e.g., Notifying the public, EIA evaluation, Bringing of materials,	Financial, human	County Government, Professional experts (e.g Planners), Residents	100M
Construction of Magadi Road	- Road width expansion, e.g., Implementation of modern safety features, Establishment of green belt buffer	Financial, technological, human	County Government, Engineering Consultants, Transport Authority, Construction Contractors, Environmental Agencies	300M
Implementation of BRT System	- Feasibility study and route planning, e.g., Construction of dedicated BRT lanes, Installation of BRT system amenities	Financial, technological, human	County Government, BRT Operators, Transport Authority, Construction Contractors, Service Providers	500M
Development of Well-Connected Corridors with waste management	- Design and construction of interconnected corridors, e.g., Establishment of formal markets, Integration of safety features and amenities, Paying the relocated residents	Financial, technological, human	Urban Planning Dept., Market Authorities, Small Business Owners, Local Government, Community Groups	400M
Project Total				1,300M

This table outlines the project costing and resource requirements for the Magadi Road Corridor project, detailing the actions, required resources, and associated costs for each activity.

5.5 Framework for Implementing Institution/Agencies

5.5.1 Role of Actors and Implementation Agencies

Role of Actors and Implementation Agencies for Magadi Road Corridor Project

Table 14: Framework for implementation

Actors	Specific Agencies	Roles
County Government	Kajiado County Government	- Overall project coordination and management- Allocation of funding- Policy formulation and enforcement
Engineering Consultants	Various contracted firms	- Design and planning of road expansion e.g. feasibility studies and technical assessments
Transport Authority	Kenya Urban Roads Authority (KURA)	- Implementation of modern safety features e.g. supervision of road construction
Environmental Agencies	National Environment Management Authority (NEMA)	- Environmental Impact Assessments (EIA)- Monitoring and ensuring compliance with environmental regulations
Community Groups	Local community organizations	- Public participation and feedback e.g. community sensitization on project benefits and impacts
BRT Operators	Nairobi Metropolitan Area Transport Authority (NaMATA)	- Operation and management of BRT system e.g. ensuring efficient and reliable public transportation
Construction Contractors	Selected construction firms	- Road construction e.g. installation of BRT lanes and amenities
Service Providers	Utility companies, e.g., Kenya Power, Water Services	- Provision and maintenance of utilities e.g. construction of solar power plant and wastewater recycling plant
Urban Planning Dept.	Kajiado County Urban Planning Department	- Design and construction of interconnected corridors e.g. establishment of formal markets
Market Authorities	Local market management bodies	- Management and regulation of formal markets e.g. ensuring proper infrastructure and services
Small Business Owners	Local entrepreneurs and vendors	- Participation in formal markets e.g. feedback on market infrastructure and services
Private Sector	Investors and private companies	- Funding and investment e.g. participation in infrastructure development
Professional Experts	Various consultants and specialists	- Technical expertise in design and construction e.g. advisory roles in environmental and engineering aspects
Residents	Local population	- Participation in public consultations e.g. feedback on project impacts and benefits

5.5.2 Government Framework

Table: Governance Framework for Magadi Road Corridor Project

Table 15: Government Framework

Plan Phase	Description
Plan Preparation	Encompasses plan inception, site assessment, visual survey, public participation, site suitability analysis, conceptual spatial plans, preferred alternative, EIA, and provision of implementation, monitoring, and evaluation schedules.
Plan Approval and Adoption	The plan is submitted to the county and set for approval by the county assembly. This involves approvals from other relevant authorities before the final approval and entails community participation where stakeholders are engaged.
Plan Implementation	Involves partnership formulation, project phasing, implementation, monitoring, evaluation, and review of existing plans. Covers three interconnected aspects: urban planning and design, urban rules and regulations, and municipal finance and financial planning. Leaders and stakeholders are involved in monitoring and review of this process.
Development Control	Following the implementation of the plan, development control aims to ensure that operations on land comply with spatial development plans. Includes policy guidelines, regulations, and standards established by the planning authority. Factors to consider include the nature, location, and distribution of water supply systems and services.
Public Participation	The plan from inception process to implementation must involve members of the public, with their opinions regarded highly. Ensures community engagement and feedback integration throughout the project lifecycle.

This governance framework outlines the stages and critical activities involved in the planning, approval, implementation, development control, and public participation for the Magadi Road Corridor project.

5.6 Monitoring and Evaluation

A crucial tool for directing the evaluation of developments and their effects on the neighborhood is monitoring and evaluation. It incorporates the identified activities, success metrics, anticipated results, monitoring tools, and M&E frequency.

Monitoring and Evaluation Schedule for Magadi Road Corridor Project

Table 16: Monitoring and Evaluation

Activities	Indicators	Means of Monitoring	Frequency of M&E
Site preparation and assessment	Completion of site surveys, EIA report	Site inspection reports, EIA documentation	Quarterly
Upgrade of Magadi Road to Four Lanes with a Green Belt Buffer	Increased road capacity, green belt established	Traffic flow analysis, environmental surveys	Biannually
Implementation of safety features	Number of safety features installed	Safety audits, inspection reports	Quarterly
Construction of dedicated BRT lanes	Length of BRT lanes completed	Construction progress reports, site visits	Monthly
Installation of BRT system amenities	Number of stations equipped, operational BRT buses	Operational audits, user feedback surveys	Quarterly
Development of interconnected corridors	Connectivity improvements, reduced travel times	Traffic flow analysis, connectivity studies	Biannually
Establishment of formal markets	Number of formal markets established, vendor satisfaction	Market audits, vendor surveys	Annually
Integration of safety features and amenities in corridors	Safety features implemented, user satisfaction	Safety audits, user feedback surveys	Quarterly
Coordination of all components	Progress of integrated project implementation	Project progress reports, stakeholder meetings	Monthly
Monitoring and adjustment of implementation	Balanced development, quality of life improvements	Quality of life surveys, environmental impact assessments	Annually

CHAPTER 6:

6.0 CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

The Magadi Road Corridor Development and Management Strategy Plan represents a comprehensive approach to addressing the significant urban challenges faced by Ongata Rongai due to rapid and uncontrolled development. By integrating transportation, environmental sustainability, and economic development, this plan aims to create a more efficient, livable, and resilient urban environment. The proposed upgrades to Magadi Road, the implementation of a Bus Rapid Transit (BRT) system, the development of well-connected corridors, and the establishment of formal markets are all critical components designed to enhance the overall quality of life for residents.

The success of this plan hinges on effective implementation, continuous monitoring, and adaptive management. Through careful planning, stakeholder engagement, and the integration of best practices in urban development, the Magadi Road Corridor can become a model for sustainable urban growth. The project's phased approach ensures that immediate needs are addressed while laying the groundwork for long-term development and resilience.

6.2 Recommendations

Several key recommendations are proposed to ensure the successful implementation and long-term sustainability of the Magadi Road Corridor Development and Management Strategy Plan. Firstly, robust stakeholder engagement is paramount. Continuous and meaningful involvement of residents, businesses, and environmental groups will foster community ownership, enhance project relevance, and ensure sustained support. This approach will also facilitate early identification of concerns and opportunities, enabling proactive adjustments to project strategies to align with community needs and aspirations.

Secondly, the plan should prioritize environmental management practices. Implementing regular ecological impact assessments (EIAs), preserving green spaces, and integrating sustainable landscaping measures are crucial for maintaining biodiversity and enhancing the corridor's ecological resilience. This not only safeguards the natural environment but also promotes the health and well-being of residents by providing recreational spaces and improving air quality.

Lastly, the strategy should focus on integrated transportation solutions and economic development initiatives. Prioritizing the implementation of the Bus Rapid Transit (BRT) system, along with enhancing cycling and pedestrian infrastructure, will alleviate traffic congestion, reduce carbon emissions, and improve accessibility. Concurrently, supporting local economic growth through infrastructure enhancements and fostering small business development will stimulate job creation, economic resilience, and community prosperity.

By implementing these recommendations, the Magadi Road Corridor Development and Management Strategy Plan can achieve its goals of creating a more sustainable, livable, and resilient urban environment for the residents of Ongata Rongai.

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ANNEX

ANNEX 1: Exhibition Posters for Oral Examination: Magadi Road Corridor Linear Development Plan

SCHOOL OF BUILT ENVIRONMENT
DEPARTMENT OF URBAN AND REGIONAL PLANNING
PLANNING DEVELOPMENT PROJECT BU R 2402
BY
CLEARENCE OGHINO OSAHO
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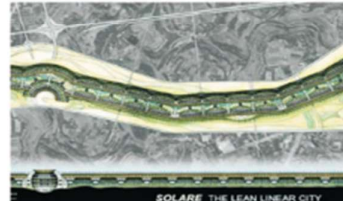
CHAPTER ONE
Background of the Planning Development Project
1.1 MAGADI ROAD CORRIDOR LINEAR DEVELOPMENT AND MANAGEMENT PLAN
Capturing an Urban Urban Pattern: The Magadi Road Corridor Development and Management Plan (2024-2034) aims to leverage the linear urban pattern along Magadi Road to optimize land use and transportation infrastructure for sustainable growth.
Enhancing Transportation Infrastructure: The project focuses on improving public transit and pedestrian infrastructure to reduce congestion, enhance accessibility, and promote mixed-use development.
Mitigating Negative Impacts: Targeted strategies and infrastructure interventions aim to address and mitigate the negative impacts of linear development, such as congestion and social disparities.
Holistic Approach: By integrating transportation planning with spatial interventions, the plan seeks to create a cohesive strategy for sustainable urban development along Magadi Road and similar corridors.
Improving Quality of Life and Economic Prosperity: The ultimate goal is to enhance the quality of life for residents and drive economic prosperity for businesses along the corridor through sustainable and inclusive development practices.
Problem Statement
• **Traffic and Infrastructure Challenges:** Rapid urbanization along the Magadi Road corridor has led to severe traffic congestion and the narrow lanes and inadequate signage, along with infrastructure deficiencies like poor road maintenance and limited public transportation, compromising accessibility and safety.
• **Land Use Conflicts:** The corridor faces significant land use conflicts, with competing interests among residential, commercial, and industrial stakeholders leading to zoning disputes and regulatory challenges that hinder infrastructure improvements.
• **Environmental and Health Concerns:** Urbanization has encroached upon natural habitats and green spaces, increasing vehicular traffic and noise-polluting air and noise pollution, thus posing threats to environmental sustainability and public health, highlighting the need for sustainable development solutions.
Location and area coverage of the planning area
Magadi Road Corridor Development and Management Plan
The study area is located in Ngong Ngong Municipality, Kapanga North Constituency within two wards Otaga Rongai ward and Makenyewa ward, Kapanga County, Kenya. The site area covers approximately a total of about 1300m running from Magadi-Guido road junction to Quarry Market.
Goals
The goal of the Magadi Road Corridor Linear Development and Management Plan is to enhance transportation efficiency and safety, promote sustainable development, and ensure equitable growth along the corridor.
OBJECTIVES
1. To review literature, principles, policies, legislations, development control guidelines and standards on sustainable linear development for enhancing liveability along road corridors.
2. To conduct contextual analysis of the Magadi road corridor planning area in Otaga Rongai Town.
3. To develop alternative plans, strategies, and policy options for the Magadi road corridor Linear Development and Management Plan in Otaga Rongai Town.
4. To assess the alternative plans and to formulate a preferred Magadi road corridor linear development and management Plan to be used in Otaga Rongai Town.
5. To formulate implementation, monitoring, and evaluation framework for Magadi road corridor Integrated Physical Land Use Plan in Otaga Rongai Town.

Data Needs and Requirements
To solve the Magadi Road project problem, a comprehensive understanding of various data needs and requirements is essential. This includes traffic assess current and future road usage, road condition surveys to determine necessary repairs and upgrades, and environmental impact assessments to ensure sustainable development. Socio-economic data is crucial to evaluate the project's impact on local communities, while geographic information system (GIS) data helps in mapping and planning. Additionally, financial data is needed to estimate costs and budget allocations, and legal and regulatory requirements must be reviewed to ensure compliance with local, regional, and national guidelines. Collecting and integrating these diverse data sets will facilitate informed decision-making and effective project implementation
• **Site Assessment Approach:** The site assessment involved a situational analysis covering demographic, environmental, infrastructural, and policy aspects, evaluated through a SWOT analysis to determine site suitability for development.
• **Methods of Evaluation of Development Option:** The Goal Achievement Matrix (GAM) was used to compare planned goals with objectives and required actions, ranking development models to select the optimal plan for the Magadi Corridor.
• **Approaches to Planning, Design, and Space Organization:** The planning and design approach focused on functionality, sustainability, and aesthetics, ensuring the space aligns with its historical, cultural, and environmental context.

CHAPTER THREE
LITERATURE REVIEW
THEORETICAL FRAMEWORK
Linear City Model
• **Origins and Concept:** The linear city concept, first theorized by Spanish engineer Arturo Soria y Mata in 1882, envisioned a central avenue lined with buildings, facilitating transportation and encouraging agricultural production along the development. The model was partially realized in a 55-km project around Madrid in the early 1890s.
• **Design Guidelines:** Soria y Mata's design included a central avenue for traffic, particularly rail, with adjacent parallel streets marking the city's edge and perpendicular streets forming blocks. The plan emphasized greenery, with single-family homes on large lots, tree-lined streets, and public amenities like flowers, benches, and fountains to enhance the urban environment.
• **Key Features and Application:** The Linear City Model centers around a transportation axis that optimizes accessibility and economic efficiency by concentrating development along this route. It emphasizes spatial concentration and linear growth, making it relevant for understanding urban patterns along major transportation routes like Magadi Road in Otaga Rongai.



The Linear Megastucture Model
Continues Linear Structure: The Linear Megastucture Model features an extended linear form along a major transportation route, designed to minimize land use while maximizing accessibility, integrating housing, workplaces, shopping, and recreational facilities into a self-contained urban environment.
• **Central Transportation Axis:** A central transportation corridor runs through the megastucture, including high-capacity public transit systems like trains or buses, as well as roads for private vehicles, facilitating efficient movement of people and goods.
• **Sustainability and Modularity:** The model emphasizes sustainability with energy-efficient buildings, green roofs, renewable energy sources, and promotes walking and cycling. It uses modular construction techniques for phased development and flexibility, enabling rapid construction and adaptation.
• **Application to Magadi Road Corridor:** For the Magadi Road Corridor, a linear megastucture parallel to the road would integrate residential, commercial, industrial, and recreational spaces, ensuring self-sufficiency. The central transportation corridor would feature sustainable public transit options, and the design would include sustainable practices and phased development for gradual urban growth and adaptation.



SOLARE THE LEAN LINEAR CITY

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Policy and Legal Framework: Overview

Urban Integrated Development Planning

- County-wide 5-year Integrated Development Planning (IDP): Comprehensive development planning for counties over five years.
- Urban-specific Integrated Development Planning (IDUP): Ten-year plans focused on urban areas.
- Urban IDUPs, 10-year Sectoral Plans, and Spatial Plans as inputs to CIDP: Components that feed into the broader county development plan.
- Annual program-based budget in a 3-year fiscal framework (MTEF): A budgeting process that aligns with development plans.
- County Governments Act 2012, Section 108 (1): Mandates integrated development planning for counties.

Urban Legislation Branch

- Legal and operational support for governments and cities: Provides frameworks for regulating urbanisation and development.
- Promotion and development of equitable, gender-responsive, and socially inclusive governance models: Ensures fair land use and access to services.

Constitutional Basis

- Article 60: Principles of land policy: Guides land use and management policies.
- Article 66: Regulation of land use: Allows government regulation of land for public interest.
- Article 67 (b): Monitoring and oversight by the National Land Commission: Provides oversight for land use planning.
- Fourth Schedule, Part I (21): General principles of land planning: Coordinates county planning.
- Fourth Schedule, Part II (4): County planning and development: Mandates county-level planning and development.
- Article 184: Governance and management of Urban Areas and Cities: Provides criteria for urban area management.

County Government Act, 2012

- Sections 102-115: Principles of planning and development facilitation: Guides county planning and development processes.

Urban Areas and Cities Act, 2012

- Section 40: Contents of Integrated Urban Area Development plans: Specifies what urban development plans should include.

Physical Planning Act Cap 286

- Matters considered in spatial planning: Provides a framework for development control and spatial planning.

Linkages with Other Plan and Policy Documents

- County Government Act 2012: Mandates integrated county development plans and budgets.
- Public Finance Management Act (PFMA), 2012: Outlines efficient management of public resources.
- Urban Areas and Cities Act, 2011: Requires cities and towns to have integrated development plans.
- Constitution of Kenya 2010: Devolves planning and development to county governments.
- Kenya Vision 2030: Long-term development policy aiming for middle-income status by 2030.
- Sustainable Development Goals (SDGs): Global targets for addressing various development challenges.
- County Spatial Plan: Provides a spatial development strategy for counties.
- CIDP II: Focuses on policy and budgetary planning for urban management.
- New Urban Agenda (NUA): Framework for inclusive and sustainable urban development.
- Joint Agenda 2030: Vision for a prosperous Africa through inclusive growth.
- East African Community Urban Development Strategy: Aims for sustainable regional urban development.

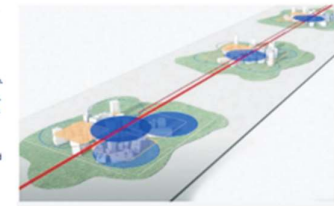
Urban Design

- Population Density Distribution:** Illustrates how people are spread across different urban areas, guiding infrastructure and service planning.
- Land Use Composition:** Breaks down urban land use into residential, commercial, industrial, recreational, and green spaces, aiming for balanced and sustainable development.
- Transportation Mode Share:** Shows the proportion of trips by walking, cycling, public transit, and private vehicles, promoting sustainable transport options to reduce congestion and pollution.
- Quality of Life Indicators:** Measures aspects like access to parks, air quality, housing affordability, and community engagement, focusing on creating healthy and inclusive urban environments.
- Integrated Approach:** Considers the interconnections between land use, transportation, and infrastructure to create cohesive and sustainable urban environments.
- Mixed Land Use:** Integrates diverse activities within neighborhoods to promote walkability and community interaction, contrasting with segregated linear developments.
- Human Scale:** Prioritizes pedestrian and cyclist experience with inviting, accessible streetscapes, in contrast to linear planning's focus on vehicular efficiency.
- Placemaking:** Enhances neighborhood identity and character through cultural and historical integration, fostering community pride and cohesion.
- Flexibility and Adaptability:** Allows for responsive urban development that can evolve over time, adapting to changing needs and conditions compared to rigid linear planning approaches.

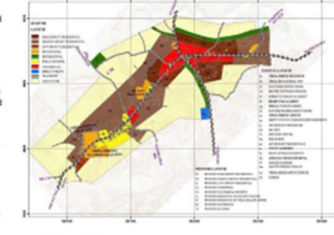
Case studies

THE LINE – THE SAUDI-ARABIAN LINEAR CITY CONCEPT AS THE PROTOTYPE OF FUTURE CITIES

- International City with Diverse Population:** The Line will host over 1 million residents from around the globe, situated across four distinct landscapes: Coastal, coastal desert, Mountain, and Upper Valley.
- Renewable Energy for Zero Carbon Emissions:** The city will rely entirely on renewable energy sources such as solar, wind, and hydrogen power, ensuring a pollution-free, carbon-neutral environment.
- Five Minute Walkability Concept:** The city will prioritize walkability, with essential services like schools, medical clinics, and recreational areas all within a five-minute walk, eliminating the need for cars.
- Hyper-Connected Modular Settlements:** The Line will feature a 170km chain of mixed-use communities connected by high-speed underground trains or hyperloop, integrating nature into urban design.
- Multi-Layered Transportation System:** The city will have three distinct transportation levels: the Pedestrian Layer for walkers, the Service Layer for daily amenities, and the Spine Layer for high-speed transit, ensuring all travel times are under 20 minutes.
- Inspiration for Sustainable Urban Development:** The Line's focus on renewable energy, walkability, innovative transportation, and integration of nature can significantly inform the Magadi Road Corridor project, promoting sustainability, improved livability, and economic growth.

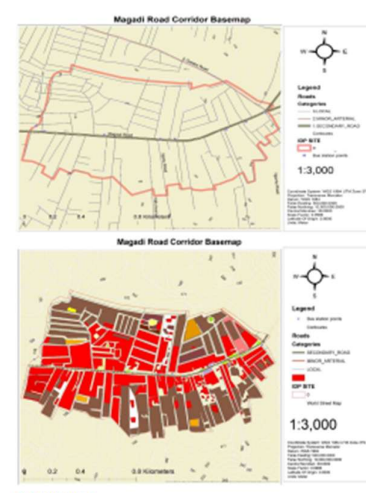
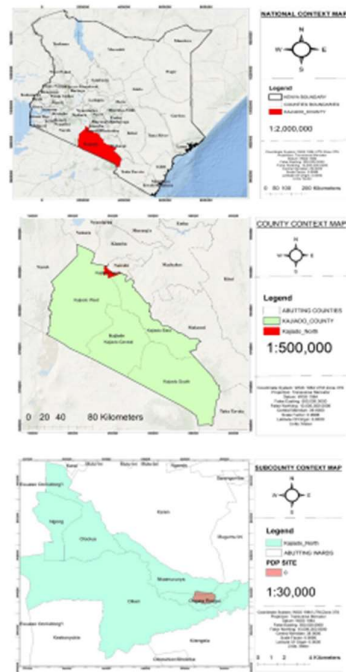


2. VIHGA MUNICIPALITY INTEGRATED DEVELOPMENT PLAN 2018-2022 Linear concept
The Vihiya Municipality Integrated Development Plan (IDP) aims to address the challenges posed by high-density urbanisation along major roads like the Kiunga-Kilimungu Road. It proposes a spatial framework to guide future development by integrating residential and commercial areas more cohesively. By enhancing connectivity and preserving rural-urban linkages, the plan seeks to improve infrastructure provision and service delivery while promoting sustainable urban growth in Vihiya Township.



CHAPTER THREE

Contextual Site Location

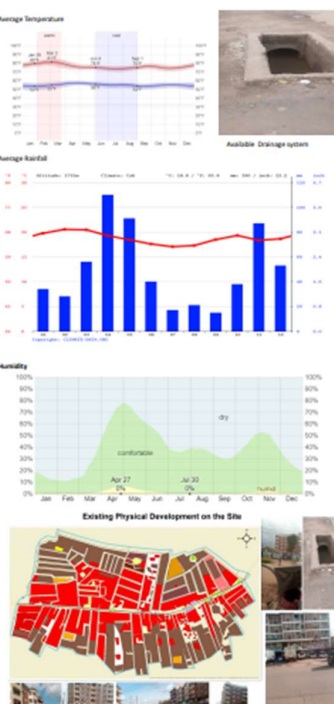


Site Viability Analysis
Population Projections

Year	2020	2025	2030	2035	2040	2045
Population	101,751	111,508	121,014	131,201	141,017	150,100

Geology and Soils
The soils in Otago Range are largely derived from the weathering of volcanic rocks and alluvial deposits, resulting in a variety of soil types.
Common soil types include black cotton soils (Vertisols), which are rich in clay and have high water retention capabilities, making them suitable for agriculture but challenging for construction due to their expansive nature.
Other prevalent soils include sandy and silty soils, which offer better drainage and are more suitable for building and infrastructure development.

Hydrology and Drainage Systems
The area's hydrology is influenced by its position within the semi-arid environment of the Great Rift Valley. Rainfall in Otago Range is seasonal and often sporadic, leading to variable water availability. The primary source of surface water includes seasonal rivers and streams that flow during the rainy season.





CHAPTER FOUR

Table 4: Planning Development Regulation

AREA OF CONCERN	REGULATIONS
Roads	For the proposed upgrade of Magadi Road in Upper Regency, Standard Urban Lane Road should be between 5.25 to 5.75 meters wide to accommodate general traffic. BBT Lane: Dedicated BBT lane should be between 3.5 to 3.75 meters wide to meet the demand for BBT traffic. Right of Way: The right of way should be 10 meters wide for the right of way and 5 meters wide for the BBT lane. Suburb lane road to building is about 3 meters. These are per the Kenya Roads Act (KRA) which governs the development and maintenance of roads in Kenya, ensuring that roads meet specific standards for safety and capacity. Kenya National Highways Authority (KANHA) and Kenya Urban Roads Authority (KURA) Guidelines: Provide specific standards for the design and construction of roads, including lane widths, road markings, and signage. National Transport and Safety Authority (NTSA) Regulations: Include safety standards for road design to ensure road users' safety and improve traffic flow.
Green belt	Physical and Land Use Planning Act, 2019 Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Urban and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Urban and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl.
Urban corridors and buildings	Physical and Land Use Planning Act, 2019 Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl.
Markets	Physical and Land Use Planning Act, 2019 Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl. Physical and Land Use Planning Act, 2019: Provides guidelines for the integration of green belts into urban planning to ensure sustainable land use and prevent uncontrolled urban sprawl.

Site Accommodation Schedule

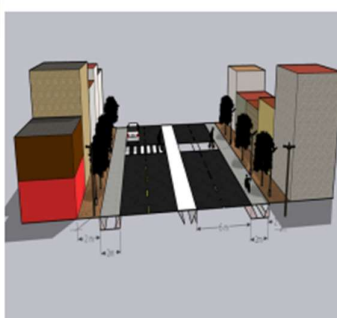
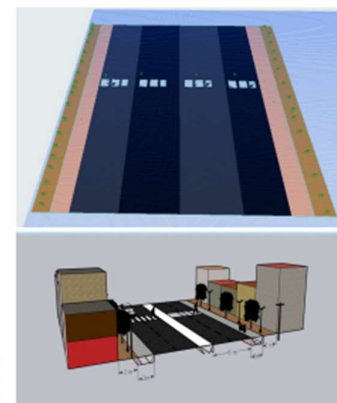
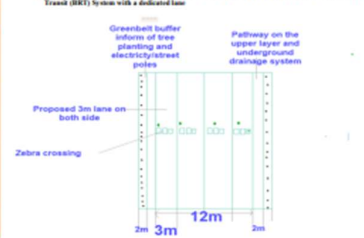
Land Use	Typology	Facilities/Provisions	Area (Ha)
Residential	High-density housing, low-density housing	Apartments, single-family homes, duplexes, commercial units, parking lots	15.8
Commercial	Retail, office spaces	Shopping centers, office buildings, restaurants, banks, parking areas, public transport	26.8
Public Purpose	Government buildings, community centers	Managed offices, community halls, public libraries, cultural centers, emergency services	8.0
Industrial	Light industrial, warehouses	Factories, distribution centers, warehouses, service facilities, loading docks	4.0
Education	Schools, colleges	Primary and secondary schools, vocational training centers, college campuses, libraries, sports facilities	8.0
Recreational/Open Space/Conservation Areas	Parks, nature reserves, sports complexes	Public parks, sports fields, sports fields, picnic areas, conservation areas	6.4
Transportation Infrastructure	Roads, public transit hubs	Roadways, bus stops, BRT lanes, parking facilities, pedestrian pathways, bike lanes	10.7
Public Utilities/Facilities	Utility services, waste management	Water treatment plants, electricity substations, waste collection and recycling centers	6.7
Agriculture	Urban farming, community gardens	Farmlands, greenhouses, community gardens, farm-to-market	6.7

Land Use Budget

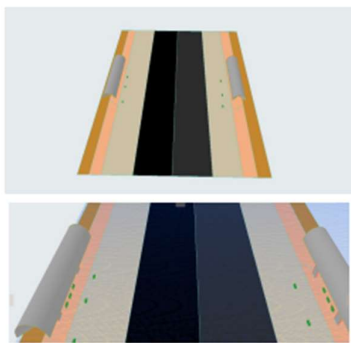
Land Use Type	Area in Ha	Percentage (%)
Residential	55.6	40
Commercial	26.8	20
Public Purpose	8.0	6
Industrial	4.0	3
Education	8.0	6
Recreational/Open Space/Conservation Areas	6.4	5
Transportation Infrastructure	10.7	8
Public Utilities/Facilities	6.7	5
Agriculture	6.7	5

Alternative 1

Upgrade of Magadi Road to Four Lanes with a Green Belt Buffer and Implementation of a Bus Rapid Transit (BRT) system with a dedicated lane



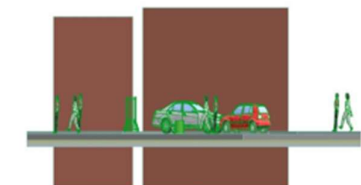
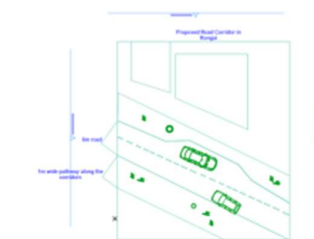
BRT STAGE



Alternative 2



The second alternative for the Oregma Rongai development project focuses on creating well-connected corridors throughout the town, complemented by the establishment of formal markets. This approach aims to facilitate efficient movement and accessibility by developing a network of interconnected roads and pathways that seamlessly link residential, commercial, industrial, and public spaces. These corridors will integrate various modes of transport, including walking, cycling, and public transit.



Alternative 3: Conventional and Community-Led Waste Management for Oregma Rongai Town

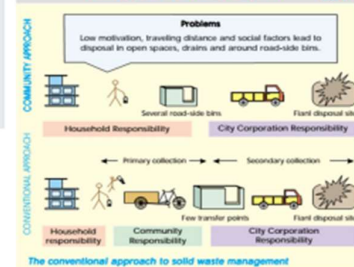
Key Components of the Community-Led Waste Management System:
Neighborhood Collection Points
Local Recycling Programs
Neighborhood Composting
Education and Awareness
Workshops and Training
Participatory Incentives



Land Acquisition and Compensation: For neighborhood collection points and composting sites, land acquisition will involve identifying suitable public spaces or negotiating with private landowners for small parcels. Fair compensation will be provided through monetary payments or community benefits, like improved infrastructure, with transparent processes and community consultation to ensure local support and minimize conflicts.



Schematic diagram showing the conventional and community-based approach to solid waste management. Box 4



Alternative 4

The integrated approach for the Magadi road corridor, Oregma Rongai development project combines upgrading Magadi Road to four lanes with a green belt, implementing a Bus Rapid Transit (BRT) system with dedicated lanes, Conventional and Community-Led Waste Management, developing well-connected corridors, and establishing formal markets. This comprehensive strategy addresses multiple urban challenges by enhancing road capacity and safety, promoting efficient public transportation, and reducing traffic congestion.

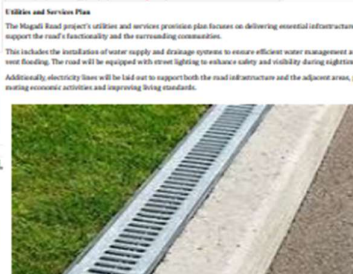
Evaluation of the Alternatives

Table 2: Goal Achievement Matrix

Objective	Weighted Sub-Objective	Alternative 1	Alternative 2	Alternative 3	Alternative 4	Weighted Sub-Objective
1	0.25	2	0.5	3	0.5	0.75
2	0.25	2	0.5	3	0.5	0.75
3	0.25	2	0.5	3	0.5	0.75
4	0.25	2	0.5	3	0.5	0.75
5	0.25	2	0.5	3	0.5	0.75
6	0.25	2	0.5	3	0.5	0.75
7	0.25	2	0.5	3	0.5	0.75
8	0.25	2	0.5	3	0.5	0.75
9	0.25	2	0.5	3	0.5	0.75
10	0.25	2	0.5	3	0.5	0.75
11	0.25	2	0.5	3	0.5	0.75
12	0.25	2	0.5	3	0.5	0.75
13	0.25	2	0.5	3	0.5	0.75
14	0.25	2	0.5	3	0.5	0.75
15	0.25	2	0.5	3	0.5	0.75
16	0.25	2	0.5	3	0.5	0.75
17	0.25	2	0.5	3	0.5	0.75
18	0.25	2	0.5	3	0.5	0.75
19	0.25	2	0.5	3	0.5	0.75
20	0.25	2	0.5	3	0.5	0.75
21	0.25	2	0.5	3	0.5	0.75
22	0.25	2	0.5	3	0.5	0.75
23	0.25	2	0.5	3	0.5	0.75
24	0.25	2	0.5	3	0.5	0.75
25	0.25	2	0.5	3	0.5	0.75
26	0.25	2	0.5	3	0.5	0.75
27	0.25	2	0.5	3	0.5	0.75
28	0.25	2	0.5	3	0.5	0.75
29	0.25	2	0.5	3	0.5	0.75
30	0.25	2	0.5	3	0.5	0.75
31	0.25	2	0.5	3	0.5	0.75
32	0.25	2	0.5	3	0.5	0.75
33	0.25	2	0.5	3	0.5	0.75
34	0.25	2	0.5	3	0.5	0.75
35	0.25	2	0.5	3	0.5	0.75
36	0.25	2	0.5	3	0.5	0.75
37	0.25	2	0.5	3	0.5	0.75
38	0.25	2	0.5	3	0.5	0.75
39	0.25	2	0.5	3	0.5	0.75
40	0.25	2	0.5	3	0.5	0.75
41	0.25	2	0.5	3	0.5	0.75
42	0.25	2	0.5	3	0.5	0.75
43	0.25	2	0.5	3	0.5	0.75
44	0.25	2	0.5	3	0.5	0.75
45	0.25	2	0.5	3	0.5	0.75
46	0.25	2	0.5	3	0.5	0.75
47	0.25	2	0.5	3	0.5	0.75
48	0.25	2	0.5	3	0.5	0.75
49	0.25	2	0.5	3	0.5	0.75
50	0.25	2	0.5	3	0.5	0.75
51	0.25	2	0.5	3	0.5	0.75
52	0.25	2	0.5	3	0.5	0.75
53	0.25	2	0.5	3	0.5	0.75
54	0.25	2	0.5	3	0.5	0.75
55	0.25	2	0.5	3	0.5	0.75
56	0.25	2	0.5	3	0.5	0.75
57	0.25	2	0.5	3	0.5	0.75
58	0.25	2	0.5	3	0.5	0.75
59	0.25	2	0.5	3	0.5	0.75
60	0.25	2	0.5	3	0.5	0.75
61	0.25	2	0.5	3	0.5	0.75
62	0.25	2	0.5	3	0.5	0.75
63	0.25	2	0.5	3	0.5	0.75
64	0.25	2	0.5	3	0.5	0.75
65	0.25	2	0.5	3	0.5	0.75
66	0.25	2	0.5	3	0.5	0.75
67	0.25	2	0.5	3	0.5	0.75
68	0.25	2	0.5	3	0.5	0.75
69	0.25	2	0.5	3	0.5	0.75
70	0.25	2	0.5	3	0.5	0.75
71	0.25	2	0.5	3	0.5	0.75
72	0.25	2	0.5	3	0.5	0.75
73	0.25	2	0.5	3	0.5	0.75
74	0.25	2	0.5	3	0.5	0.75
75	0.25	2	0.5	3	0.5	0.75
76	0.25	2	0.5	3	0.5	0.75
77	0.25	2	0.5	3	0.5	0.75
78	0.25	2	0.5	3	0.5	0.75
79	0.25	2	0.5	3	0.5	0.75
80	0.25	2	0.5	3	0.5	0.75
81	0.25	2	0.5	3	0.5	0.75
82	0.25	2	0.5	3	0.5	0.75
83	0.25	2	0.5	3	0.5	0.75
84	0.25	2	0.5	3	0.5	0.75
85	0.25	2	0.5	3	0.5	0.75
86	0.25	2	0.5	3	0.5	0.75
87	0.25	2	0.5	3	0.5	0.75
88	0.25	2	0.5	3	0.5	0.75
89	0.25	2	0.5	3	0.5	0.75
90	0.25	2	0.5	3	0.5	0.75
91	0.25	2	0.5	3	0.5	0.75
92	0.25	2	0.5	3	0.5	0.75
93	0.25	2	0.5	3	0.5	0.75
94	0.25	2	0.5	3	0.5	0.75
95	0.25	2	0.5	3	0.5	0.75
96	0.25	2	0.5	3	0.5	0.75
97	0.25	2	0.5	3	0.5	0.75
98	0.25	2	0.5	3	0.5	0.75
99	0.25	2	0.5	3	0.5	0.75
100	0.25	2	0.5	3	0.5	0.75

Legend: 1=High, 2=Medium, 3=Low

Detailed Design Formulation



Six Environmental Management Plan

Structured environmental management plan for the Magadi Road Corridor Development and Management Strategy Plan in Oregma Rongai Town.

Activity	Location	Impact	Responsible Agency	Control Measures	Monitoring	Notes
Construction	Magadi Road	Dust, noise, vibration	Construction Company	Use of dust suppressants, noise barriers, and vibration monitors.	Regular monitoring	Regular monitoring required
Construction	Magadi Road	Public access, safety	Construction Company	Use of safety barriers, signage, and traffic management.	Public access management	Public access management required
Construction	Magadi Road	Water pollution	Construction Company	Use of silt fences, sediment traps, and proper disposal of waste.	Water quality monitoring	Focus on high-risk areas
Construction	Magadi Road	Land use, vegetation	Construction Company	Use of erosion control measures, revegetation, and proper disposal of waste.	Land use management	Regular inspection
Construction	Magadi Road	Soil erosion, sediment	Construction Company	Use of silt fences, sediment traps, and proper disposal of waste.	Soil erosion management	Community participation required
Construction	Magadi Road	Public access, safety	Construction Company	Use of safety barriers, signage, and traffic management.	Public access management	Provision of public access
Construction	Magadi Road	Water pollution	Construction Company	Use of silt fences, sediment traps, and proper disposal of waste.	Water quality monitoring	Use of water pollution
Construction	Magadi Road	Land use, vegetation	Construction Company	Use of erosion control measures, revegetation, and proper disposal of waste.	Land use management	Regular inspection
Construction	Magadi Road	Soil erosion, sediment	Construction Company	Use of silt fences, sediment traps, and proper disposal of waste.	Soil erosion management	Regular monitoring and maintenance
Construction	Magadi Road	Public access, safety	Construction Company	Use of safety barriers, signage, and traffic management.	Public access management	Provision of public access
Construction	Magadi Road	Water pollution	Construction Company	Use of silt fences, sediment traps, and proper disposal of waste.	Water quality monitoring	Use of water pollution
Construction	Magadi Road	Land use, vegetation	Construction Company	Use of erosion control measures, revegetation, and proper disposal of waste.	Land use management	Regular inspection
Construction	Magadi Road	Soil erosion, sediment	Construction Company	Use of silt fences, sediment traps, and proper disposal of waste.	Soil erosion management	Regular monitoring and maintenance
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Construction	Magadi Road	Water pollution	Construction Company	Use of silt fences, sediment traps, and proper disposal of waste.	Water quality monitoring	Use of water pollution
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The table outlines the development control guidelines for the Magadi Road Corridor project in Ongata Rongai Town, aligned with Kenya's development control guidelines:

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	Infrastructure Management	Construction Projects	Civil Engineering Dept.	Design, construction, inspection
Public Involvement	Public Facilities	Public Works Projects	Municipal Planning Authority	Facilities plans, construction, feedback
	Accessibility	Barrier removal projects	Building Control Authority	Accessibility audits and certification
Transportation	Public Transit	Public Transport	Transport Authority	Public transit usage data
	Non-Motorized Transport	Cycling paths and pedestrian crossings	Urban Planning Dept.	Path lengths and user surveys
Safety	Fire Safety	Fire safety audits	Building Control Authority	Fire safety inspections and certification
	Traffic Safety	Traffic safety audits	Traffic Police Dept.	Traffic safety audits and accident reports
Economic Development	Support for Small Business	Small Business Support	Economic Development Dept.	Business licenses and market opportunities
	Investment in Green Infrastructure	Green infrastructure projects	Environment Agency	Number of green business certifications
Community Engagement	Public Participation	Community consultation	Municipal Planning Authority	Public consultation surveys and feedback
	Community Monitoring	Monitoring and reporting	Environment/SDGs	Group activity reports and outcomes

Frequency of police and administrative	
visits to the premises	

Environmental impacts	The green building process encourages the use of energy, materials, products, and services that are environmentally sound and sustainable. Reduction in greenhouse gas emissions and mitigation of urban heat island and other factors through the BRT system. Reduction in water consumption and energy consumption, including the use of low-voltage lighting. Use of sustainable materials. Use of environmentally safe, durable, and reliable materials for interiors.
Social impacts	Increased pedestrian safety, mobility and access to essential services. Increased access to essential services for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children.
Economic impacts	Increased and sustained employment opportunities for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children.
Urban development impacts	Increased and sustained employment opportunities for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children.
Overall impacts	Increased and sustained employment opportunities for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children. Increased and sustained employment opportunities for people with disabilities, the elderly, and children.

Table 1. Project Feature Matrix

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Source: U.S. Census Bureau, *Marriage, Divorce, Remarriage in the 1990s*, Washington, D.C., 1995.

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		Potential problem situations	
		Public Notification of Project	
		Conducting an EIS	
		Project Planning and Coding	
Design	Design Stage	Project submission for approval	
		Design plan for Maggot Road expansion	
		Design proposal for the Market	
		Design proposal for public transport facilities	
		Proposal for road infrastructure upgrades	
		Promotion of greenery along the route	
Long term	Construction Stage	Construction of the proposed transport infrastructure	
		Relief stations of existing markets	
		Tree planting along the Maggot Road	
		Expansion of Maggot Road to four lanes	
		Construction of dedicated Bus Rapid Transit (BRT) lanes	
		Construction of the solar power plant	
		Construction of the wastewater recycling plant	
		Expansion of access roads to neighbourhoods	
		Development control enforcement	
Post design and construction	Post design and construction	Development control enforcement	
		Relief stations and maintenance of infrastructure	
		Site management	
Post design and construction	Post design and construction	Regulation and management of the project	
		Public consultation on the sustainability of the development	

Activity	Activity
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Site preparation and assessment	• Feasibility study, e.g., Soil, Seismicity, and Geology • Environmental impact, EIA evaluation • Socio-economic assessment	Financial, human	County Government, County Council, County Engineer's Office, County Surveyors, Residents	100%
Construction of Magadi Road	• Road map preparation, e.g., Implementation of road on safety basis, Establishment of gully lines before fill	Financial, technological, human	County Government, Engineering Contractors, Transport Authority, Contractors, Engineers, Environmental Engineers	100%
Implementation of BRT System	• Feasibility study of new lines • e.g., Construction of new lines, Installation of BRT lanes, Installation of BRT system operation	Financial, technological, human	County Government, Engineering Contractors, Transport Authority, Contractors, Engineers, Environmental Engineers	500%
Development of BRT corridors with route management	• Design and construction of dedicated corridors, e.g., Establishment of formal routes, Organization of staff and vehicles, Pricing the allocated routes	Financial, technological, human	Urban Planning, Urban Engineers, Surveyors, Road Builders, Road Contractors, Community Groups	400%
Project Total				1,000%

Arkans	Specific A
County	County

[illegible]

Activities	Indicators
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[illegible]

Infrastructure Stress Mitigation: The research findings informed strategies within the

Service Delivery Optimization: Insights from the research guide the Magadi Road Corridor plan in optimizing service delivery mechanisms. It can highlight areas where service delivery gaps are prominent due to linear development and propose measures to enhance service provision efficiency, such as zoning regulations, public transport improvements, and community facilities.

planning challenges (i.e., haphazard growth, land use conflicts, and environmental degradation) has helped the corridor plan adopt more sustainable and organized development practices. This includes promoting mixed-use developments, preserving green spaces, and integrating smart city technologies.

Policy and Regulatory Frameworks: Insights from the research has influence the policy and regulatory frameworks proposed in the Magadi Road Corridor plan. It can advocate for stricter land-use regulations, environmental impact assessments, and infrastructure standards to manage and control linear development's effects more effectively.

knowledge and recommendations that has guided the formulation and implementation of the Magadi Road Corridor Development and Management Plan. By addressing the challenges identified in the research, the plan aims to create a more sustainable, resilient, and well-managed urban corridor that meets the needs of its residents and supports its long-term economic and social development.

6.1 Conclusion

represents a comprehensive approach to addressing the significant urban challenges faced by Ongata Rongai due to rapid and uncontrolled development. By integrating transportation, environmental sustainability, and economic development, this plan aims to create a more efficient, livable, and resilient urban environment. The proposed upgrades to Magadi Road, the implementation of a Bus Rapid Transit (BRT) system, the development of well-connected corridors, and the establishment of formal markets are all critical components designed to enhance the overall quality of life for residents.

The success of this plan hinges on effective implementation, continuous monitoring, and adaptive management. Through careful planning, stakeholder engagement, and the integration of best practices in urban development, the Magadi Road Corridor can become a model for sustainable urban growth. The project's phased approach ensures that immediate needs are addressed while laying the groundwork for long-term development and resilience.

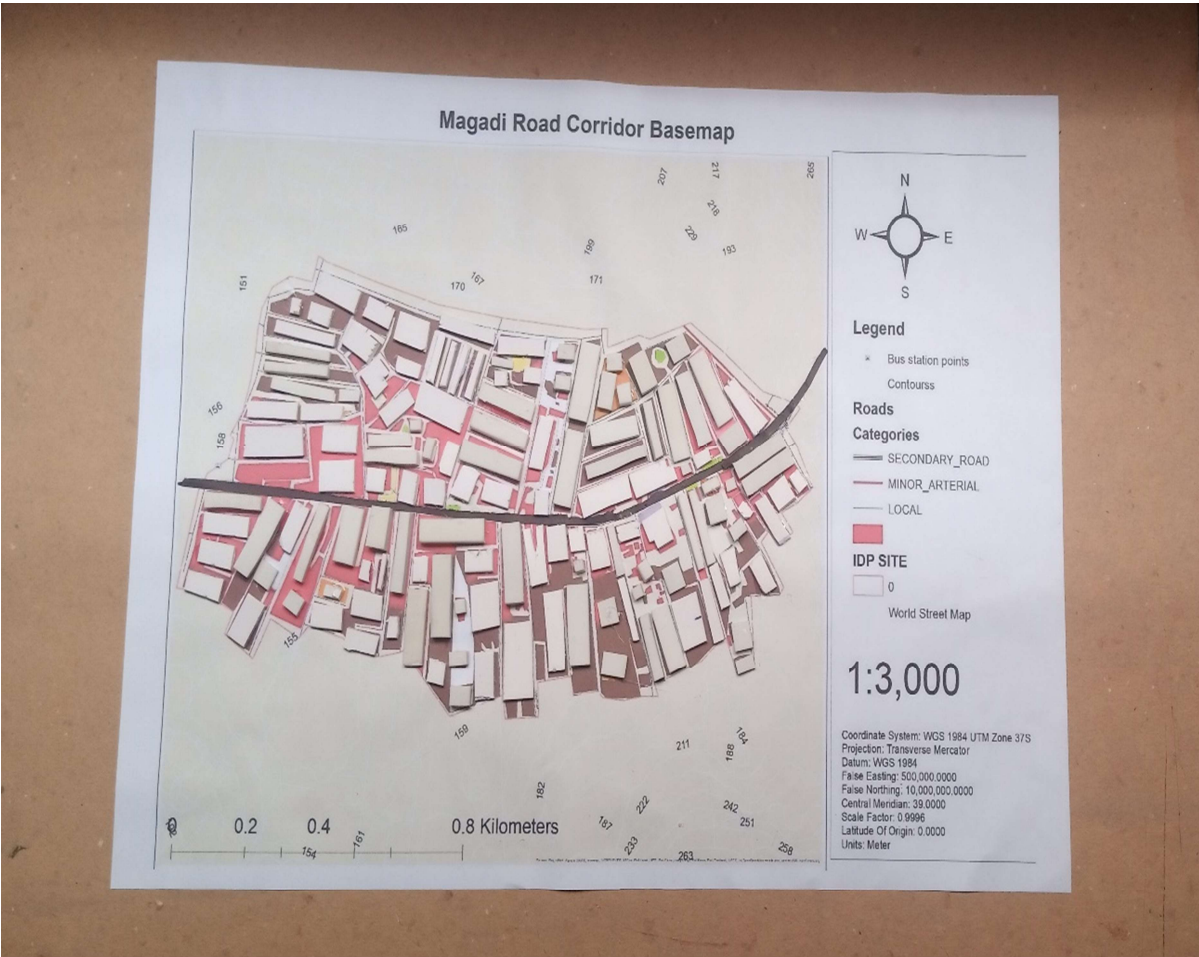
Several key recommendations are proposed to ensure the successful implementation and long-term sustainability of the Magadi Road Corridor Development and Management Strategy Plan. Firstly, robust stakeholder engagement is paramount. Continuous and meaningful involvement of residents, businesses, and environmental groups will foster community ownership, enhance project relevance, and ensure sustained support. This approach will also facilitate early identification of concerns and opportunities, enabling proactive adjustments to project strategies to align with community needs and aspirations.

Secondly, the plan should prioritize environmental management practices. Implementing regular ecological impact assessments (EIAs), preserving green spaces, and integrating sustainable landscaping measures are crucial for maintaining ecological balance and enhancing biodiversity. Additionally, the plan should include provisions for environmental education and awareness campaigns targeting residents and businesses. This not only safeguards the natural environment but also promotes the health and well-being of residents by providing recreational spaces and improving air quality.

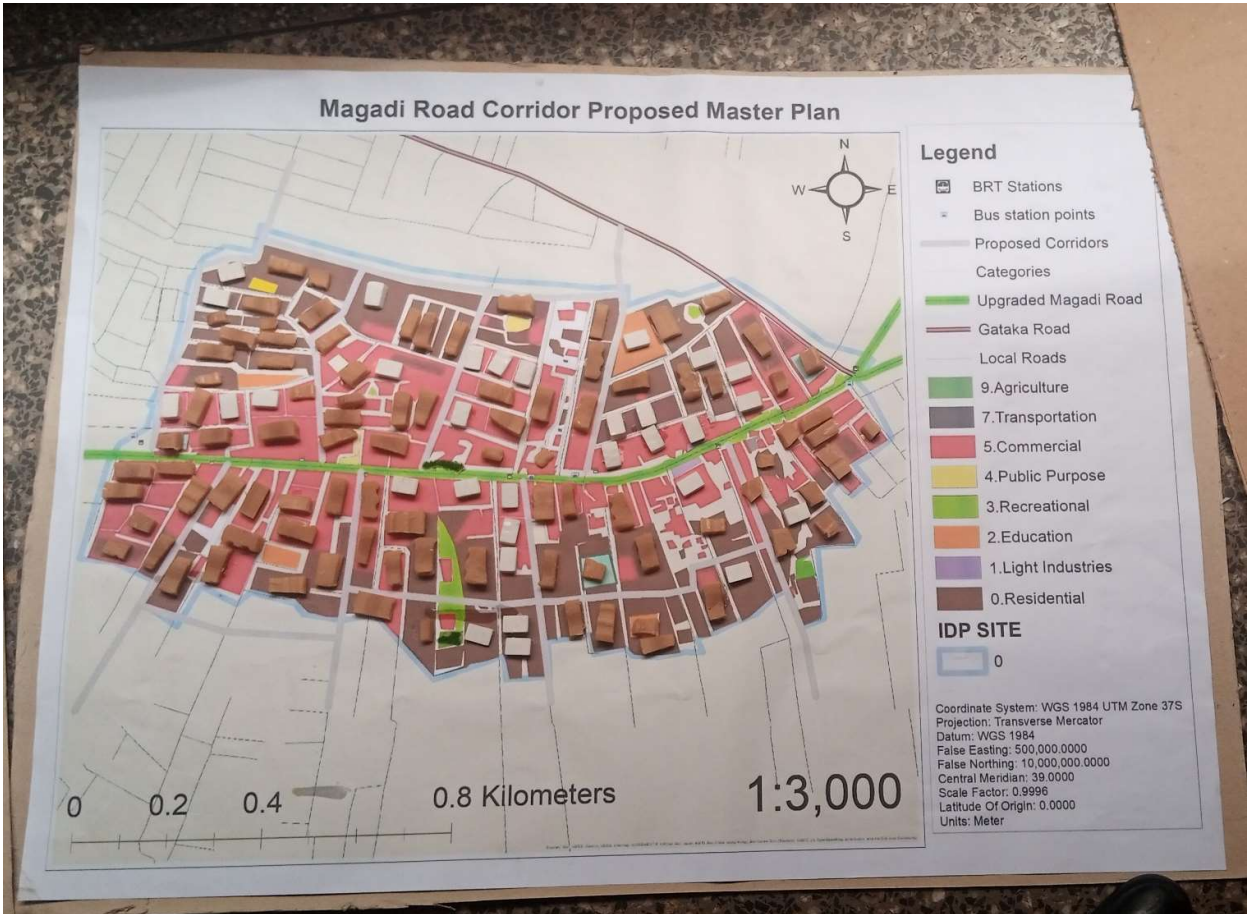
Thirdly, the plan should focus on integrated transportation solutions and economic development initiatives. Prioritizing the implementation of the Bus Rapid Transit (BRT) system, along with enhancing cycling and pedestrian infrastructure, can significantly reduce traffic congestion and improve public transit efficiency. Concurrently, supporting local economic growth through infrastructure enhancements and fostering small business development will stimulate job creation, economic resilience, and community well-being.

By implementing these recommendations, the Magad Road Corridor Development and Management Strategy Plan can achieve its goals of creating a more sustainable, livable, and resilient urban environment for the residents of Oshana Region.

ANNEX 2A: Image of 3D Physical Model of the Existing Situation of Magadi Road Corridor



ANNEX 2B: Image of 3D Physical Model of Proposed Master Plan



ANNEX 2C: Image of 3D Physical Model of Action Plan

